Technical Preprint — Version 1

S³ LIGHT-TRISPHERE

A Technical Atlas of an Interactive First-Principles Scientific Universe

Architecture, Scientific Contracts, Laboratories, Typed Instruments, Provenance, Navigation and Machine Access

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Scope. An architectural and operational map of the executable Atlas. Mathematical identities, catalogue entries, reduced models, conditional reconstructions and proposed bridges retain distinct evidential status. This manuscript does not replace the interactive system.

Abstract

S³ LIGHT-TRISPHERE is an interactive three-dimensional scientific atlas organized into seven worlds, a registry of 133 laboratories and 145 typed computational instruments in release 4.352.0. It combines astronomical catalogue layers, mathematical constructions, reduced physical models, a conditional compact-three-sphere reconstruction and a golden-ratio scale ladder. Laboratories supply spatial contexts and inspectors; typed instruments supply declared inputs, domains, outputs and verification references. These entities are connected in two different ways: Invariant Nexus represents explicitly typed scientific relationships, while the quantity bus transfers compatible numerical quantities under declared conversion and refusal rules. Provenance, formulas, model boundaries, navigation routes and machine-readable descriptions make the system addressable by researchers, developers and autonomous agents.

This preprint documents that architecture, the human navigation model, representative laboratory computations, source classifications, exports, validation mechanisms and the WebXR control layer. It provides complete laboratory and instrument indexes, a principal formula register, source-tag records and reproducible machine-access examples. The generated release manifest supplies the catalogue census. Scientific contracts separate native mathematical structure, parameter semantics, provenance and display conventions, while explicit status fields distinguish declared metadata from unresolved entries. Source provenance, numerical verification and empirical validation retain distinct meanings. The document provides an architectural and operational map of the Atlas, whose executable calculations, parameter exploration, spatial navigation and device-dependent interaction form the working research environment.

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1 Introduction

S³ LIGHT-TRISPHERE provides a common research interface for scientific objects that ordinarily live in separate tools: ephemerides, geometric spaces, field solvers, compact-object models, topological invariants and computational contracts. Its unifying object is an interface, not a claim that all represented systems have one physical mechanism. A laboratory can share mathematics with another laboratory while using different units, assumptions, observables and empirical evidence.

The Atlas solves an organizational problem at the boundary between scientific calculation and inspection. A user should be able to ask what is being drawn, what quantity a control changes, which equation produces a displayed value, where that equation is valid and how the result can be exported. Spatial organization can make a fibre, orbit, phase loop or interior profile inspectable in ways a static plot cannot. The corresponding risk is that attractive geometry appears to establish more than the calculation supports. This preprint makes the semantic boundaries explicit.

1.1 Audience and reading paths

Researchers should begin with scientific discipline, the seven worlds and the detailed laboratory walkthroughs. Developers should read the architecture, navigation, contracts and quantity bus. Agents should use the machine-interface section together with the complete registries and the API reference. Reviewers should compare the scientific-status taxonomy, source inventory, reconstruction assumptions and validation limitations. An exact identifier is more reliable than a translated title when passing from the manuscript to the software.

1.2 Release identity and scientific records

The documented release is 4.352.0, build `geometry-evidence-2026.10.02.1`, at repository commit `a9a219c6de92e1a9db2ad3f898ccdd626c7af3db`. Its executable entry point is `index.html`, with Git blob SHA `50404d8e2fdfb8170b3ea6c78259eb9c3c9af2df`. The release manifest defines the world, laboratory and instrument census; the scientific contract defines datum statuses and relation semantics; the agent descriptor identifies machine-access resources and transports. [1, 2, 4, 5]

The Atlas maintains complementary scientific records: registry entries identify available contexts and operations, instrument contracts specify inputs and outputs, formula records state calculations and source tags, and verifier references identify numerical or mathematical checks. These records support traceability; empirical validation additionally requires a model-specific comparison with observations.

Table 1: Release census and snapshot distinctions.

Source	Worlds	Labs	Instruments	Interpretation
Structural catalogue	7	133	145	Complete names; narrative contains older counts.
HTML registries	7	133	145	Static declarations including generated additions.
Release manifest	7	133	145	127 computational and 6 parametric laboratories.
Nexus graph	—	132	—	408 typed edges; a different registry.

1.3 Discrepancies that affect interpretation

Earlier descriptive material lists Nexus as 69 nodes and 144 relations; the release registry contains 132 nodes and 408 relations. The HTML's first formula-array literal contains 126 entries, followed by seven appended definitions: the effective principal registry has 133. The release manifest reports 57 quantity-bus links and 39 refusals; neither is a Nexus edge count. The package file retains version 4.270.0 while the HTML, manifest and `version.json` identify release 4.352.0. Embedded integration metadata also retains a 4.150.0 source-manifest stamp. These identifiers refer to different records and must be distinguished when reproducing a calculation.

The earlier S³ fiducial radius and horizon values are superseded by a parameterized conditional reconstruction in this release (section 18). The premium-rendering pipeline is enabled by default with a user opt-out. The executable HTML imports local core and visual modules, so local deployment requires the repository's module layout rather than an isolated HTML file.

2 Scientific and Epistemic Discipline

The scientific contract separates a datum's evidential status from its software verification status. `measured`, `inferred`, `derived`, `simulated`, `hypothetical` and `illustrative` are datum-level categories. Laboratory badges such as `VERIFIED`, `THEOREM`, `MODEL` or `CONDITIONAL` serve a different purpose. A laboratory labelled VERIFIED can expose an internally checked mathematical identity without supplying observational evidence. [4]

Table 2: Epistemic classes and admissible reading.

Class	What supports it	What does not follow
Observation / catalogue	Named measurement or published catalogue, frame and epoch	Every displayed component is directly observed.
Standard / literature model	Cited assumptions and equations	Accuracy outside its calibrated or declared regime.
Mathematical identity	Defined spaces, conventions and derivation	Identity of distinct physical realizations.
Numerical experiment	Algorithm, discretization, seed, residual and tolerance	General theorem or empirical confirmation.
Reduced model	Explicit omitted terms and limited degrees of freedom	Full engineering, stellar or cosmological simulation.
Conditional S³	Round compact spatial geometry and conditioned curvature inputs	Detection of cosmic topology.
FBS3R	Published ladder construction and its internal anchors	Replacement of standard cosmology.
Visualization	Projection, semantic marker or layout contract	A metric or measured scalar.
Proposed bridge	Typed hypothesis and a review/admission requirement	An established relation merely because it ranks highly.

A reconstruction screen is the boundary of a chosen observable or reconstructible domain. It is not an edge of the universe: the mathematical carrier satisfies $\partial S^3_R = \emptyset$. A quasi-local region can be approximately Euclidean while embedded in a globally curved mathematical space. Neither statement selects the topology of the observed universe.

A useful audit asks five questions of a value: what is the quantity kind, in which unit, in which frame and clock, under which model, and from which source or computation? Scientific precision means answering those questions, not merely displaying more decimal places. A residual tests a specified identity or algorithm. It does not automatically assess systematic error, observational calibration or model adequacy.

Missing metadata is an explicit result. `UNDECLARED` must not be filled by recognizing a familiar symbol or reading a colour. A beta in an impedance map and a beta in a skyrmion model may both be dimensionless yet denote different coordinates. Source tags similarly carry scoped provenance: `DERIVED` identifies computation, whereas a catalogue tag identifies a data lineage. Accuracy envelopes belong to the relevant approximation and epoch, not to the Atlas as a whole.

3 System Architecture

World. A top-level navigation and scene context with its own scale, controls and default entry. There are seven worlds.

Laboratory. A registered scientific context, typically a native construction or solver with a view, inspector, controls, route and model boundary.

Typed instrument. An inspectable computational specification with identifier, inputs, domains, outputs and status. A laboratory may host several instruments, and an instrument need not be a laboratory.

Visible object. A rendered or labelled selectable entity whose card can expose quantities, source tags and actions. Object count is not laboratory count.

Relation. A typed scientific claim between contexts; it need not execute a calculation.

Quantity-bus link. A declared operational connection carrying an actual source output into a compatible destination input.

Contract. Machine-readable semantics and constraints; a contract can truthfully contain undeclared fields.

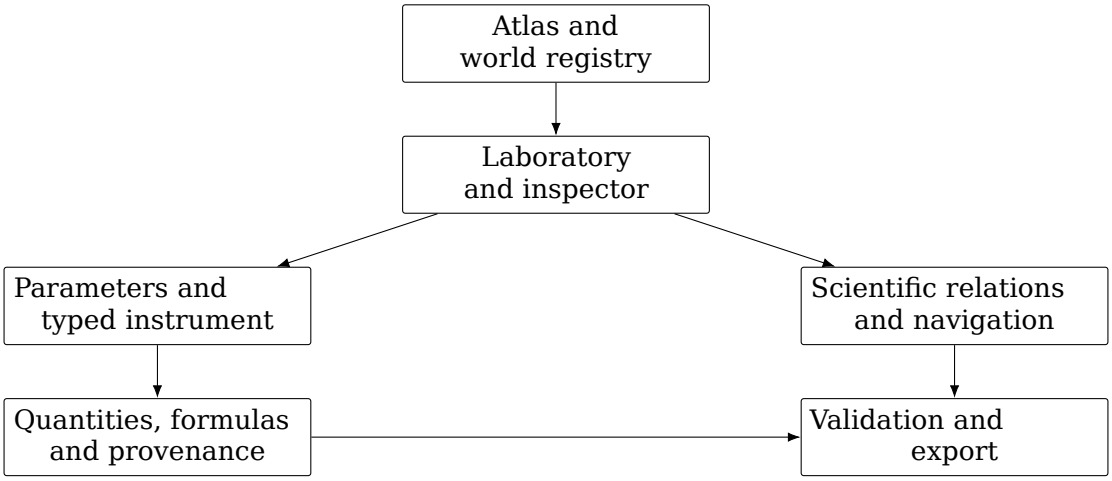


Figure 1: Conceptual architecture derived from the registries. Arrows indicate inspection and data dependencies, not physical causality.

The browser layer combines Three.js scenes, DOM controls and object registries. Core modules provide reusable computations and machine entry points. The public

static deployment exposes files and client-side computation; a separately launched server supplies HTTP/MCP computation. These transports must not be conflated. A field called `lab` in an instrument description means its associated interface context; historically named schema `hcc.laboratory/1` describes an instrument, so schema names alone cannot determine entity type.

The architecture has several complementary indexes: `WORLD_REGISTRY`, `LAB_REGISTRY`, `HCC_API_SPECS`, `FORMULA_REGISTRY`, `SOURCE_MAP`, Nexus relations and the bus wiring. A change to one registry need not change the others. For example, adding a conceptual laboratory can enlarge navigation before it has a typed evaluator. The manifest's capability classification makes this distinction machine-visible: the six parametric laboratories are Nexus, Symmetry Discovery, eigenmodes, rings, Center Lab and DRD.

The information path in figure 1 is also a debugging path. If a displayed quantity is suspicious, identify its object or inspector, its instrument/output name, the evaluator and inputs, then the formula and source. If a requested computation is unavailable, return that absence at the appropriate layer rather than trying to recover a number from its graphic.

4 The Seven Worlds

The seven worlds are operational entry points, not seven mutually exclusive scientific disciplines. Their exact routes and default entries are listed in appendix A.

4.1 Solar Atlas / Sun to Cosmos

Entry. Open this world; registry default `solar`.

The Solar world begins with a heliocentric ecliptic J2000 frame and one scene unit per astronomical unit. Planetary positions come from the implemented Keplerian ephemeris, not from drawing orbital ellipses at arbitrary phases. Body-size exaggeration is separate from orbital distance; the true-scale option removes that exaggeration. Reference planes, motion vectors and the hierarchy locator communicate orientation, but the locator explicitly uses semantic marker spacing.

Controls include date and time rate, pause and direction, heliocentric/geocentric framing, orbital lines, sky catalogues, true scale and scale hand-offs. Selecting a body opens a scientific card; Focus moves the camera, and supported bodies provide a surface view. JPL, lunar, astrometric and catalogue tags must be inspected at the chosen epoch. The world extends outward through galactic and extragalactic layers; a navigation hand-off is not a physical boundary.

The implemented sky is substantially richer than the older DOCX summary: the live interface describes all 88 constellations, 15,544 Hipparcos stars to $V = 7$, and separate galaxy and quasar catalogues. These are release-specific catalogue claims, not a completeness claim about the sky. Machine instruments concerning local flow, nesting, stellar distances and angular scales are attached to relevant contexts; the complete instrument index records their actual hosts. Exports preserve application state and source/quantity information. Approximate ephemerides and distance-quality filters set the scientific limits.

4.2 Cycles and Events

Entry. Open this world; registry default `cyc`.

Cycles places lunar, annual, precessional, galactic and historical clocks on a shared epoch while retaining their distinct meanings. Its display plane is explicitly a non-metric arrangement of instruments. A galactic orbit, a precession cone and a season dial cannot be read as one physical coordinate system just because they are simultaneously visible.

The user chooses clocks, reverses or advances the epoch, compares normalized phases, searches for bounded commensurabilities and visits event or chronology presets. A phase coincidence is a numerical search result; eclipse classification requires the separate ephemeris and shadow geometry. Ancient Chronometry is the one laboratory formally parented to this world. Imported cycle modules supply additional phase and galactic-event models. Calendar definitions remain source-qualified, and event dates retain the timing assumptions of their solver. Reproducibility requires the epoch, each period/anchor and the search bound, not merely a screenshot of aligned needles.

4.3 Observable Universe / Horizons

Entry. Open this world; registry default `obs`.

The Horizons world uses one scene unit per gigalightyear and distinguishes age/light-travel screens, last scattering, particle horizons and other distance concepts. Its purpose is to make the scale and boundary of a reconstructed observable domain intelligible. The displayed domain lies inside a conditional carrier; its screens do not establish the global topology.

Controls adjust layers and framing, and scale navigation leads toward the Solar astronomical layer or the S³ carrier. The Cosmic Web laboratory is scientifically associated with this world but formally registered under S³: navigation ownership and scientific family are different classifications. Its deterministic Zel'dovich construction must be distinguished from observed catalogue structures. Source tags, current reconstruction parameters and model-specific distances are necessary export context. The current build derives its carrier constants; historical fixed screen values must be distinguished from the parameterized reconstruction.

4.4 S³ Carrier and Hopf Structure

Entry. Open this world; registry default `nsflow`.

This world is the mathematical carrier and the principal laboratory host. Its global scene convention is one unit per 100 Gly, whereas individual laboratories can use normalized, phase-space or otherwise non-metric native constructions. The registry declares `nsflow` as its default entry, but preserved state and legacy entry controls can reopen another subview. An explicit laboratory route is therefore the reproducible entry point.

Controls include section/angle and mode parameters, observer-center constructions, Hopf fibres, laboratory selection, relation inspection and Multiview. The 132 S³ laboratory records cover much more than cosmology: quantum, wave, thermodynamic and astrophysical examples coexist here. Native dimensionality is not inferred from the rendered 3D surface. Mathematical residuals are evaluated before projection

where the station contract says so. The S³ world exports model state, laboratory results and graph/contract records, with conditional-cosmology caveats attached.

4.5 FBS3R φ-Ladder

Entry. Open this world; registry default `fbs`.

FBS3R maps scale to a logarithmic level N anchored to a Planck length. Its nested-shell rendering makes many orders of magnitude navigable. Level, traversal speed/range, density exponent, exact Fibonacci comparison and sensitivity controls have distinct roles. A shell's presentation radius is not automatically a directly measured physical radius.

The published recombination label $N_{\mathrm{rec}} = 266$, the independent CoScale clock and a physical-length conversion must be read separately. Exploratory changes to the ratio or Planck anchor are Tier 2 scenarios and hide canonical anchors rather than relabelling them as observed. The `ladder` and capacity-related instruments provide machine contracts; their hosts differ from ordinary laboratory routes. State and experiment exports should include the active anchors and exponent. The ladder does not establish that catalogued scales prefer integer golden-ratio rungs; the current source explicitly includes a non-significant catalogue test.

4.6 Field Laboratory

Entry. Open this world; registry default `field`.

Field Laboratory offers reduced lattice/PDE and Monte Carlo experiments rather than a universal field simulator. The current world chooser distinguishes a solver lattice, an SI-dimension lattice and a Sedov–Taylor blast-wave view. This is broader than the earlier seven-equation description. Only one of these three fills the shared display volume at a time; changing the visible instrument does not assert a physical coupling between them.

Within the solver, select the equation, seed, boundary treatment, rate and parameters, then compare the field with an analytic reference or conserved diagnostic. Heat, damped waves, Gray–Scott, Poisson-type electric/gravitational examples, Schrödinger packets, FitzHugh–Nagumo and Ising modes use reduced/reference assumptions. A computed field array, solver time and boundary are the scientific state. The displayed density and post-processing quality belong to presentation. Exports and bus use require named quantities and declared semantics; medical or engineering predictions cannot be inferred from these demonstrations.

4.7 Fractal Explorer

Entry. Open this world; registry default `fractal`.

Fractal Explorer treats iteration and geometry as mathematical objects. The GPU path uses distance-estimation/raymarch techniques for modes such as Mandelbulb; other modes construct true geometry, including orbitals, 4-polytopes, bifurcation structures and Mandelbrot–Julia fibre views. Iteration depth and zoom are not physical time or length unless a separate model explicitly supplies that meaning.

Controls are mode-dependent: powers, iterations, slices, camera movement and geometric parameters cannot be described by one universal solver contract. The cur-

rent world description emphasizes measured box-counting dimensions and the stated fit window. That estimate is sensitive to finite resolution and sampling; visual self-similarity is not an exact proof. True-geometry modes can enter XR, while mono fullscreen raymarch modes are replaced by a supported geometry mode. Machine access should discover attached instruments such as `polytope`, `divalg` and `mixmaster`, not assume a typed instrument named fractal exists.

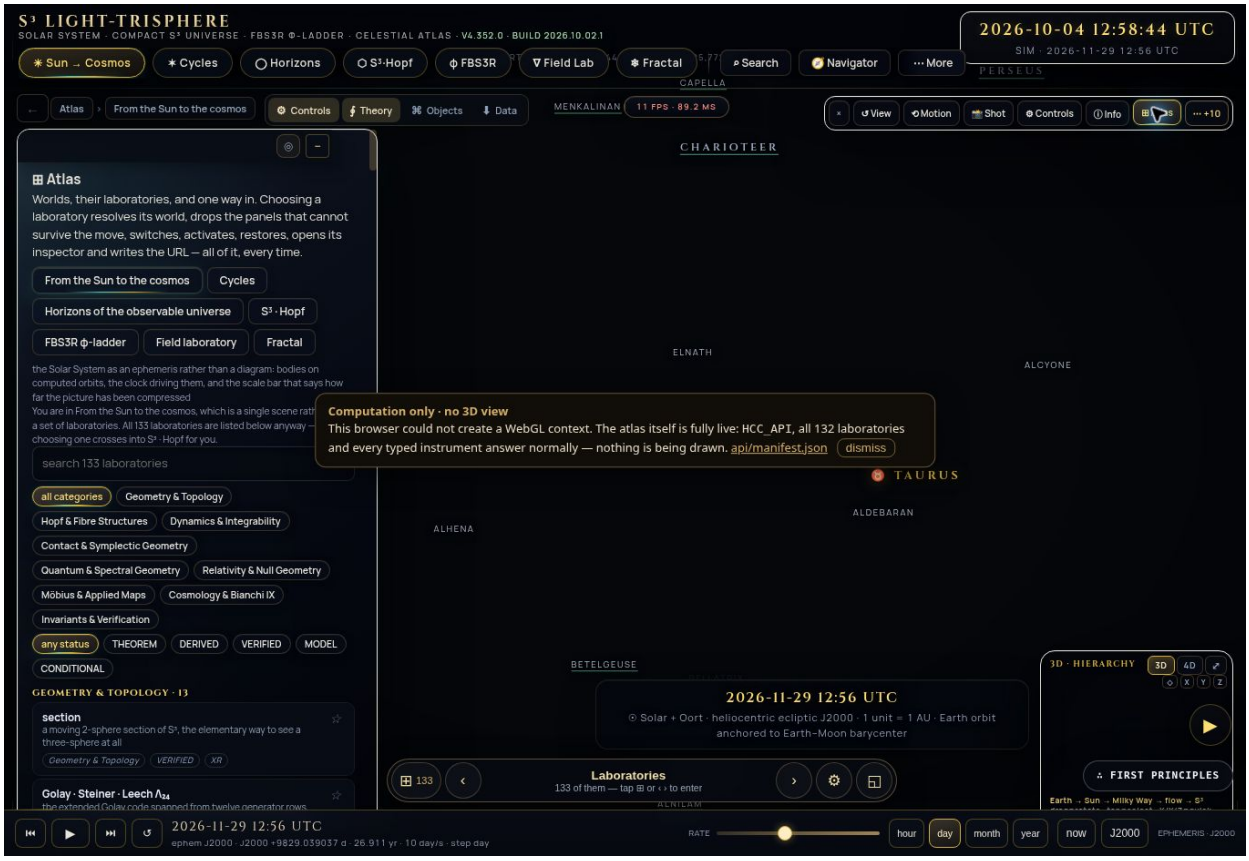


Figure 2: The seven-world switcher and Atlas catalogue, with category and status controls.

5 Human Navigation Model

The implemented hierarchy is Atlas → World → Laboratory → Inspector. The navigation registry resolves a laboratory to its parent world before activating its view and writing the route. The source explicitly replaces earlier scattered entry points, hidden laboratory buttons and DOM-class inference. Old audit descriptions of zero-size buttons or absent hashes describe the problem the current architecture addressed; they are not instructions for using this build.

The world switcher chooses a scene context. The Atlas catalogue lists laboratories with category/status filters, search and favourites; the lower laboratory rail provides previous/next movement and comparison access. Atlas Navigator and the First-Principles navigator add journeys and object discovery. Universal search accepts names, identifiers and scientific terms; it is distinct from a laboratory's local controls filter and the formula search.

The inspector presents Controls, Theory, Objects and Data for the active context. A selected-object card is narrower: it describes a particular planet, fibre, screen, lab-

oratory proxy or other object and supplies relevant focus/actions and provenance. A double activation can approach a target; it does not run a laboratory's numerical evaluator by itself. Surface landing and object-following have camera semantics separate from changing scientific parameters.

Use the breadcrumb and Back to recover context. A world route has the form `#/world/solar`; a laboratory route is `#/world/s3/lab/hol`. Ancient Chronometry uses `#/world/cyc/lab/chronometry`. Reloadable routes identify contexts, but they do not encode every slider, station, camera and solver buffer. Exported state is required when those details matter.

Panels have scopes and are removed when they cannot apply to the new world. A pinned panel retains its declared behaviour; it is not a guarantee that any control remains scientifically meaningful everywhere. Desktop shortcuts, touch bottom sheets and horizontal mode rows expose the same conceptual hierarchy through different layouts. figure 2 shows the entry catalogue; appendix I supplies a compact interaction reference.

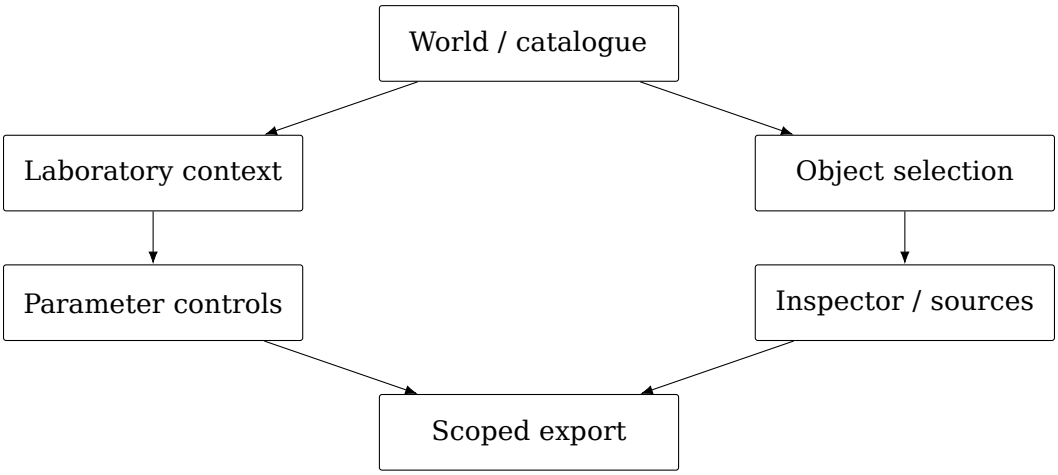


Figure 3: Human navigation separates laboratory state from object focus; both contribute to an interpretable export.

6 Laboratory Architecture and Scientific Families

A well-specified laboratory joins three layers. First is the native scientific object: a sphere, spinor space, phase space, grid, categorical state space or reduced stellar interior. Second is its computational experiment: input parameters, a formula or solver, diagnostics, tolerances and validity limits. Third is its display: projection, camera, geometry, controls and selection. Confusing these layers can make a correct picture communicate an incorrect scientific claim.

An analytic anchor is most useful when the residual is evaluated independently. Subtracting an expression from a value built by the identical expression can confirm serialization or wiring but does not independently validate the mathematics. The Holonomy polygon and analytic solid angle illustrate a stronger comparison. Finite discretizations still require convergence and tolerance reasoning. Validation of a native invariant should precede stereographic projection or visual simplification.

The manifest classification is operational: a computational laboratory has an attached typed instrument; a parametric one has controls but no attached typed e-

valuator. The current six parametric records are `nexus`, `syd`, `eig`, `ring`, `lab` and `drd`. Parametric does not mean scientifically empty: Nexus and Symmetry Discovery perform substantial internal computations. It means their public typed-instrument association is absent in this catalogue.

Navigation categories and scientific families are different. The nine navigation categories are optimized for finding views; the six Nexus clusters organize relation geography. For explanatory purposes the following families overlap both.

Table 3: Scientific families and representative contexts.

Family	Representative laboratory identifiers
Geometry and topology	`topo`, `dfx`, `wil`, `leech`, `s3gamma`
Quantum and matter	`qm`, `spin`, `atom`, `qcd`, `he3`, `nuosc`
Waves and materials	`wave`, `disp`, `imp`, `sc`, `te`, `pv`
Dynamics and phase space	`grav`, `chaos`, `kam`, `poin`, `psp`, `act`
Thermodynamics and dense matter	`kin`, `he`, `bb`, `eos`, `wd`, `ns`
Astrophysics and observation	`psr`, `qso`, `adisk`, `phot`, `resolve`, `photon`
Cosmology and structure	`cmb`, `cosmicweb`, `cosmo`, `lam`, `mimg`
Stellar evolution and compact objects	`mainseq`, `seis`, `wind`, `sn`, `bht`, `bhr`, `bbh`
S³ fluids and spectra	`s3shell`, `s3torus`, `nsflow`, `nsgal`, `triad`, `vstretch`
Symmetry and transport	`syd`, `hol`, `nul`, `noe`, `cps`
CIVP and braiding	Seven `civp*` stations, `emb`, `anyon`, `jonesq`, `anyzoo`
Chronometry	`chronometry`, `osc`, `clock`, `omnistrobe`

The complete registry in appendix B preserves every identifier and title. Its route is authoritative even where a short description is historically misleading: for example, the Center Lab description calls it an entry bench while the world registry now selects `nsflow` by default. It is better to report this divergence than silently edit a registry's wording into a new scientific contract.

7 Representative Laboratories: Detailed Technical Walkthroughs

The following cases use the same audit sequence: purpose and question; native construction; inputs and controls; calculation and diagnostic; visual encoding; provenance and relations; machine access and export; limitations and a reproducible workflow. Native spaces described mathematically below must not be confused with completeness of the strict metadata fields: a source can implement a construction while its `first_principles.native_space` remains `UNDECLARED`.

7.1 Spinor and Light-Cone Observatory

Purpose and question. exact CP¹, SL(2,C), null-tetrad, spin-weight and radial conformal-infinity identities. How does a two-spinor determine a null direction, and what survives a change of phase or Lorentz frame?

Native construction. Normalized spinors occupy $S^3 \subset \mathbb{C}^2$; quotienting a common phase gives $\mathbb{CP}^1$. Minkowski vectors are represented by Hermitian 2×2 matrices. These are native algebraic constructions, not three-dimensional screen coordinates.

Table 4: Declared input contract for nul. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`station`	null	null|lorentz| celestial	string
`theta`	0.7	rad	[0.01, 3.13]
`phi`	1.1	rad	[0, 2π]
`gauge_phase`	1.137	rad	[0, 2π]
`boost_rapidity`	0.6	dimension-less	[-4, 4]
`boost_angle`	0.35	rad	[0, 2π]

Controls and computation. Compare $\det X$ before and after $X \mapsto AXA^\dagger$ and compare the null ray before and after a common phase. A double-cover sign changes the spinor without changing the induced Lorentz map. The laboratory controls include speed, θ, φ, Delta u, sigma(u), Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `nullity`, `future_directed`, `spinor_norm_residual`, `determinant_modulus`, `phase_drift`, `interval_before`, `interval_after`, `interval_residual`, `double_cover_residual`, `cross_ratio_residual`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `nul` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (every station is a representation identity in flat spacetime, in closed form; the residuals below are floating point and are returned rather than asserted away). The source declares `docs/verify-atlas-instruments.cjs(nullity,phaseinvariance,theSL(2,C)interval,thedoublecover,thecrossratio)`.

Machine access and export. Open Spinor and Light-Cone Observatory. Use `HCC_API.describe("nul")` before `HCC_API.evaluate("nul",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. The UI additionally presents ray-bundle and causal-diamond material; the typed core describes three primary stations. Do not treat five UI presentations as five identical API station values. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.2 Holonomy Observatory

Purpose and question. closed-path return maps across round-sphere transport, Berry phase, SU(2), Lorentz boosts and SL(2,R) Möbius actions. What return map remains after a closed path, and which comparison is invariant under a change of frame?

Native construction. The stations use tangent transport on S^2, a quantum phase line, $SU(2)$, Lorentz boosts and $SL(2,\mathbb{R})$ respectively. A common loop metaphor does not merge these spaces.

Table 5: Declared input contract for hol. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`station`	sphere	sphere| berry|su2| wigner| mobius	string
`a`	0.55	dimension-less	[0, 1]
`b`	0.65	dimension-less	[0, 1]

Controls and computation. For sphere transport compare the wrapped angle with enclosed solid angle; for Berry phase compare with minus one-half the oriented solid angle. Group stations compare conjugacy invariants rather than a plotted arrow alone. The laboratory controls include speed, colatitude θ, sphere radius R, Delta u, sigma(u), Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `phase`, `closed_form`, `residual`, `tolerance`, `within_tolerance`, `solid_angle`, `invariant`, `gauge_drift`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `hol` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (each station has a closed form; the discrete transport, the 257-vertex Wilson loop and the matrix products are its discretisations and every residual is returned rather than asserted away). The source declares `docs/verify-atlas-instruments.cjs(solidanglevstransport,Berryvs-Omega/2,gaugeinvariance,Lorentzclosure,Mobiusconjugacy)`.

Machine access and export. Open Holonomy Observatory. Use `HCC_API.describe("hol")` before `HCC_API.evaluate("hol",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. Use angles modulo the declared period. A large raw difference across a branch cut can represent a small physical residual. Reset the station,

retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

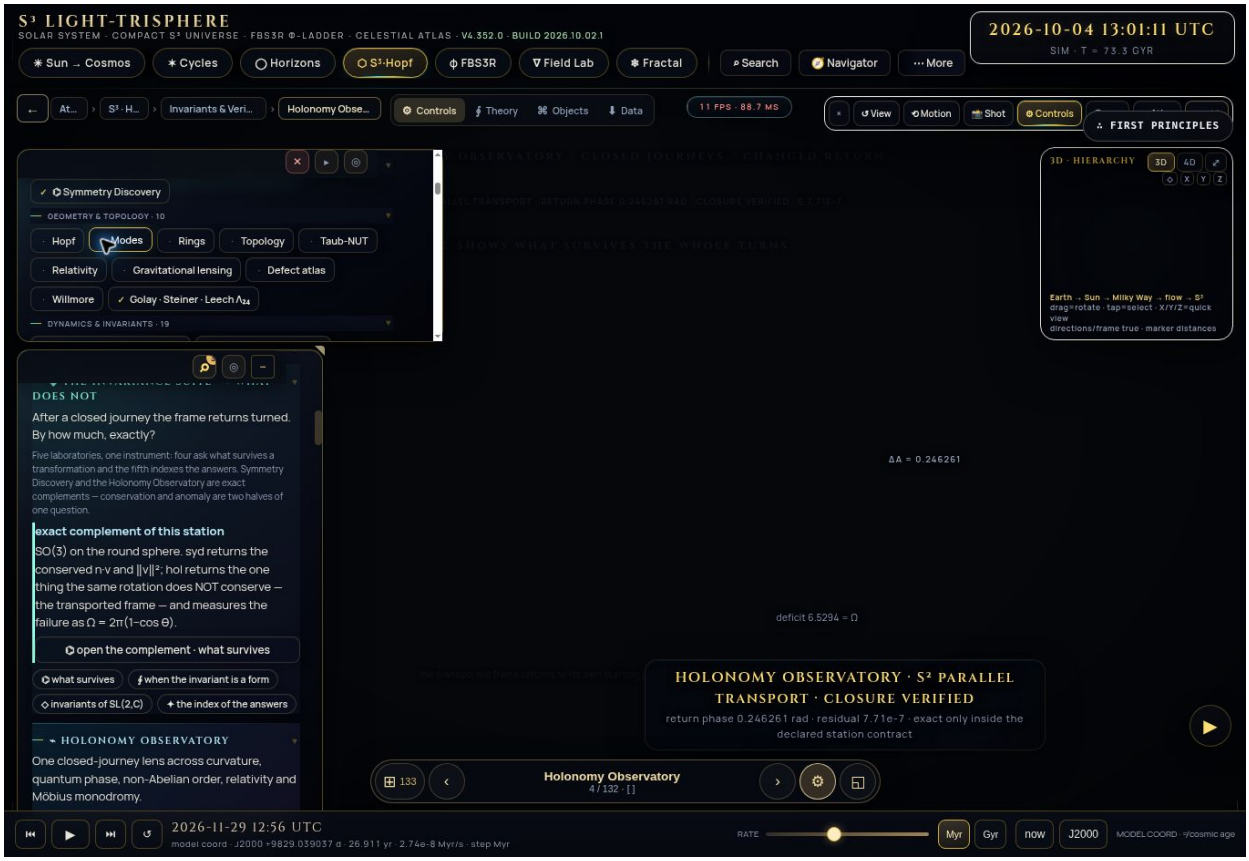


Figure 4: Holonomy Observatory interface: station controls and native diagnostic readouts.

7.3 Contact and Action Observatory

Purpose and question. contact dynamics on an S³ energy shell, exact Reeb return, Legendrian curves, KS regularization and Maslov action. How do a contact form, its Reeb flow and a phase quotient constrain motion on an energy shell?

Native construction. The unit S^3 is embedded in $\mathbb{R}^4$, with $\lambda = \frac{1}{2}\sum_j (x_j\,dy_j - y_j\,dx_j)$ and Reeb field $R = 2Jx$. The contact plane is $\ker\lambda$.

Table 6: Declared input contract for act. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`station`	reeb	reeb| legendrian| ks	string
`eta`	0.7	rad	[0.05, 1.5]
`p`	2	dimension-less	[1, 9]

Continued from previous page

Input	Default	Unit	Domain
`q`	3	dimensionless	[1, 9]
`rho`	0.8	dimensionless	[0.05, 5]
`gauge_angle`	0.4	rad	[0, 2π]

Controls and computation. Check $\lambda(R) = 1$ and contraction into $d\lambda$ on the contact plane. For a Legendrian curve test $\lambda(\dot{\gamma}) = 0$ before projection; for the KS map test its radius relation. The laboratory controls include speed, Clifford latitude η, relative phase δ, Delta u, sigma(u), Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `sphere_residual`, `lambda_R_residual`, `contraction_residual`, `hopf_drift`, `legendrian_residual`, `coprime_p`, `coprime_q`, `ks_radius`, `ks_identity_residual`, `ks_gauge_residual`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `act` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (the contact identities are evaluated in R^4 in closed form and hold to the last bit; the linking number is a finite Gauss sum after stereographic projection and carries its own discretisation, which is returned separately). The source declares `docs/verify-atlas-instruments.cjs(lambda(R)=1,thecontraction,theHopfimage,theLegendriantangency,theKSidentity)`.

Machine access and export. Open Contact and Action Observatory. Use `HCC_API.describe("act")` before `HCC_API.evaluate("act",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. The broader UI includes return/action and Maslov presentations, while the typed contract has its own narrower station domain. A projected curve is not a contact-invariant measuring ruler. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.4 Radiation-MHD and Helicity Observatory

Purpose and question. five reduced analytic and kinematic stations for flux freezing, magnetic helicity, reconnection, shocks and magnetized Rayleigh–Taylor instability. Which ideal invariants and instability scalings survive in separately declared reduced models?

Native construction. The five stations are analytic or kinematic constructions for flux freezing, helicity, reconnection, shocks and magnetized Rayleigh–Taylor instability.

Table 7: Declared input contract for rmhd. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
mach	8	dimensionless	[1, 100000]
gamma_gas	1.6666666666666667	dimensionless	[1.05, 2]
B	1	dimensionless	[0.01, 100]
rho	1	dimensionless	[0.01, 100]
log_lundquist	4	dimensionless	[1, 14]
atwood	0.55	dimensionless	[0, 1]
wavenumber	1.4	dimensionless	[0.01, 20]
tension	0.28	dimensionless	[0, 5]

Controls and computation. Compare each output with its station-specific identity or asymptotic law. A flux-freezing comparison has a different validity domain from a reconnection model, in which ideal flux conservation is deliberately relaxed. The laboratory controls include Speed, B / ρ, B / ρ, Cauchy shear, $\log_{10}$ Lundquist S, Shock Mach, RT A / k / tension. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `compression`, `compression_ceiling`, `compression_headroom`, `pressure_jump`, `temperature_jump`, `alfven_speed`, `reconnection_rate`, `plasmoid_unstable`, `rt_growth_rate`, `rt_stable`, `rt_critical_wavenumber`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `rmhd` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (the Rankine-Hugoniot jump conditions, the Alfven speed, the Sweet-Parker rate and the magnetised Rayleigh-Taylor growth rate are all closed form; they are reduced analytic stations and not a self-consistent radiation-GRMHD evolution, which is what the limits are about). The source declares `docs/verify-atlas-instruments.cjs(thestrong-shockceiling,theAlfvenscaling,S^-1/2,theRTstabilityboundary)`.

Machine access and export. Open Radiation-MHD and Helicity Observatory. Use `HCC_API.describe("rmhd")` before `HCC_API.evaluate("rmhd",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. The name is not evidence of a coupled radiation-MHD PDE

solver. No global energy-conserving radiation/field/fluid evolution is established by these five demonstrations. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.5 White-Dwarf Degeneracy Laboratory

Purpose and question. Nauenberg zero-temperature composition-dependent mass–radius family and Chandrasekhar endpoint. How does the zero-temperature mass–radius curve depend on composition and approach its endpoint?

Native construction. This is a reduced stellar-structure relation, with electron molecular weight controlling the Chandrasekhar scale.

Table 8: Declared input contract for wd. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`mass`	0.85	M_sun	[0.05, 1.45]
`mu_e`	2	dimension-less	[1, 3]

Controls and computation. Use the Nauenberg family $R \propto [(M_{\mathrm{Ch}}/M)^{2/3} - (M/M_{\mathrm{Ch}})^{2/3}]^{1/2}$ within its allowed mass range. Test composition sensitivity and approach the endpoint from below. The laboratory controls include Mass M/M⊙, μ_e=A/Z, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `radius_km`, `chandrasekhar_mass`, `mean_density`, `compactness`, `surface_redshift`, `escape_velocity`, `distance_to_limit`, `radius_earths`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `wd` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is MODEL (Nauenberg 1972 zero-temperature analytic fit to the Chandrasekhar equation of state; the limiting mass is exact for the model and the radius is a fit, so the status is MODEL and not EXACT). The source declares `docs/verify-atlas-instruments.cjs(thelimit,thenon-relativisticscaling,monotonicity,theSiriusBcomparison)`.

Machine access and export. Open White-Dwarf Degeneracy Laboratory. Use `HCC_API.describe("wd")` before `HCC_API.evaluate("wd",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. Thermal structure, rotation, magnetic support and detailed atmospheres are outside this relation. The mathematical radius limit is not a complete collapse simulation. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.6 Black-Hole Thermodynamics

Purpose and question. Kerr horizon radius, ergoregion, area entropy, angular velocity and a separate Schwarzschild-only evaporation worldtube. How do horizon geometry, entropy and temperature scale with mass and spin under the declared stationary model?

Native construction. Kerr geometry supplies the horizon and ergoregion; a separate evaporation presentation is explicitly Schwarzschild-only.

Table 9: Declared input contract for bht. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`M`	1.98892e+30	kg	[1e-08, 1e+45]
`chi`	0	dimensionless	[0, 0.999999]

Controls and computation. For $a_* = cJ/(GM^2)$, $r_+ = (GM/c^2)(1+\sqrt{1-a_*^2})$. Compare the zero-spin limit and check the extremal boundary without extending past $|a_*| = 1$. Entropy uses horizon area, not the apparent on-screen silhouette. The laboratory controls include Dimensionless spin χ, log₁₀(M/kg), Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `r_plus`, `area`, `T_H`, `Omega_H`, `entropy`, `spin_root`, `t_evap_yr`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `bht` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (closed form for the Kerr family; the evaporation time is the Schwarzschild Page result and is labelled as such). The source declares `docs/verify-atlas-instruments.cjs(independentroutes)`.

Machine access and export. Open Black-Hole Thermodynamics. Use `HCC_API.describe("bht")` before `HCC_API.evaluate("bht",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. Do not combine the spinning horizon with a Schwarzschild evaporation lifetime as if the latter integrated Kerr spin evolution. Hawking reference curves are theoretical model outputs, not observations of evaporation. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.7 Neutron-Star / TOV Laboratory

Purpose and question. reduced Tolman–Oppenheimer–Volkoff interior with declared equation-of-state dependence. What equilibrium profile follows from a chosen

equation of state and central condition?

Native construction. A radial hydrostatic model integrates the Tolman–Oppenheimer–Volkoff equations with explicit equation-of-state dependence.

Table 10: Declared input contract for ns. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`central_density`	0.00128	code	[1e-06, 0.02]
`step`	0.002	code	[0.0001, 0.01]

Controls and computation. Inspect central regularization, the pressure-zero stopping condition and the mass profile. Compare sequences under a controlled equation-of-state change rather than treating one returned radius as universal. The laboratory controls include $\log_{10}$ ρ_c, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `mass`, `radius_km`, `radius_code`, `compactness`, `buchdahl_margin`, `surface_redshift`, `newtonian_radius_ratio`, `is_past_maximum`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `ns` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is MEASURED (a real fourth-order integration of the Tolman-Oppenheimer-Volkoff equation for a Gamma = 2, K = 100 polytrope; the equation is exact and the numbers are its converged solution, verified against the exactly known Newtonian limit). The source declares `docs/verify-atlas-instruments.cjs(theNewtoniann=1radius,stepconvergence,themaximum,theBuchdahlbound)`.

Machine access and export. Open Neutron-Star / TOV Laboratory. Use `HCC_API.describe("ns")` before `HCC_API.evaluate("ns",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. A reduced TOV model is static and spherical; rotation, magnetic stresses, temperature and an observational mass–radius likelihood require additional modelling. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.8 Supernova Laboratory

Purpose and question. Bateman abundances, grey gamma escape, stable Arnett diffusion, a conditional photosphere, a smooth ejecta/Sedov match and a radius-continuous radiative phase switch. How do decay power, diffusion, leakage and expansion phases shape a reduced transient?

Native construction. The source combines Bateman abundances, grey gamma escape, stable Arnett diffusion, a conditional photosphere and matched expansion phases.

Table 11: Declared input contract for sn. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`days`	50	d	[0, 2000]
`nickel_mass`	0.6	M_sun	[0.001, 5]

Controls and computation. Check abundance balances and phase-boundary continuity separately. A smooth ejecta/Sedov match and radius-continuous radiative switch impose specific continuity properties, not all derivatives or conservation laws automatically. The laboratory controls include Speed, Epoch, M(^{56}Ni), M_ejecta, v_ejecta, κ_opt, n_0 ISM. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `fraction_Ni`, `fraction_Co`, `fraction_Fe`, `fraction_sum`, `luminosity`, `luminosity_Ni`, `luminosity_Co`, `cobalt_dominates`, `decay_timescale_days`, `magnitudes_per_100d`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `sn` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT (the Bateman solution of a two-step decay chain is closed form, and the three fractions sum to 1 identically at every time; the half-lives and the decay energies are measured quantities and are quoted as such). The source declares `docs/verify-atlas-instruments.cjs(thesum,the1/epoint,thepeakofthecobaltcurve,thelate-timeslope)`.

Machine access and export. Open Supernova Laboratory. Use `HCC_API.describe("sn")` before `HCC_API.evaluate("sn",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. The displayed light curve depends on reduced opacity and deposition assumptions. It is not a multidimensional explosion, nucleosynthesis or full radiative-transfer calculation. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.9 Pulsar Laboratory

Purpose and question. magnetic-dipole lighthouse geometry and spin-down reference relations. How do a rotating magnetic geometry and dipole spin-down reference relations affect observables?

Native construction. The lighthouse representation is a viewing geometry; spin-down quantities belong to a separate reduced torque prescription.

Table 12: Declared input contract for psr. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`period`	0.0334	s	[0.001, 100]
`period_derivative`	4.21e-13	s/s	[1e-22, 1e-08]
`alpha`	0.6	rad	[0.01, 3.13]
`zeta`	0.8	rad	[0.01, 3.13]

Controls and computation. Vary the line of sight and magnetic inclination to distinguish a beam crossing from an intrinsic luminosity change. Check the period/period-derivative assumptions before interpreting characteristic age or inferred field. The laboratory controls include $\log_{10}$P (s), $\log_{10}\dot{P}$, Tilt α / view ζ, Tilt α / view ζ, Beam width, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `B_surface`, `characteristic_age`, `spin_down_luminosity`, `rotational_energy`, `rvm_max_slope`, `rvm_slope_measured`, `rvm_swing`, `death_line_ratio`, `is_millisecond`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `psr` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is MODEL (the magnetic dipole braking model with n = 3, R = 10 km, I = 1e45 g cm^2 and orthogonal rotation; every formula is exact for that model and the model is what the limits are about). The source declares `docs/verify-atlas-instruments.cjs(theCrabagainstitscatalogue values,theRVMslopeinclosed form,thescalings)`.

Machine access and export. Open Pulsar Laboratory. Use `HCC_API.describe("psr")` before `HCC_API.evaluate("psr",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. A missing visible pulse need not mean the source stopped emitting. A characteristic age is not an independently measured chronological age. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.10 Quasar Laboratory

Purpose and question. Eddington scaling, thin-disk efficiency and relativistic Doppler-beaming reference model. What luminosity, accretion and beaming scales follow from a black-hole mass and chosen efficiency?

Native construction. Eddington scaling, thin-disc efficiency and relativistic Doppler beaming provide separate reference relations.

Table 13: Declared input contract for qso. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`mass`	100000000	M_sun	[1, 100000000000]
`eddington_ratio`	0.1	dimensionless	[1e-06, 3]
`radiative_efficiency`	0.0571	dimensionless	[0.001, 0.42]

Controls and computation. Compare $L_{\mathrm{Edd}} = 4\pi GMm_pc/\sigma_T$ under its ionized-hydrogen assumptions and $L = \eta\dot{M}c^2$. Change viewing angle while retaining the intrinsic source parameters to isolate beaming. The laboratory controls include $\log_{10}$ M_BH, Eddington λ, Viewing angle θ, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `L_Edd`, `luminosity`, `accretion_rate`, `schwarzschild_radius_au`, `isco_radius_au`, `salpeter_time_yr`, `isco_efficiency_closed_form`, `super_eddington`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `qso` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT for the limit (L_Edd = 4 pi G M m_p c/sigma_T is closed form and depends only on the mass); MODEL for the accretion efficiency, which is the Schwarzschild ISCO value and assumes a thin disc around a non-spinning hole. The source declares `docs/verify-atlas-instruments.cjs(linearityinmass,theISCOefficiencyinclosedform,theSalpetertime)`.

Machine access and export. Open Quasar Laboratory. Use `HCC_API.describe("qso")` before `HCC_API.evaluate("qso",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. Efficiency is an input, not a unique consequence of mass. Disc geometry, inclination and accretion regime constrain interpretation; an animated jet is not a resolved GRMHD calculation. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.11 Impedance, Möbius and Smith Geometry

Purpose and question. fractional-linear impedance maps, reflection coefficient and resonance matching. Which apparently nonlinear plane transformation is a rigid action on the Riemann sphere, and how does this relate to reflection coordinates?

Native construction. The typed instrument is `smith`, acting on $\mathbb{CP}^1$. The wider impedance UI also uses $\Gamma = (Z - Z_0)/(Z + Z_0)$ and inverse impedance reconstruction.

Table 14: Declared input contract for smith. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`z_re`	0.5	dimensionless	[-1000000, 1000000]
`z_im`	0.6	dimensionless	[-1000000, 1000000]
`theta`	1.5707963267948966	rad	[-2π, 2π]

Controls and computation. Compare the direct fractional-linear map with stereographic lift, rotation and projection in the chordal metric. A Euclidean plane residual is ill-conditioned near the pole. The laboratory controls include r = Re z, x = Im z, Line loss α, Speed, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `w_re`, `w_im`, `route_agreement`, `unitarity`, `determinant_residual`, `on_sphere`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `imp` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is THEOREM (the map is a rigid rotation; both routes agree identically) + VERIFIED to the chordal metric on CP^1. The source declares `docs/verify-smith-mobius-unrolling.cjs(14/14)`, `docs/verify-hopf-splitting.cjs(7/7)`.

Machine access and export. Open Impedance, Möbius and Smith Geometry. Use `HCC_API.describe("smith")` before `HCC_API.evaluate("smith",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. An infinite plane coordinate is a legitimate point of the compact sphere. Distinguish this representation check from measured Touchstone data or an inverse RLC fit. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.12 Fibonacci Anyon Braiding

Purpose and question. Fibonacci anyons, braided by hand: worldlines you weave and the quantum state they compute. How does a braid word act in a declared fusion basis, and which category identities constrain it?

Native construction. The fusion rule is $\tau \times \tau = 1 + \tau$, with $d_\tau = \varphi$. A finite fusion space carries unitary braid matrices.

Table 15: Declared input contract for anyon. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`n`	8	anyons	[1, 80]
`braid`	121	word in 1,2,a	string

Controls and computation. Check unitarity and the braid relation before comparing words. Preserve chirality and basis: the core uses phases conjugate to the display convention discussed in section 15. The laboratory controls include Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `d_tau`, `D_total`, `gamma_topological`, `dim_to_tau`, `dim_to_vacuum`, `dim_total`, `log_dim_rate`, `braid_trace_re`, `braid_trace_im`, `yang_baxter_residual`, `central_charge`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `anyon` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is VERIFIED CATEGORY (F, R, braid, S, T and c = 14/5 all checked against each other). It is NOT a claim that the manuscript's boundary sector IS a Fibonacci liquid — a shared φ is a hint, not an identification.. The source declares `docs/verify-fibonacci-anyons.cjs(14/14)`.

Machine access and export. Open Fibonacci Anyon Braiding. Use `HCC_API.describe("anyon")` before `HCC_API.evaluate("anyon",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. An exact Fibonacci category is not evidence that a proposed boundary sector realizes a physical anyon liquid. A finite random-walk approximation to a gate is not a general compilation theorem. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.13 S³ Spectral and Navier–Stokes Laboratory

Purpose and question. an exact non-stationary solution put back into the equation and its residual measured, thousands of tracers carried by it on S³ measuring its energy, and the ledger of what is proven, conditional and open. Can an explicitly constructed time-dependent field be substituted into the declared equation and tested without mistaking a special solution for a general theorem?

Native construction. The native carrier is the round S^3_R, with Killing and curl-eigenfield components and a specified viscous operator.

Table 16: Declared input contract for nsflow. Defaults and bounds are from the pinned manifest.

Input	Default	Unit	Domain
`k`	1	1	[1, 4]
`sigma`	1	1	[-1, 1]
`gain`	0.55	1	[0, 2]
`amp`	3.2	1	[0.01, 10]
`nu`	0.02	1	[0.0001, 1]
`t`	0.37	1	[0, 100]

Controls and computation. Compute the curl of the momentum-equation residual and compare tracer-sampled energy with the analytic expression. The pressure-eliminated residual tests the declared construction under its smoothness and topology assumptions. The laboratory controls include shell k, Killing rotation g, shell amplitude a, viscosity ν, speed, Grid intensity. UI controls can include presentation settings absent from the typed input object.

Outputs and diagnostic. The typed contract returns `stokes_eigenvalue`, `shell_dimension`, `curl_level`, `carrier_left`, `carrier_right`, `shell_energy_fraction`, `mean_square_speed`, `energy_decay_time`, `residual_curl`, `residual_relative`, `ricci_extinction_time`. Interpret each using its quantity kind and unit in the companion manifest; a residual, status and physical observable are different output kinds.

Visualization and relations. The spatial view locates the construction and its selected state. The native checks above precede display projection. Use the Nexus neighbourhood of `nsflow` to inspect a relation claim and type; use the bus only when an explicit compatible output–input link is present.

Provenance. The release badge is EXACT solution, MEASURED against the partial differential equation: the residual $\partial_t u + \nabla_u u + \nu A u$ is computed by finite differences on S³ and its curl — zero exactly when it is a pressure gradient — is published beside the curl of the convective term alone.. The source declares `docs/verify-navier-stokes-on-s3-integrated.cjs`, `docs/verify-the-fluid-modes-are-spins.cjs`.

Machine access and export. Open S³ Spectral and Navier–Stokes Laboratory. Use `HCC_API.describe("nsflow")` before `HCC_API.evaluate("nsflow",input)`. Preserve the returned object with the explicit inputs and release identity; the common report/Markdown surfaces provide a portable account.

Validity and workflow. An exact special solution does not prove global regularity for arbitrary data. The related `s3lock` instrument explicitly retains an open terminal obstruction. Reset the station, retain its explicit inputs, vary one model parameter within bounds, inspect its diagnostic, compare a known limit and export both runs. Keep the view configuration separately if the comparison includes images.

7.14 Invariant Nexus

Purpose and question. Find which explicitly stated relationship connects two scientific contexts. The question is semantic before it is geometric: identity, representation, invariant, limit, coupling, causal succession, analogy and contrast have different evidential force.

Native space and inputs. A finite typed graph, 132 nodes and 408 edges in this snapshot. Focus node, relation filter, lens and path endpoints are graph-query inputs; screen coordinates are a deterministic non-metric embedding.

Computation and outputs. Neighbourhood and path operations expose node/edge identifiers and the associated claims. The spectral lens uses three non-trivial eigenvectors of normalized adjacency. Resource-allocation candidate scores rank common-neighbour structure; held-edge retrieval is an internal graph-reconstruction diagnostic.

Visualization, provenance and machine access. Read the edge inspector and its source/status before using a conduit. The `nexus` laboratory is parametric and has no same-ID typed instrument. Graph exports and diagnostics retain semantics, spectral metadata and unadmitted candidates. See section 12 for the full mechanism.

Workflow and limit. Choose a source node, restrict a relation type, inspect a path and export its edge records. Do not treat geometric distance as scientific similarity or a proposed edge as an admitted law.

7.15 Symmetry Discovery Chamber

Purpose and question. Test candidate transformations against finite samples and ask which diagnostics remain stable. The source covers $SO(3)$, Lorentz, $U(1)$, symplectic and Möbius families.

Native construction and inputs. Each family has its own action and invariant. A seed state, candidate action and finite orbit are mathematical inputs; animation speed and camera are presentation controls. The strict native-space metadata is not thereby completed.

Computation and diagnostic. Apply a candidate to samples and compare a declared invariant or closure residual. Finite agreement is evidence about those samples under the tested tolerance; it is not a proof that an unknown continuous symmetry group has been discovered.

Interface, provenance and export. The route is Symmetry Discovery. Its registry kind is parametric and it has no typed computational instrument. Retain the route, family, seed, candidate and diagnostic from the inspector/state surfaces; do not invent `evaluate("syd")`.

Workflow and relations. Start with a known preserving action, then a deliberately violating candidate, and compare the same diagnostic. Links to spin, holonomy and Möbius constructions are representation or invariant comparisons, not physical identifications.

7.16 Cosmic Web Laboratory

Purpose and native model. One deterministic Zel'dovich displacement field connects initial modes with sheets, filaments, nodes and voids. Lagrangian coordinates $\mathbf{q}$ map to $\mathbf{x} = \mathbf{q} - D\nabla\Phi(\mathbf{q})$, with a chosen sign convention for the potential. The Jacobian's eigenvalues classify the deformation before shell crossing.

Inputs, computation and diagnostics. Record the seed, mode content, growth parameter and smoothing choices exposed by the scene. Compare the same particles through growth, inspect collapse along principal directions and test the Jacobian interpretation inside the single-stream regime. The spatial web is a model realization,

not a fitted catalogue merely because familiar structures are nearby in the Atlas.

Machine interface and provenance. The manifest assigns the laboratory `cosmicweb` to instrument `nesting`. That instrument tests catalogue containment in a common J2000 frame; it does not expose the Zel'dovich evolution as an interchangeable compute call. Its centre/shape refusal rules matter: walls and unknown centroids cannot be turned into spherical membership claims.

Exports, relations and workflow. Save scene parameters for a model comparison and a separate `nesting` report for catalogue containment. Use the common frame and source tags when comparing contexts. A successful containment predicate is not a binding-energy result, and a morphological analogy is not a measured dynamical history.

7.17 CIVP: a Seven-Station Conditional Chain

The chain separates mathematical gates from claims about physical realization. Its normalized controls span $[0, 1]$ but resolve into different station-specific quantities. Query each contract for those transformations; a shared slider domain is not a shared physical unit.

7.17.1 CIVP CP¹ evaluation lock — a station of the trace-free de Sitter locking chain

Question and computation. RΓ(CP¹, L_q(−Z)) ≃ 0 if and only if N_emb = q_ind. The kernel and cokernel of the confluent evaluation matrix are computed by elimination and compared with h^0 and h^1 of the residual bundle O(q−1−N); the two routes share no code.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `delta_lock`, `h0`, `h1`, `rank`, `kernel_dim`, `cokernel_dim`, `det_agreement`, `bergman`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. This is a theorem about a line bundle. It does not derive certificate C_E — that the physical embadon support realises the evaluation frame — from CIVP dynamics.

Workflow, interface and evidence. Open civplock, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.2 CIVP molecular cut — a station of the trace-free de Sitter locking chain

Question and computation. q_can = Σγ_i and $\bar{\gamma}$ = q_can/N_emb. The equality q_can = N_emb holds exactly when $\bar{\gamma} = 1$; pointwise unit capacity is the stronger condition γ_i = 1, which is certificate C_U.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `q_can`, `gamma_bar`, `weight_variance`, `certificate_C_U`, `gluing_compatible`, `max_mass_defect`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. Raising the spread with the mean pinned at one keeps q_can = N_emb while C_U fails. The count is the capacity by accident there, not by a theorem.

Workflow, interface and evidence. Open civpcut, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.3 CIVP Jones ladder — a station of the trace-free de Sitter locking chain

Question and computation. $1/3 < \lambda^* < 1/2$ is $2 < [M:N] < 3$, and the small-index spectrum $\{4\cos^2(\pi/m)\}$ has exactly one value there, so $\mathrm{Ind} = \varphi^2$. The interval (1,2) is EMPTY, which is what forces a divisible endomorphic representation to index one.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `index`, `in_window`, `forced_index`, `forced_m`, `divisibility_root`, `fuzzy_lambda`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. The window is a theorem gate, not an assumption about the type of the glued CIVP algebra. Nothing here shows a physical defect realises it.

Workflow, interface and evidence. Open civpidx, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.4 CIVP A_4 transfer — a station of the trace-free de Sitter locking chain

Question and computation. $BB^T = N_\tau^2 = [[1,1],[1,2]]$, with spectrum $\{\varphi^2, \varphi^{-2}\}$ and determinant 1. A^n is the Fibonacci matrix and $\mathrm{Tr}\, A^n = L_{2n}$ — an exact integral shadow of an irrational eigenvalue.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `trace`, `lucas_2n`, `F_2n`, `spectrum_residual`, `quantum_dimension`, `global_dimension`, `fusion_dimension`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. $d_\tau = \varphi$, $I_\tau = \varphi^2$ and $\mu_{Fib} = \varphi + 2$ are three different invariants. A Fibonacci fibre over the configuration space is not the molecular count, and fusion rules do not fix chirality.

Workflow, interface and evidence. Open civpa4, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.5 CIVP UV selector — a station of the trace-free de Sitter locking chain

Question and computation. $\kappa_q = \Delta^2 \log I_q$ is invariant under $I_q \to C\, e^{aq} I_q$ and complete modulo it. $\Delta^2 \Gamma_q = \log((q+2)/(q+1)) - \kappa_q$, and strict convexity with one adjacent sign change gives a unique maximiser.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `q_star`, `argmax_direct`, `c_star`, `m_star`, `kappa_invariance_residual`, `identity_residual`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. The same reweighting that leaves κ at machine zero moves $q_\star$ by tens of sectors. That is why removing certificate C_UV leaves the location undetermined and the shape intact.

Workflow, interface and evidence. Open civpsel, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.6 CIVP Berry carrier and Capelli transfer — a station of the trace-free de Sitter locking chain

Question and computation. $c_1(L_N) = N-1$ and ind $\bar{\partial} = N$ differ by exactly one. Ordinary moments see only distinct spectral points; the rational transfer $R_\varepsilon(u) = P(u+\varepsilon)/P(u)$ recovers every multiplicity by $m_n = -\Sigma_\{j\geq n\}\ d_j$.

Inputs and outputs. Inputs: `a`, `b`, `c`. Outputs: `chern_number`, `index_dbar`, `off_by_one`, `spectral_rank`, `hankel_rank`, `capelli_recovered_rank`, `stinespring_leakage`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. Central data known only modulo $C_k \to A\ B\hat{}k\ C_k$ cannot determine the rank: $r\cdot m^k$ has the same cross ratios for every r. One absolute rank-sensitive datum is unavoidable.

Workflow, interface and evidence. Open civpcar, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

7.17.7 CIVP conditional closure — a station of the trace-free de Sitter locking chain

Question and computation. $R_q = \ell_P\sqrt{(q/\pi)}$ and $\Lambda_q = 3\pi/(q\ \ell_P^2)$ is an exact kinematic map from a selected capacity. Remove any one of C_X, C_win, C_U, C_E, C_UV, C_top and a conclusion is no longer determined. C_UV and C_top are the two halves of what used to be one certificate: convexity with a unique crossing, and the separate demand that the topology-sensitive Casimir remainder move neither.

Inputs and outputs. Inputs: `a`, `b`, `c`, `q`. Outputs: `R_q`, `Lambda_q`, `horizon_area`, `roundtrip_residual`, `delta_S_gen`, `certificates_held`, `shape_dimension`. The complete defaults, units and bounds are retained in the companion manifest.

Validity boundary. Switching a certificate on in an input is not deriving it. Without a derived $q_\star$ the cosmological constant shown is arithmetic on an unknown, and this station says so on every run. C_top is undischarged everywhere in this atlas: showing that a mechanism removes sensitivity to a CONSTANT vacuum shift controls only the constant part of $\ln(\zeta_\partial\hat{}q\ T_q)$, and the topology-sensitive remainder is what moves the crossing.

Workflow, interface and evidence. Open civpclo, describe the same instrument identifier, evaluate a baseline, then remove or vary one assumption. Export the returned diagnostics and gate statuses. Interpret the scene as the station construction; an enabled certificate is an input assumption, not its derivation.

The final closure now distinguishes six certificates, $C_X, C_{\text{win}}, C_U, C_E, C_{UV}, C_{\text{top}}$. Older five-certificate wording is stale. The topology-sensitive remainder is a separate unresolved issue. The quantity bus contains a selector-to-closure transfer; the seven visual stations are not thereby a fully automated seven-stage computational pipeline.

7.18 S³ Center Lab and Hopf Fibres

Purpose and space. The central bench, route centre lab, is a parametric entry context. Hopf fibres are circles on $S^3 \subset \mathbb{C}^2$ over points of $S^2 \cong \mathbb{CP}^1$, not concentric shells around a privileged physical centre.

Controls and calculation. Choose base points, fibre phase and projection; compare a native point with its Hopf image. The common phase leaves the base fixed. Distinct regular fibres have linking number one in the standard orientation. The projection can create apparent stretching or crossings that must be interpreted in the native space.

Machine interface and provenance. The bench has no computational instrument. The manifest assigns `hopf` to `anyon`; this host assignment is not a standalone Hopf-linking calculator. The `spin` instrument separately computes the single-qubit Hopf construction and returns sampled-fibre diagnostics. Describe the actual instrument before choosing it.

Workflow and limitations. Keep two fibres and vary the projection to distinguish topological linking from a screen crossing. Export a spin/Hopf report when using that typed calculation, plus scene state for the fibre view. Neither the central bench nor a common fibre geometry establishes a physical centre of the universe. See section 19 for metrics and projections.

7.19 Field and Fractal Workflows

Field purpose, inputs and native model. Choose a declared lattice solver, its coefficients, initial data, boundary and time step. These are a discretized field problem, not a generic particle animation. Solver-lattice, dimension-lattice and blast-wave views have different operations.

Field computation and evidence. Compare with a mode-appropriate analytic reference or balance. Record stability settings and refine the grid/time step before claiming convergence. Export the model state and diagnostics when offered; neither the world name nor a rendered colour provides an instrument identifier. Section 21 documents the current world and heat/spreading example.

Fractal purpose, inputs and native model. Choose the mathematical family, iteration limit, power or projection and camera. Mandelbulb distance estimation, orbital geometry and projected polytopes are distinct constructions with different numerical assumptions.

Fractal computation and evidence. Retain the stopping criteria and sampling choices. A box-counting estimate needs its fit window; a visual zoom is not a physical evolution. Save the selected mode and view state. XR may switch unsupported raymarch modes to supported geometry, so record the post-entry mode. Section 22 gives the rendering boundary.

Relations and workflow. Use dimensional analysis for field quantities and mathematical relations for fractal constructions. A visual resemblance supplies no automatic bus link. Start from reset, vary one declared parameter, inspect the appropriate diagnostic and retain state plus provenance for both views.

8 Typed Instruments

Typed instruments are callable computational contracts. The registry `HCC_API_SPECS` stores them by identifier; `hccApiRegister` also supplies quantity kinds from an explicitly declared coupling table where a field has not already declared one. This inheritance is based on admitted connections, not on guessing from a symbol. An instrument includes a title, world/host context, status, verifier references, formulas, input descriptors, output descriptors, limits and an evaluator.

`HCC_API.describe(id)` returns the public descriptor using the historically named schema `hcc.laboratory/1`. Input fields expose name, type, unit, quantity kind, default, lower/upper bounds and documentation. Output fields expose name, unit, quantity kind and documentation. `HCC_API.instruments.list()` correctly maps field names rather than array indices; comments preserve the earlier bug that motivated this check.

8.1 Refusal and validation semantics

The shared resolver converts number inputs to numbers, rejects non-finite values and refuses values outside declared bounds. Enumerated string alternatives are checked when a pipe-separated alternative set is declared in the unit field. Booleans are coerced. These mechanisms do not establish that every cross-field constraint is checked: evaluators perform further restrictions, and some documented limitations remain a caller responsibility. A numeric input later rounded to an integer must be distinguished from an exact integer-input contract.

For example, the white-dwarf instrument accepts a composition and mass but must also respect the composition-dependent Chandrasekhar limit. The Kerr instrument excludes extremality by bounding `chi` below one. The spherical-harmonic evaluator checks $|m| \leq \ell$ after its declared numeric inputs. The Einstein-ring instrument documents that its three angular-diameter distances do not simply add; individually admissible numbers can still describe an inconsistent cosmological configuration.

8.2 A concrete inspection and evaluation

The following signatures belong to the browser API. Execute them only in a page whose core is ready; this is a browser API, not a public HTTP endpoint.

```
await HCC_API.ready({timeout: 10000});
const spec = HCC_API.describe('bht');
const result = HCC_API.evaluate('bht', {
  M: 1.98847e30, chi: 0.5
});
```

The inputs are a mass in kilograms and a dimensionless spin. The result includes horizon quantities; the separately returned evaporation time is a Schwarzschild reference at that mass, not the rotating hole's evaporation prediction. Inspecting the description and limitations is therefore part of executing the instrument scientifically.

appendix C includes all 145 instruments and their host associations. The accompanying JSON preserves complete manifest input/output metadata, including nulls, while

the source extraction retains formulas and limits. A host mismatch is not repaired by renaming: `s3ns`, for instance, is an instrument associated with the `s3shell` laboratory.

9 Machine-Readable Interface and Agent Access

There are three distinct execution surfaces. The public static site serves `api/manifest.json`, `api/agent.json`, other research/contract resources and the GPU-independent `agent.html` workspace. The browser page exposes `HCC_API` and navigation interfaces. A locally launched server offers HTTP/MCP services. The agent-discovery document explicitly declares `public_http_compute:false`, no public MCP endpoint, static reads and local ESM computation. A static JSON catalogue is not a server-side evaluator. [5]

9.1 Readiness, health and capability discovery

`ready(opts)` polls until the instrument and laboratory registries exist. It resolves with the API or rejects on timeout with a TIMEOUT code and health information. It does not certify that all scientific tests, assets or XR features have passed. `health()` reports core status, release/build, counts, render mode and capabilities. `capabilities()` reports rendering/headless state, reason, WebGL2, presence of XR, worker and storage support. The presence of `navigator.xr` is weaker than support for an immersive session.

`HCC_API.list()` lists instruments. `HCC_API.labs.list()` lists laboratories with world, category, route, description, operational kind and instrument association. `labs.get(id)` returns a record or null; `instruments.has(id)` and `count()` support discovery without trial execution. The headless route `index.html?render=0` retains core operation without requiring the visual renderer. It still needs the application's imports; headless does not make an isolated HTML file self-sufficient.

9.2 Static SDK and scientific workspace

The agent workspace provides catalogue, dependency, forecast-audit and agent-access tabs. Its SDK example imports `connectAtlas` from `api/agent-client.mjs`, then calls `search`, `describe`, `forecast` and `audit`. A forecast is a conditional scaling scenario; chronological holdout metrics or synthetic examples do not supply independent empirical confidence by themselves.

```
import { connectAtlas } from './api/agent-client.mjs';
const atlas = await connectAtlas(
  'https://4vmp5pkwz4-pixel.github.io/HCC/'
);
atlas.search({query: 'anyon'});
atlas.describe('anyon');
atlas.forecast('bht.M', 0.1);
// await atlas.audit(input); // input must satisfy its schema
```

This example uses the SDK exposed by the agent workspace. In Node, use the local module from a repository checkout. The server command `nodeserver/server.mjs` and its default origin `http://127.0.0.1:8974` are declared in agent discovery. The named

server tools include listing/describing/running/sweeping laboratories, run validation, cancellation and artifact export. Their schemas must be read from the server contract before invocation.

9.3 Safe operational sequence

Discover release and access capabilities; enumerate worlds/laboratories; resolve the exact route and attached instrument; inspect its descriptor; validate units, domain and cross-field conditions; evaluate on an available execution surface; inspect diagnostic/status fields; follow provenance and relations; then export inputs and results together. Stop on unknown capability, undeclared semantics or refusal. A retry may change an input only as a new declared experiment, never as a hidden substitute for the refused request. section 29 develops this into an agent procedure.

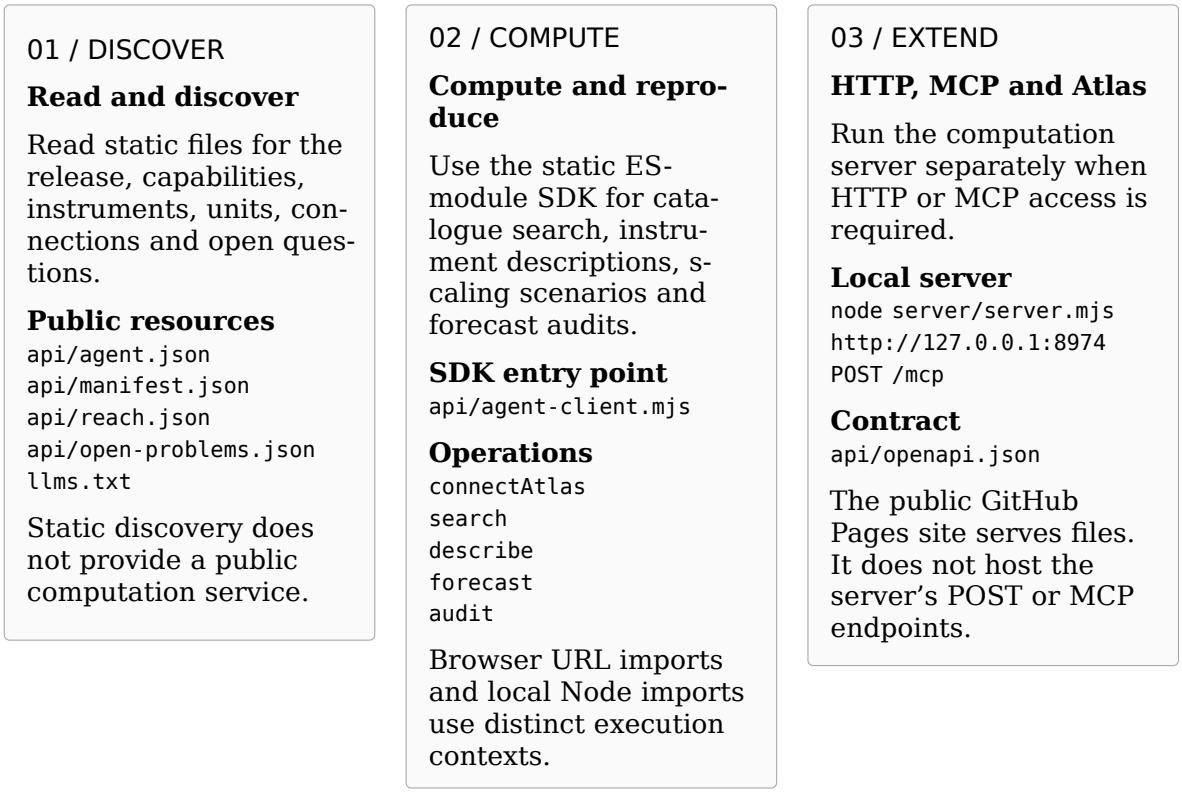


Figure 5: The agent workspace's three access modes in release 4.352.0: static discovery, local ESM computation and a separately launched HTTP/MCP server. Schematic overview.

10 Scientific Contracts and First-Principles Metadata

The first-principles schema is `hcc.first-principles/1`. Its laboratory records distinguish native space, native dimension, state dimension, display dimension, projection, metric/form, coordinates, domain, source status and parameter semantics. The display description explicitly says that a 3D Atlas render surface is not evidence of native physical dimensionality. The parameter descriptor adds symbol, role, quantity kind, unit, dimensional signature, domain, source status, exact value and default when available.

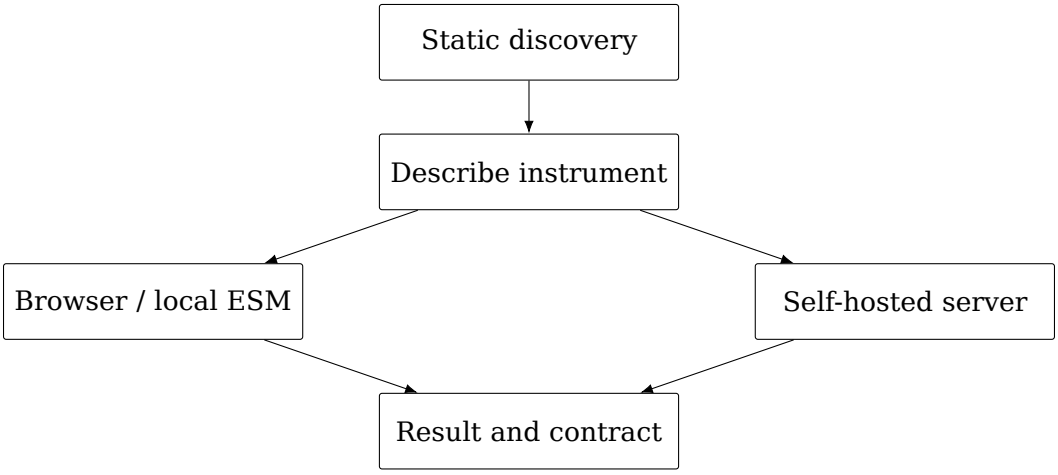


Figure 6: Machine access has distinct transports. A public static descriptor does not create a public compute endpoint.

Contract presence is not contract completeness. The release manifest reports 133 first-principles laboratory contracts, only two with declared native dimension, and 527 parameters with zero fully declared semantic records. The two native-dimension overrides are `anyzoo` and `jonesq`, whose state spaces are categorical/representation-dependent. The remaining omissions are not filled from educational prose. This is a material boundary on machine reasoning in release 4.352.0.

An instrument can nevertheless declare useful typed inputs and outputs while its laboratory-level first-principles record is incomplete. The two schemas serve different layers. A Holonomy computation can expose a station, phase and residual without the broad laboratory record declaring a unique native space for all five stations. The live Formula lens indeed displays VERIFIED alongside UNDECLARED native space and projection (figure 7); this combination is meaningful, not contradictory.

Implementation anchor: `hccFirstPrinciplesForLab;hccFpParameterDescriptor;HCC_FP_DIMENSION_OVERRIDES`.

A downstream agent should preserve null and UNDECLARED separately from numerical zero. Unknown unit is not dimensionless. A missing projection contract does not authorize reconstructing a metric from vertices. A categorical state dimension cannot be replaced by the three coordinates of its on-screen marker. These refusals protect the distinction between a scientific model and its display.

11 Provenance and Source Architecture

`SOURCE_MAP` supplies internal keys, visible tags and explanatory source labels. `FORMULA_REGISTRY` binds expressions to units and source keys. Selection cards and inspectors expose source tags, while formula-location functions connect a formula to a producer, world or laboratory. This creates a useful chain: displayed value → quantity/output → evaluator → expression → source and model assumptions.

The 41 source-map entries include astronomical catalogues and standards, external mathematical literature, project-specific constructions and internal derived quantities. JPL approximate ephemerides, Meeus lunar theory, CODATA and IAU tags must not be read as one common accuracy class. The JPL tag explicitly describes the Earth

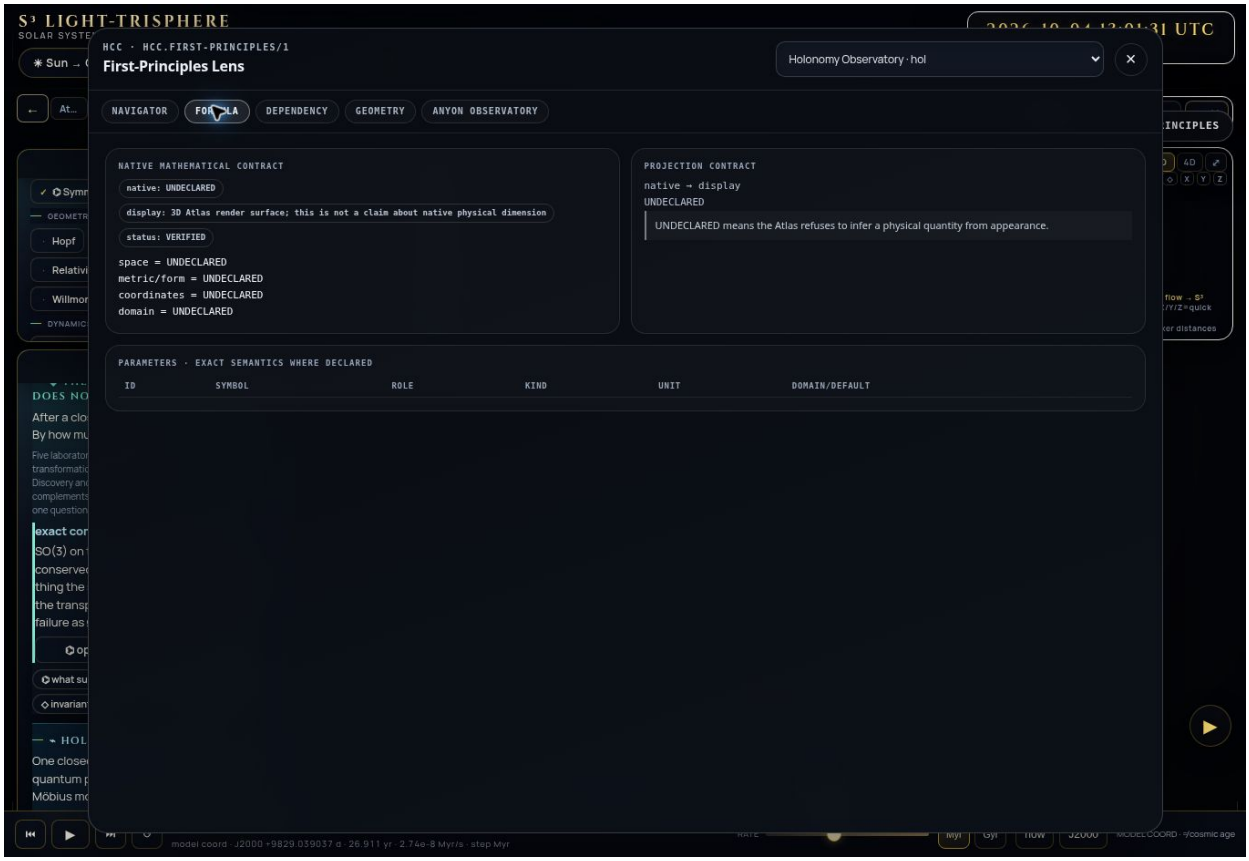


Figure 7: First-Principles Formula lens for Holonomy. The laboratory status `VERIFIED` and native metadata marked `UNDECLARED` refer to distinct contract layers.

row as an Earth–Moon-barycentre approximation. The IERS tag distinguishes mean-precession display from precision Earth orientation. The FBS source label explicitly separates N_{rec} from the CoScale clock. appendix G preserves the full internal labels.

There is also a newer research-catalogue layer, imported from `core/research/catalog.mjs` and surfaced through `api/research.json`. Agent discovery names geometry-evidence operations for explicit FLRW/CPL geometry, DESI covariance, a supplied-reconstruction null test, round-S³ modes and reference lensing. Its stated boundary is no fitted posterior and no topology detection. That newer layer should not be confused with the older S3-v38 tag still present in `SOURCE_MAP`.

11.1 Formula status and provenance do not collapse

The effective principal registry contains 133 formulas: 126 initial definitions and seven appended predictive/Nexus/RLC/thermal-peak records. The formula census reports 126 computed somewhere and seven stated only. Coincident counts in different stages do not prove they are the same subsets. The Formula Portal's larger programme corpus is another collection and must not replace this application-level register.

Use `formulaSite(id)`, `formulasHere(ctx)`, `formulaCensus()` and `formulaFind(q)` to locate implementation sites. A statement with a source tag but no producer is still a statement; the interface can explicitly say "stated, not computed here". Provenance gives traceability, not automatic correctness. The bibliography and source index distinguish external references, internal source identifiers and derived quantities.

12 Invariant Nexus

Invariant Nexus is a typed graph over laboratory contexts. The release registry contains 132 nodes and 408 edges, grouped into geometry/topology/information, quantum/fields/matter, waves/response/materials, dynamics/chaos/phase space, energy/thermodynamics/dense matter and astrophysics/observation. It includes Ancient Chronometry; the Nexus context itself is not a second scientific node representing its own graph. Counts are determined from the effective declarations and are corroborated by the live HUD.

Every edge has endpoints, an operational type, a readable label, a plain-language claim, status, identifier and direction flag. Eight classes distinguish exact identity, representation, shared invariant, limit/reduction, coupling/input, causal succession, analogy and contrast. The broader scientific contract has eleven canonical types; the eight-class UI groups some of them. appendix D records the mapping and the data supplement retains all 408 claims.

12.1 Layouts and lenses

Disciplinary geography positions groups for orientation. The final spectral implementation uses three non-trivial eigenvectors of the unweighted normalized adjacency matrix $D^{-1/2}AD^{-1/2}$, where A records adjacency and D holds vertex degrees. The eigenvectors organize a registry, not physical space. Edge categories, their descriptions and filtering carry semantics that layout alone cannot carry. An earlier export implementation describes a projection of relation-type counts; the later wrapper re-

places that description with the spectral contract. The later code is the effective implementation.

A local lens suppresses unrelated paths around a focus; a traced-path lens isolates a comparison; global topology is an explicit choice. Candidate bridges use a leave-one-declared-edge-out structural audit and resource-allocation scores. A high score says a graph could recover a declared adjacency; it does not establish a missing law or an empirical relation. The structural audit uses 404 eligible held edges while the full registry contains 408: an audit denominator is not the registry size.

12.2 Paths and the epistemic firewall

A shortest typed path is a shortest path in the chosen relation graph and filter. It is not the shortest causal history, strongest evidence chain or smallest physical separation. Causal edges retain their declared direction even when comparison traversal can examine them in either direction. An analogy on any step prevents a path from being summarized as physical identity.

A Nexus path from spin to black-hole thermodynamics passes through the anyon zoo and information laboratory, beginning with an analogy and continuing through coupling claims. That sequence is useful for comparison but does not identify a spin, a topological fusion channel and a horizon as one physical system. Up to four consecutive nodes can be composed into Multiview. Graph export `fbs3r_invariant_nexus.json` preserves typed claims and semantic disclaimers. *Implementation anchor:* `NEXUS_TYPES;NEXUS_RELATIONS;nexusExportData`.



Figure 8: Invariant Nexus interface: 132 laboratory nodes, 408 typed relations, relation filters and graph-navigation controls.

13 Quantity Bus and Computational Coupling

A Nexus relation says that two contexts have a declared scientific relationship. A bus connection says that a particular output can be supplied to a particular input. The first is a claim; the second is an executable wiring decision. A Nexus coupling edge explicitly does not guarantee a unit-safe solver connection.

The manifest supplies 57 admitted links, 39 written refusals and 97 isolated instruments. Each admitted row names `from`, `to`, unit, quantity kind, conversion scale and whether conversion occurs. Isolation is relative to the current admitted bus, not proof that an instrument is scientifically unrelated. The live Nexus HUD's 35 bus routes is another aggregation, not a replacement for the manifest's 57 endpoint-level links.

Compatibility has several layers: dimensional conversion, quantity-kind agreement, acceptable destination domain and an explicit scientific interpretation. Matching text such as "beta" is insufficient. The source records a refused impedance/skyrmion beta identification, and the newer quantity-kind table makes that refusal structural. A value can have the correct dimension but the wrong clock, reference frame or physical role.

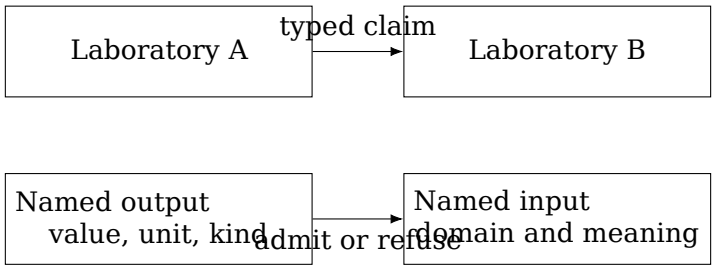


Figure 9: Nexus and quantity bus answer different questions. Neither arrow implies an empirical causal effect without an additional model.

13.1 Examples and failure modes

The implemented CIVP connection is `civpsel.q_star` to `civpclo.q`. This carries a selected sector into a conditional capacity map. It does not turn all seven CIVP stations into an automatically executed pipeline and does not discharge the closure certificates. Those assumptions must still be explicitly inspected.

The chain `bht.T_H→bb.T→wave.wavelength` shows why a link can become operational only in a particular upstream context: changing black-hole mass supplies temperatures for which the blackbody peak lies inside the wave instrument's admissible range. The source also records `ns.radius_km→nuosc.baseline` with zero response in the oscillation-length output, because that particular output does not depend on baseline. A valid transfer and a nonzero influence are different properties.

Contested inputs require one selected driver. The source explains that admissible luminosities from several laboratories can compete for one photometer input; declaration order can resolve this choice, and refused alternatives carry reasons. Reproduction therefore needs the selected wiring and driver priority, not just a list of dimensionally possible pairs. Read `bus.refusals()` and the neighbourhood/candidate surfaces before assuming automatic composition. *Implementation anchor:* `HCC_COUPLING_KINDS;hccBusDeclareLinks;HCC_NOT_LINKS`.

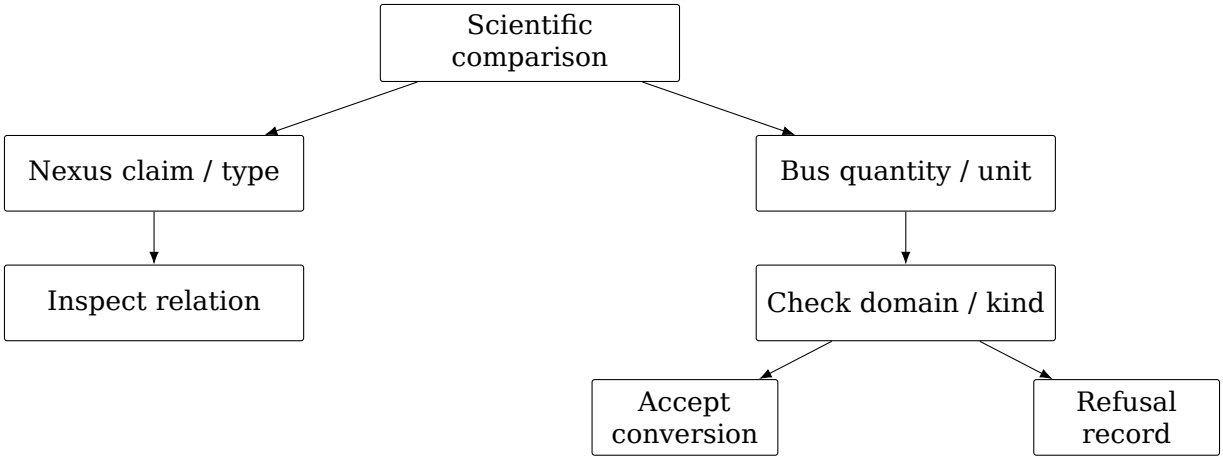


Figure 10: A typed scientific relationship and an executable quantity transfer require different checks. The refusal branch is part of the bus contract.

14 Multiview

Multiview compares up to four contexts at once. It can be entered from the laboratory rail, a prepared scientific comparison, or consecutive nodes on a Nexus path. The purpose is to inspect a question across native spaces, not to merge their coordinates into a new physical scene. A null-ray construction, lensing calculation, wave detector response and conditional multiple-path geometry can share an interface without sharing all assumptions.

Each tile retains its laboratory identity and relevant state. The selected tile controls which inspector is active. Shared timing/orientation is an operational aid; it does not establish clock equivalence, synchronization of physical systems or a numerical bus connection. Prepared presets expose their view lists, observed bus quantities and comparison contract through `HCC_API.multiviewPresets()`.

The single-camera cinematic stage is disabled in Multiview because its camera-facing construction is inappropriate for four cameras. Geometry-lifetime management retains active comparison contexts instead of disposing of them as inactive. These rendering decisions do not authorize changes to scientific sample counts, residuals or exported values. figure 11 documents the actual four-tile control state obtainable here; no claim about stereo or GPU performance follows from it.

15 Core Mathematics

This section gives equations their operational meaning. The complete principal expression index in appendix E preserves source strings and declared units; the equations below use standard mathematical typesetting without silently repairing source conventions.

15.1 Round S³ geometry and observable fraction

The curvature-radius conversion used by the conditional reconstruction is

$$R = \frac{c}{H_0\sqrt{|\Omega_K|}}. \tag{1}$$

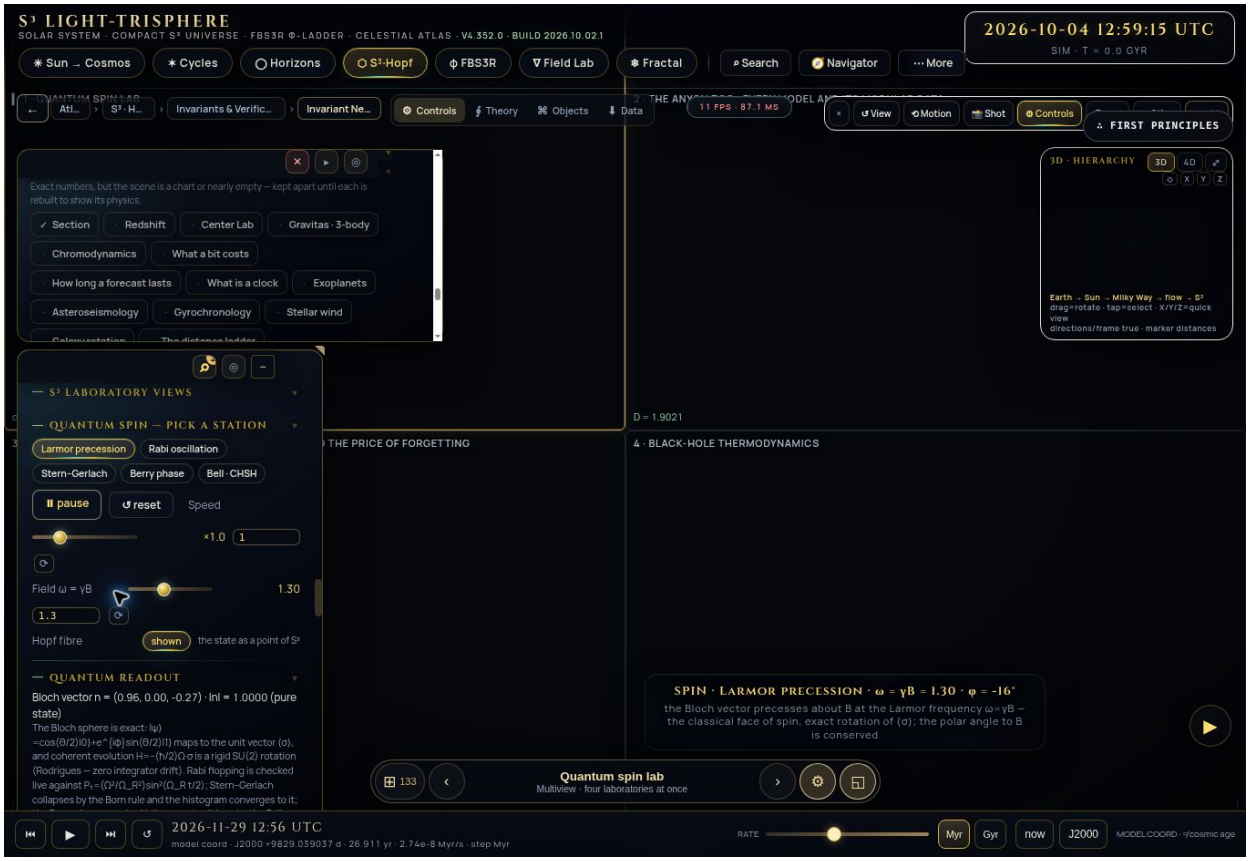


Figure 11: Multiview interface for a four-context comparison along a Nexus path: spin, anyon zoo, information and black-hole thermodynamics.

Here c is a speed, H_0 an inverse time and Ω_K dimensionless, so R is a length. A round closed spatial model requires the negative curvature-density sign under the convention used in the code; taking a modulus alone does not establish that sign. The current conditioning procedure is described in section 18.

For that mathematical carrier,

$$V(S^3_R) = 2\pi^2 R^3, \qquad C_{\text{great}} = 2\pi R, \qquad D_{\text{antipodal}} = \pi R, \tag{2}$$

$$A(\chi) = 4\pi R^2 \sin^2\chi, \qquad \Gamma(\chi) = \frac{\chi - \frac{1}{2}\sin(2\chi)}{\pi}. \tag{3}$$

The dimensionless angle χ is geodesic radius divided by R; A has area units and Γ is the enclosed fraction of total volume for $0 \le \chi \le \pi$. These are exact round-metric identities. Their application to an observable universe inherits the conditional model, horizon definition and input radius. The full great-circle circumference and antipodal separation are different lengths.

15.2 Scalar and vector spectra

The scalar Laplacian convention is

$$\lambda_\beta = \frac{\beta^2 - 1}{R^2}, \qquad g_\beta = \beta^2, \qquad \beta = 1, 2, \ldots . \tag{4}$$

With $n = \beta - 1$, this is $n(n+2)/R^2$ and multiplicity $(n+1)^2$. The eigenvalue has inverse-area units. Vector curl shells use a different index and multiplicity:

$$\operatorname{curl}\phi = \mu\phi, \quad \mu = \sigma(k+2)/R, \quad \sigma \in \{-1, +1\}, \tag{5}$$

$$\dim E_{k,\sigma} = (k+1)(k+3), \quad A_{\text{Stokes}} = \operatorname{curl}^2 - 4/R^2, \tag{6}$$

$$u(t) = \exp[-\nu k(k+4)t/R^2]u(0). \tag{7}$$

In the last line, u is the velocity field and ν the kinematic viscosity; $\nu t/R^2$ is dimensionless. The source's operator convention leaves the Killing shell undamped. It must not be replaced by an unrelated Euclidean Laplacian convention. The sharp band-rank statement $N_K = (K+1)(K+2)(2K+9)/3$ belongs to the declared finite shell sum, not to the scalar spectrum.

15.3 Golden-ratio scale and time coordinates

$$\varphi = (1+\sqrt{5})/2, \qquad R_N = \ell_P \varphi^N, \qquad N(R) = \frac{\ln(R/\ell_P)}{\ln\varphi}, \tag{8}$$

$$\rho_N/\rho_P = \varphi^{-3N}, \qquad t_N = t_P\varphi^N, \qquad R_N/t_N = c. \tag{9}$$

The last identity uses the declared Planck-anchor relation $t_P = \ell_P/c$; it does not derive c. N and φ are dimensionless, R_N a length, t_N a time and the density ratio dimensionless. The density exponent is a model prescription that can be varied in exploratory controls. Exact Fibonacci arithmetic uses

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \psi = (1-\sqrt{5})/2, \tag{10}$$

with integer recurrence as an independent exact implementation for supported indices. Dropping the second term is an asymptotic approximation, not the exact value at small n. A rendered shell index may be continuous even when a Fibonacci index is integer.

15.4 Holonomy, spinors and contact geometry

For the declared oriented spherical cap,

$$\Delta\alpha = \int K\, dA = \Omega \pmod{2\pi}, \qquad \gamma_B = -\Omega/2 \pmod{2\pi}. \tag{11}$$

K is Gaussian curvature, dA area and Ω solid angle; phases are dimensionless radians. The Berry expression applies to the declared spin-half convention, not every quantum state. Non-Abelian return is characterized by

$$C = U_x U_y U_x^{-1} U_y^{-1}, \qquad W = \tfrac{1}{2}\operatorname{Tr} C, \tag{12}$$

where the U are SU(2) rotations and W is conjugacy invariant. For the orthogonal-boost convention the source gives $\Omega_W = -2\arctan[\tanh(\alpha/2)\tanh(\beta/2)]$, with dimensionless rapidities. These are different connections and models despite their common return-map organization.

A normalized spinor $\psi \in \mathbb{C}^2$ gives $k^\mu = \psi^\dagger \sigma^\mu \psi$ and $k \cdot k = 0$. A common phase leaves the ray fixed. For $X = x^\mu \sigma_\mu \in \mathrm{Herm}(2)$, $X \mapsto AXA^\dagger$ with $A \in \mathrm{SL}(2,\mathbb{C})$ preserves the determinant/Minkowski interval. Contact normalization on unit S³ is explicitly

$$\lambda_0 = \tfrac{1}{2}\sum_j (x_j\, dy_j - y_j\, dx_j), \qquad R_{\mathrm{Reeb}} = 2Jx, \qquad \lambda_0(R_{\mathrm{Reeb}}) = 1. \tag{13}$$

Changing the factor of one half changes the flow period; it cannot be omitted as cosmetic notation.

15.5 Representative astrophysical and field equations

The geometric-unit TOV model uses

$$\frac{dP}{dr} = -\frac{(\epsilon + P)(m + 4\pi r^3 P)}{r(r - 2m)}, \qquad \frac{dm}{dr} = 4\pi r^2 \epsilon, \tag{14}$$

with $G = c = 1$ and a declared polytropic EOS. Its code units and conversion to kilometres/solar masses are part of the instrument; the graph is not a measured neutron-star interior. The cold white-dwarf family uses

$$M_{\mathrm{Ch}} = 5.816\, M_\odot/\mu_e^2, \quad R = \frac{0.0225 R_\odot}{\mu_e}\frac{\sqrt{1 - x^{4/3}}}{x^{1/3}}, \quad x = M/M_{\mathrm{Ch}} < 1. \tag{15}$$

Here μ_e is mean molecular weight per electron, and the zero-temperature approximation excludes finite-temperature corrections.

The heat reference is $\partial_t u = \alpha \nabla^2 u$, with diffusivity of dimension length squared per time. The Gray–Scott pair is $\partial_t u = D_u \nabla^2 u - uv^2 + F(1-u)$ and $\partial_t v = D_v \nabla^2 v + uv^2 - (F+k)v$ in the declared reduced units. A reaction pattern does not identify a chemical material without additional parameter calibration. These equations anchor the role of solvers; their grid/boundary choices remain essential.

16 The Solar and Astronomical Data Layer

The Solar model solves Kepler's equation $M = E - e\sin E$ by Newton iteration using J2000 elements and secular rates. Here M is mean anomaly, E eccentric anomaly

and e eccentricity. Angles are carried in the implementation's declared units. This is an approximate ephemeris, not a full numerical solar-system integration. The source tag gives an approximate 1800–2050 validity envelope and identifies the Earth row with the Earth–Moon barycentre. Outside the stated epoch range, extrapolation must remain visible.

The lunar layer uses a truncated/principal-term theory and eclipse calculations use ephemeris geometry. The current interface has a computed prediction observatory alongside the NASA catalogue used as a benchmark. The prediction observatory compares computed event timing and type with the reference catalogue. Delta-T and long-epoch lunar truncation limitations are explicitly described. Catalogue lookup, computed event search and a visually suggestive syzygy must therefore be distinguished.

The celestial layer combines catalogue directions and, where quality allows, distances and space motion. Gaia, Hipparcos and SIMBAD have different provenance and uncertainty characteristics. The interface applies a five-standard-error parallax gate for the finite-distance Hipparcos subset; excluded stars remain directions rather than invented nearby 3D positions. Constellation stick figures are line-of-sight patterns, not bound structures. The zodiac's true-distance layer is separate from the interpretive precession Age Wheel.

Galactic-plane alignment, the ecliptic, spin axes and velocity vectors specify frames. A vector relative to the local standard of rest is not the CMB-frame velocity; adding or comparing them requires the correct transformation. The Solar hierarchy locator uses true directions with semantic distances and explicitly does not establish metric separation. The world-scale chain preserves relevant direction and radius conventions at declared seams, while camera framing and body-size exaggeration remain separate display operations.

Research workflow. Set an epoch inside the desired ephemeris regime, choose a reference frame, inspect a body's source tag, compare true scale with exaggerated size, and export the epoch/frame with any computed quantity. For a catalogue distance, retain the source, quality gate and proper-motion convention.

17 Cycles and Chronometry

The shared Atlas epoch coordinates displays without making all clocks physically equivalent. A phase $\theta(t)$ requires a period and an anchor; an offset in days from J2000 is not automatically cosmological age, conformal time, simulation time or a laboratory's dimensionless evolution variable. Imported time registries distinguish domains, adapters and state scopes. Conversion is a declared operation, not a change of label.

The galactic clock uses a literature-scale solar orbital period near 230 Myr; the source's precession reference includes the 25,772-year Great Year and more recent engine choices. The current Solar controls expose Vondrák long-term precession alongside an IAU linear display model. A displayed astrological age is an interpretive layer, not a modern constellation boundary or a dynamically privileged epoch. Seasons are root-found against solar longitude, while a Saros/Metonic comparison is a bounded rational approximation between cycles.

The normalized phase solver distinguishes relative phase beat from bounded commensurability. A close period ratio is neither a detected dynamical resonance nor proof that two historical traditions refer to one phenomenon. The galactic

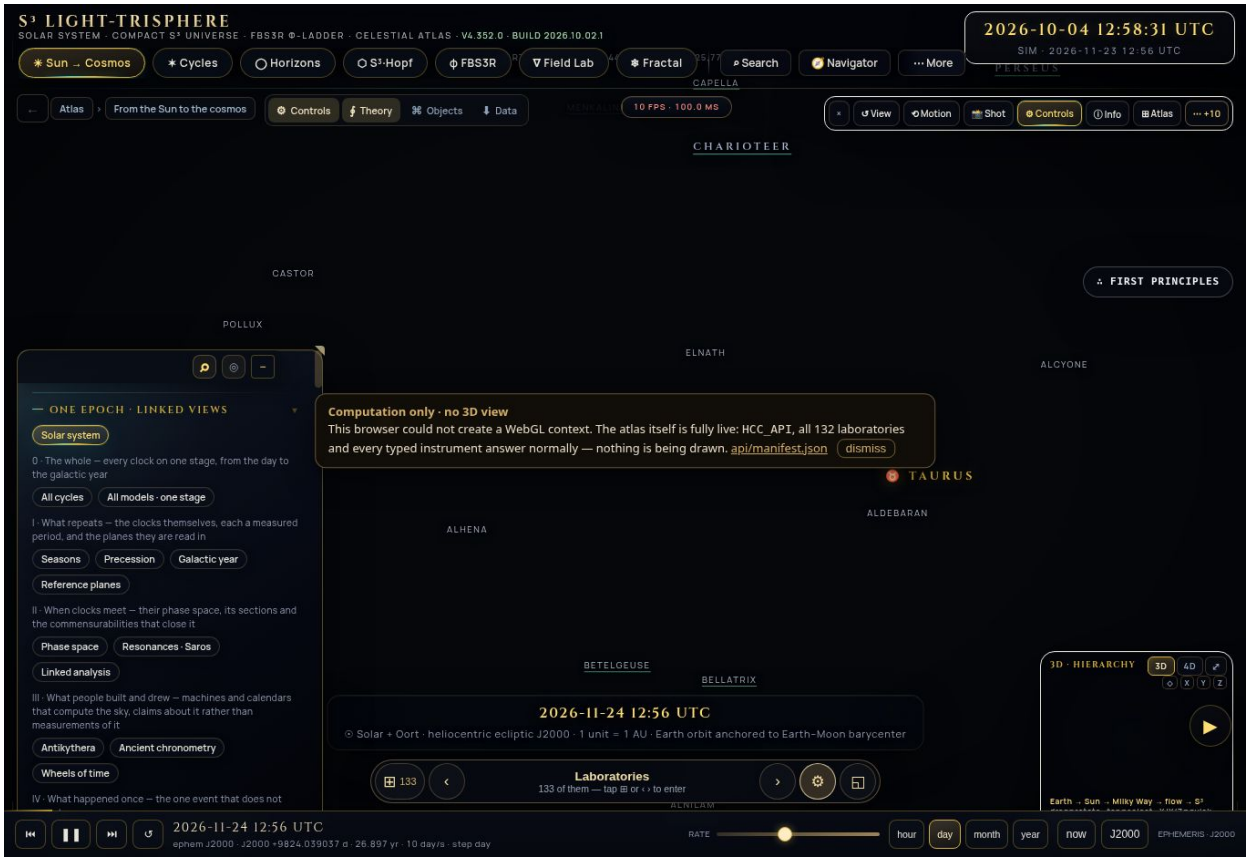


Figure 12: Solar interface: world navigation, epoch controls and astronomical catalogue labels.

butterfly/Fermi-bubble context is an event dated under a rate model, unlike a recurring phase clock. Their common timeline supports comparison but not epistemic equivalence.

17.1 Ancient Chronometry Observatory

The `chronometry` instrument takes a source-qualified `case_id` and `epoch_days` within $\pm 3{,}650{,}000$ days. It returns adjudicated status, structural-dependence flags, comparison state, closure quantities and several calendar phases. Its status explicitly leaves astronomy comparisons at `PENDING_EPOCH_CORRECTION`. A source-specific definition of a small time unit must not be averaged with another tradition's definition merely because the same word is used.

The Jain avasarpini identity in the declaration is $4K + 3K + 2K + (K - 42000\,y) + 21000\,y + 21000\,y = 10K$. The 42,000-year component is structurally dependent on twice 21,000; counting it as a second independent numerical coincidence would overstate evidence. Calendar-phase outputs also distinguish the sexagenary day cycle from the sixty-year wheel with a similar name. Export the selected textual case, anchor, epoch and gate state with any comparison.

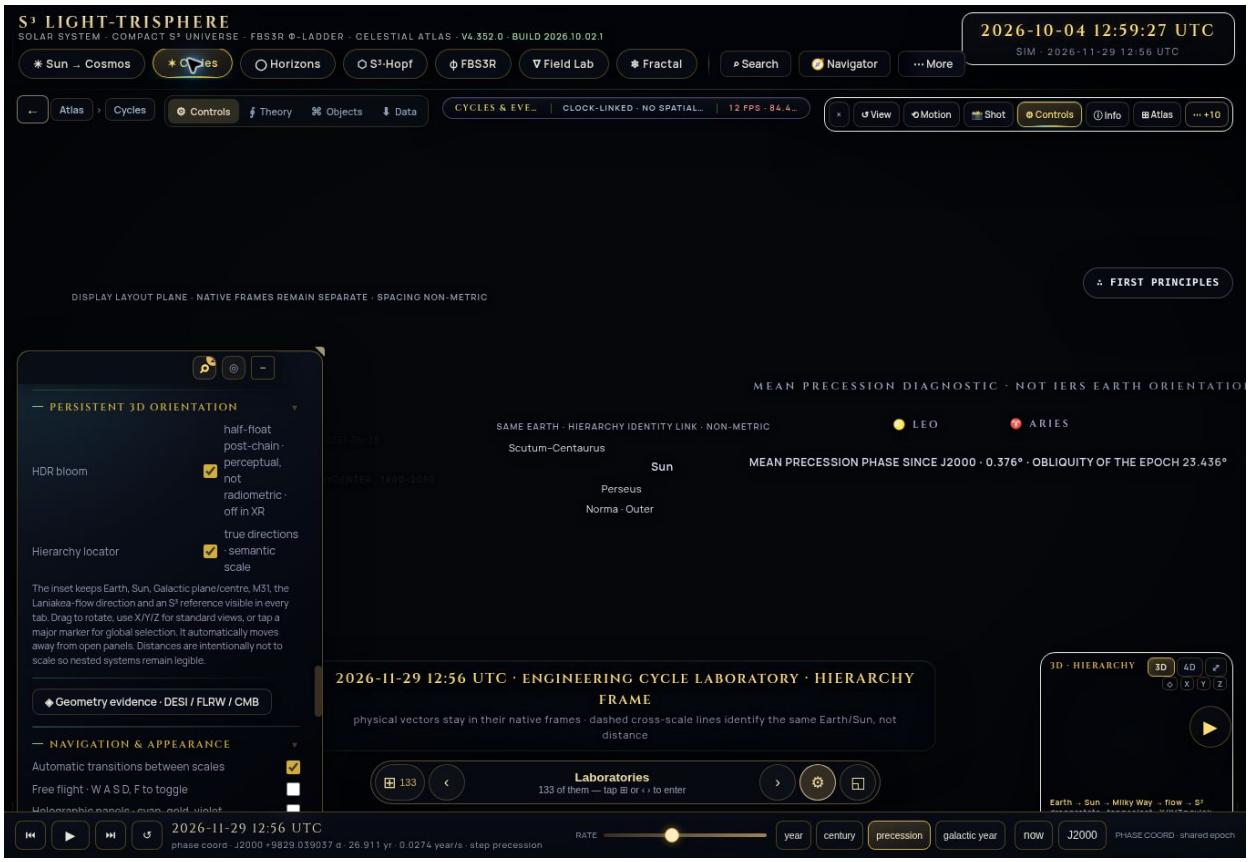


Figure 13: Cycles interface at a shared paused epoch. The labels and controls refer to distinct native clocks; their arrangement is non-metric.

18 Observable Universe and Cosmology

The current carrier reconstruction supersedes the earlier v38 fiducial. Source comments identify a one-dimensional Gaussian curvature marginal $10^3\Omega_K = 2.20 \pm 1.00$ from the cited 2026 analysis, then condition it on the closed-sign branch $\Omega_K < 0$. This curvature marginal is an input to the conditional reconstruction; branch conditioning is distinct from fitting a new cosmological likelihood.

For $X \sim \mathcal{N}(\mu_K, \sigma_K^2)$, the implemented closed-branch mass is

$$p_{\text{closed}} = \Phi(-\mu_K/\sigma_K), \qquad F_{X|X<0}(x) = \frac{\Phi((x-\mu_K)/\sigma_K)}{\Phi(-\mu_K/\sigma_K)}, \quad x < 0. \tag{16}$$

The source reports about 1.390% surviving mass, conditional median $\Omega_K \simeq -2.5975 \times 10^{-4}$ and $R_c \simeq 886.59$ Gly. The unconditioned central curvature has the open sign; these numbers are not a measurement of closure. Taking an absolute value of the central marginal would be a different operation.

18.1 A conditional background and distinct horizons

`hccS3Reconstruct` converts the declared H_0 to inverse seconds, derives curvature and volume, and evaluates a closed-Λ background using

$$E(a)^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_K a^{-2} + \Omega_X. \tag{17}$$

The effective radiation bookkeeping is labelled convention-dependent. Logarithmic Simpson integration yields age and particle/event-horizon integrals. In explanatory integral notation, $t_0 = H_0^{-1} \int da/[aE(a)]$ and $D_p = (c/H_0) \int da/[a^2 E(a)]$ over the declared finite integration domain. Event horizon uses the future interval. These quantities differ from luminosity distance, angular-diameter distance, comoving radial distance and ct_0.

The conditional density remains finite near $\Omega_K = 0^-$ while $R_c^m \propto |\Omega_K|^{-m/2}$. Consequently the tail integral converges only for $m < 2$ under this approximation. In particular the conditional mean volume is divergent. A median and quantile band are meaningful summaries; a finite "mean total universe volume" would silently add a cutoff or different prior. Global inventories $Q = 2\pi^2 R_c^3 \bar{q}$ inherit the same conditional geometry and sensitivity. Finite total mass does not by itself define a finite object count without a mass function and lower threshold.

18.2 Cosmic Web Laboratory

The `cosmicweb` scene uses one deterministic periodic potential, with a Lagrangian lattice displaced by its gradient and morphological classes evaluated from the associated deformation tensor. Nodes and connections are tied to the same reduced Zel'dovich field rather than a random decorative cloud. Growth coordinate, node census and density proxy can be published as live quantities. The manifest classifies the context as computational but its preferred attached instrument must be read from the registry; the assigned instrument is `nesting`, with its own catalogue-containment contract. It is not an N-body simulation or a reconstructed observed survey, and none of its structures is evidence for the conditional S³ topology.

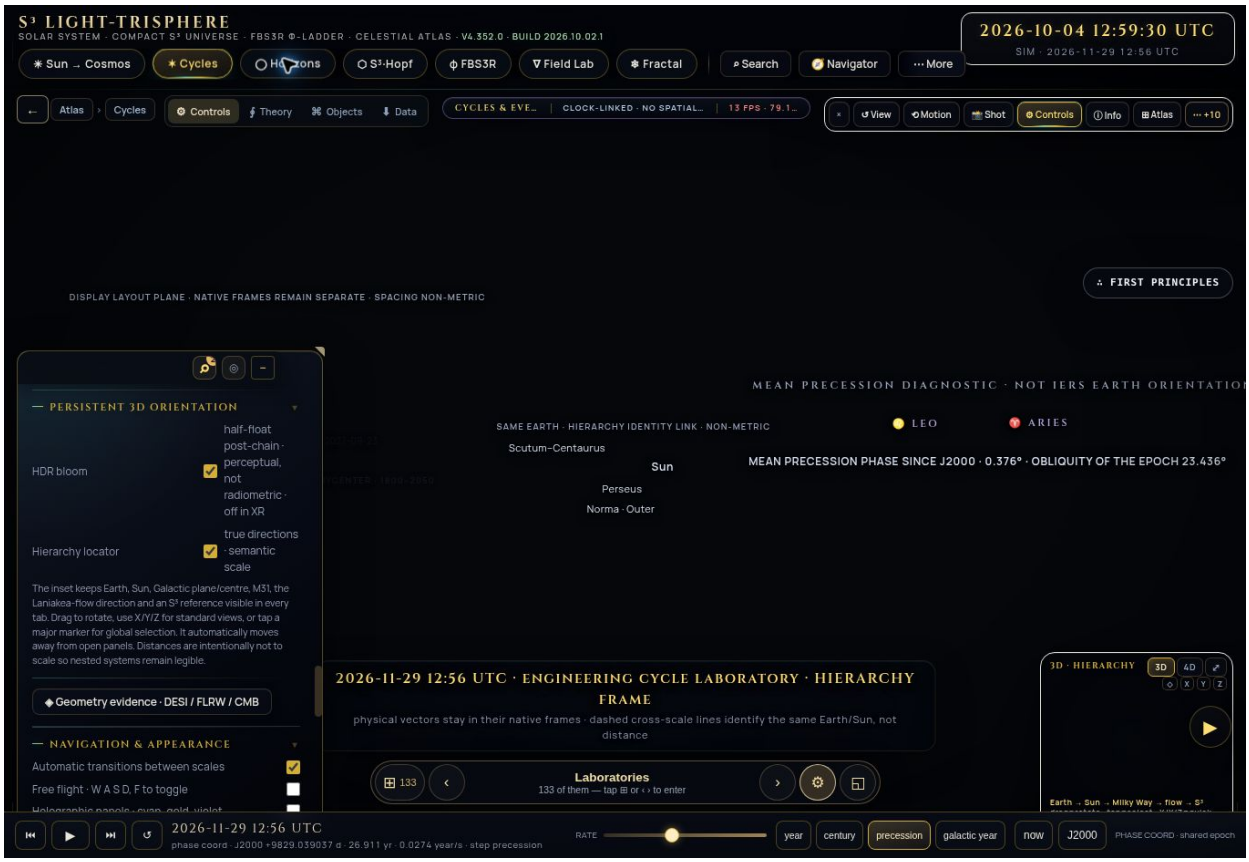


Figure 14: Horizons interface and observable-volume readout for the conditional reconstruction.

19 S³ Carrier and Hopf Structure

19.1 Native carrier, observer and projection

The round three-sphere is $S^3_R = \{x \in \mathbb{R}^4 : \|x\| = R\}$. It is three-dimensional intrinsically and embedded in four-dimensional Euclidean space. A point p chosen as observer/basepoint has antipode $-p$. For two carrier points the intrinsic separation is

$$d(p,q) = R \arccos\Big(\frac{p \cdot q}{R^2}\Big), \qquad 0 \le d \le \pi R. \tag{18}$$

Normalization and floating-point endpoint handling belong to the implementation. The displayed chord or stereographic straight-line distance is not this geodesic. A great two-sphere is a central hyperplane section; a χ-screen centered at a basepoint is a geodesic-distance level set. Neither is a wall of S³.

Center Lab supplies arbitrary unit-quaternion centers, coordinate presets, two screens, overlap diagnostics, tangent frame, antipode and geodesic arc. The controls can use quaternion, Hopf or polar coordinates. These are mathematical basepoints in a conditional reconstruction, not inferred positions of external observers. Stereographic projection makes many constructions legible but excludes a projection pole and distorts distances. A round projected screen must not be used to recover its native radius without the projection contract.

19.2 Hopf fibres and representations

A unit spinor $(z_1, z_2) \in \mathbb{C}^2$ with $|z_1|^2 + |z_2|^2 = 1$ represents S³. Common multiplication by $e^{i\alpha}$ traces a U(1) fibre with constant base point in $\mathbb{CP}^1 \simeq S^2$. Hopf coordinates separate a mixing angle from two phases. The Atlas exposes fibres, their base map and linked families; pairwise linking belongs to the mathematical fibration, not a literal set of cosmic strings.

Unit quaternions realize SU(2), with a double cover of SO(3). Spinor sign, global phase, Bloch direction and physical rotation therefore require separate meanings. The null-spinor laboratory adds an SL(2,C) representation of proper orthochronous Lorentz transformations. Contact/Reeb and KS stations reuse related maps in different dynamical contexts. A representation chain establishes a mathematical relationship under its conventions; it does not prove that qubits, photons and gravitational waves are the same system.

19.3 Spectral and fluid laboratories

The scalar spectrum in equation (4) and the signed curl-shell system in equation (7) organize different fields. `s3shell` exposes shell dimensions, decay and projector constants; `s3torus` studies an invariant two-scalar maximal-torus family; `nsflow` inserts an exact non-stationary construction back into its equation; `nsgal` evolves a finite Spin(4) spectral system. The triad laboratory relates finite Beltrami interactions to a rigid-body phase portrait; `vstretch` checks a Vandermonde form of vortex stretching. Spectral escape, tube coordinates, chiral Killing–Beltrami identities and terminal-lock probes retain separate hypotheses.

A truncated Galerkin computation does not establish global regularity for the full Navier–Stokes equation. An exact invariant family does not classify every solution. The source tag for the S³ Navier–Stokes programme explicitly separates intrinsic

round-sphere results from a relative forced singular branch requiring an external Euclidean package of assumptions. The terminal lock remains open. The forced-blow-up diagnostic laboratory can examine declared exponent scenarios without supplying the missing package or settling the unforced problem.

Research workflow. Start at an explicit laboratory route, record radius / normalization and chirality, inspect native identities before projection, compare exact and truncated cases, retain the finite band and viscosity in exports, and keep every open or external assumption in the resulting claim.

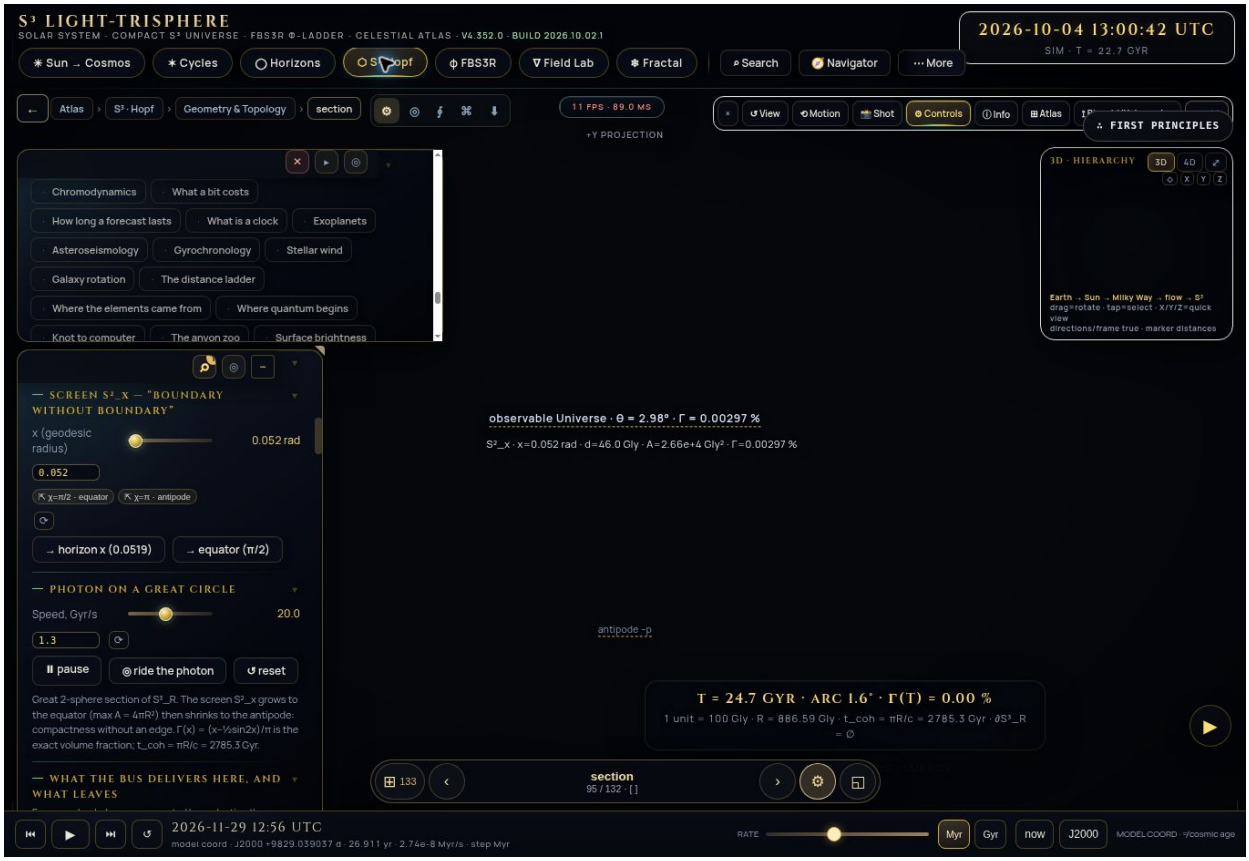


Figure 15: S³ section-view interface, with radius and coherence-time readouts.

20 FBS3R φ-Ladder

The ladder $R_N = \ell_P \varphi^N$ is an exponential scale coordinate anchored by the declared Planck-length constant. Its inverse gives a continuous level for any positive input length, but locating a length on a logarithmic ladder is not evidence that physical quantities are quantized at integer levels. The controls make level travel, local shell context and scale comparison operational.

The source fixes $\ell_P = 1.616255 \times 10^{-35}$ m, $t_P = 5.391247 \times 10^{-44}$ s and the published label $N_{\mathrm{rec}} = 266$. It derives the S³-radius and particle-horizon levels from the current reconstructed lengths. Comments retaining older approximate levels are not numerical authorities after the radius changes. A parameterized radius should be passed through the inverse map instead of keeping a previously quoted level.

20.1 Independent anchors and clocks

CoScale uses $t_N = t_P \varphi^N$ and the ratio identity in equation (9). The internal reference $N_\star = 207$ is described as approximately a light-second scale, not recombination. The recombination label is published independently of this clock. Re-quantization, epoch markers and shell traversal are separate constructs; a shared index does not make their times interchangeable.

Density controls distinguish the published cubic scaling from altered exponent experiments. Zero-point energy and Casimir-style density scalings have their own formulas, including a quartic density power, and must not be collapsed into φ^{-3N}. The exact BigInt Fibonacci path preserves integer values where floating Binet evaluation becomes an approximation. Energy proportionality to Fibonacci numbers remains a model relation whose normalization and units must be retained.

20.2 Sensitivity and falsifiability

The sensitivity view varies H_0 and curvature magnitude and propagates them into radius, level and observable fraction. Its result is a conditional response to the chosen input model. Changing φ or the Planck anchor is explicitly Tier 2 exploration. Canonical markers are hidden in that mode so their appearance cannot imply published support for the altered scenario.

The current source also tests fractional ladder offsets of 113 catalogued scales using a circular-uniformity diagnostic. It reports resultant length 0.135 and $p = 0.129$, explicitly no significant golden-rung resonance in that test. This is a catalogue-dependent finite test, not proof against every possible ladder hypothesis, and not confirmation of one. An exported experiment should preserve the catalogue, anchors, test statistic and null interpretation.

21 Field Laboratory

The field world implements reduced reference experiments on a finite lattice. Heat measures diffusion; damped waves add inertial propagation and damping; Gray–Scott generates reaction–diffusion patterns; electric and gravitational Poisson modes use elliptic reference problems; Schrödinger packets test complex-wave evolution; FitzHugh–Nagumo explores an excitable medium; and 3D Ising Monte Carlo samples a lattice statistical model. Selecting a coloured family does not specify a real material or biological system.

A reproducible solver state consists of the equation/mode, array resolution, spacing, boundary, initialization/seed, integration or relaxation step, elapsed solver time and physical/reduced parameters. The reference implementation owns these choices. A smaller visible point cloud is not evidence of a coarser solver. Conversely, changing a grid parameter is a scientific change even if the picture remains similar. Numerical stability, convergence, conservation and comparison with a known solution answer different questions.

The heat display provides a particularly useful diagnostic: it compares the measured RMS radius of the live lattice field with $\sqrt{\langle r^2 \rangle_0 + 6\alpha t}$. Once the pulse reaches absorbing faces, the infinite-domain reference ceases to describe the bounded computation. The divergence is a visible validity boundary, not necessarily a faulty integrator.

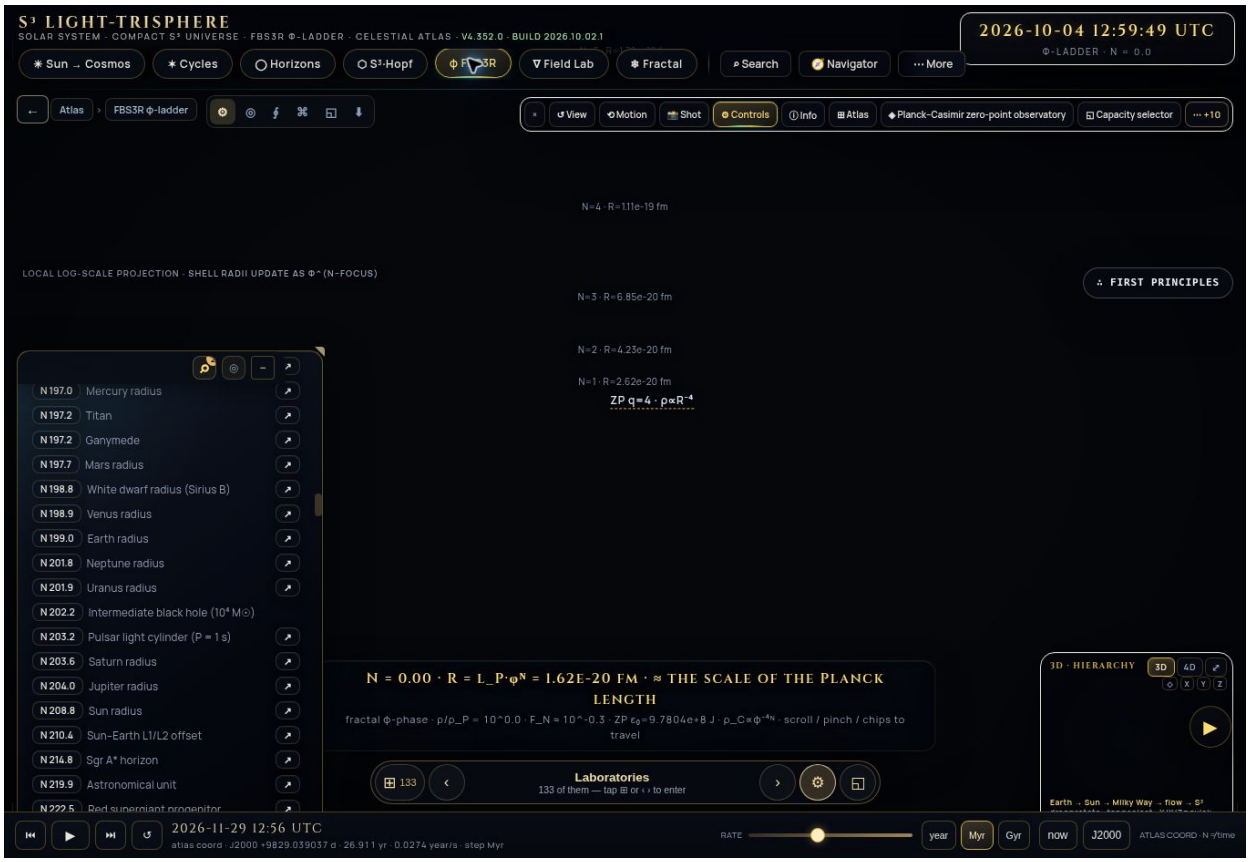


Figure 16: FBS3R interface at level zero, with scale navigation and density readouts.

The current three-way instrument chooser separates Solver lattice, Dimension lattice and Blast wave. The SI lattice organizes quantities by exponent vectors; Buckingham-style dimension analysis identifies possible scaling powers only up to dimensionless constants. A unique dimensional group cannot supply the constant determined by gas dynamics. The Sedov–Taylor view uses $R \propto (Et^2/\rho)^{1/5}$ with a similarity-solution coefficient, and is not a replacement for radiative, inhomogeneous or relativistic blast modelling.

Workflow. Reset a declared mode, record its seed and boundary, advance in controlled steps, inspect a mode-appropriate invariant/residual, change one physical parameter, compare with the analytic reference inside its domain, and export state plus diagnostics. A bus transfer is admissible only for a named output with compatible kind/unit/domain, not for an arbitrary field colour.

22 Fractal Explorer

A fractal view is a mathematical visualization with finite computation. In the Mandelbulb path a distance estimator guides ray steps through a chosen iteration/power construction; the rendered surface depends on estimator, stopping threshold, iteration cap and camera. An image that appears intricate does not prove exact fractal dimension or physical relevance.

Other implemented families produce actual scene geometry: hydrogen-orbital representations, four-dimensional polytopes under projection, a 3D bifurcation construction and Mandelbrot–Julia fibre relations. These have different native spaces and

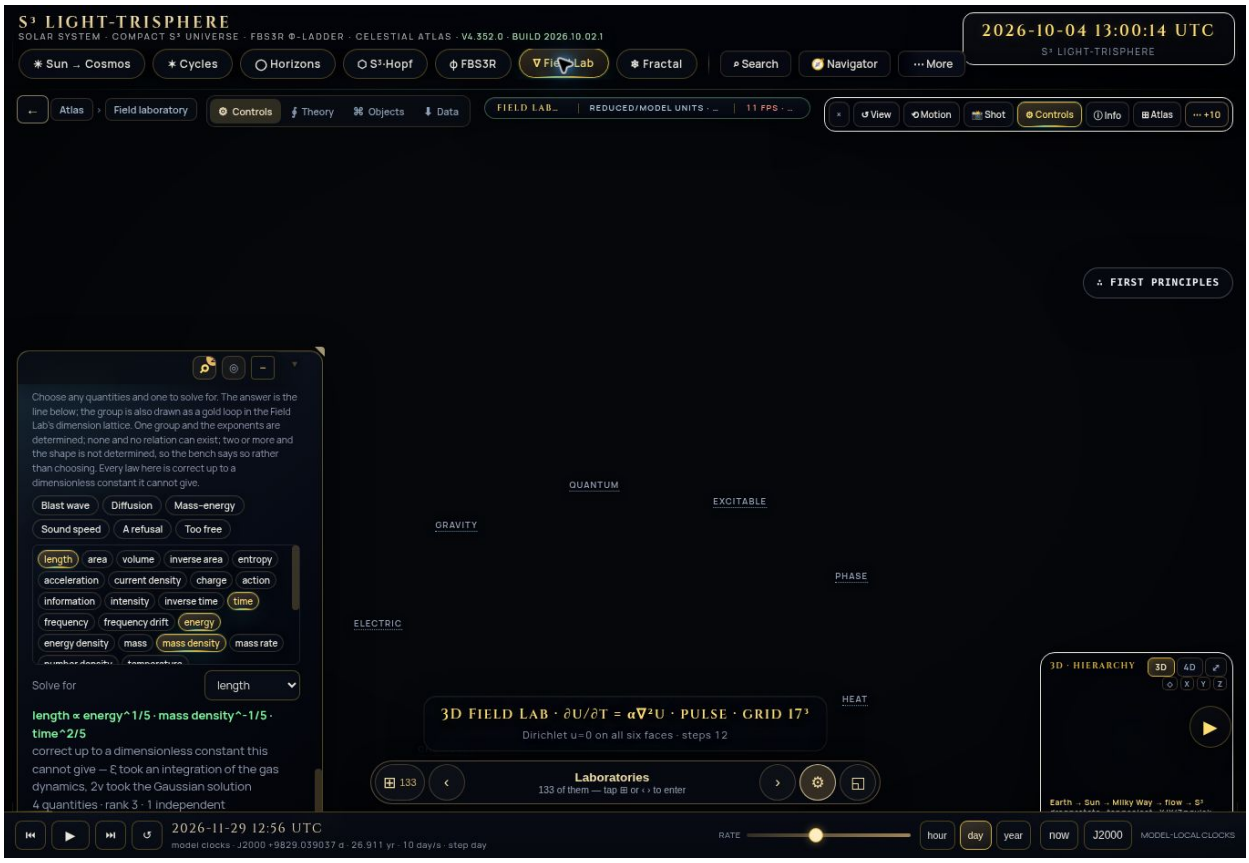


Figure 17: Field Laboratory interface: solver-family selection, controls and diagnostic readouts.

should not all be classified as the same fractal algorithm. The polytopes and division-algebra instruments expose their own exact/combinatorial contracts. The current world description additionally treats box counting as an estimate with a fit window; any reported dimension should retain that window and sampling resolution.

Fractal iteration depth, displayed flight and animation phase are not cosmic evolution. Navigation gestures can zoom or fly through a mathematical object without defining a physically measurable trajectory. Mono fullscreen raymarch modes are not stereo-safe. Entering XR on an excluded mode switches to supported orbital geometry, preserving access to the world while changing its visual mode. This fallback must be disclosed in a reproduction record.

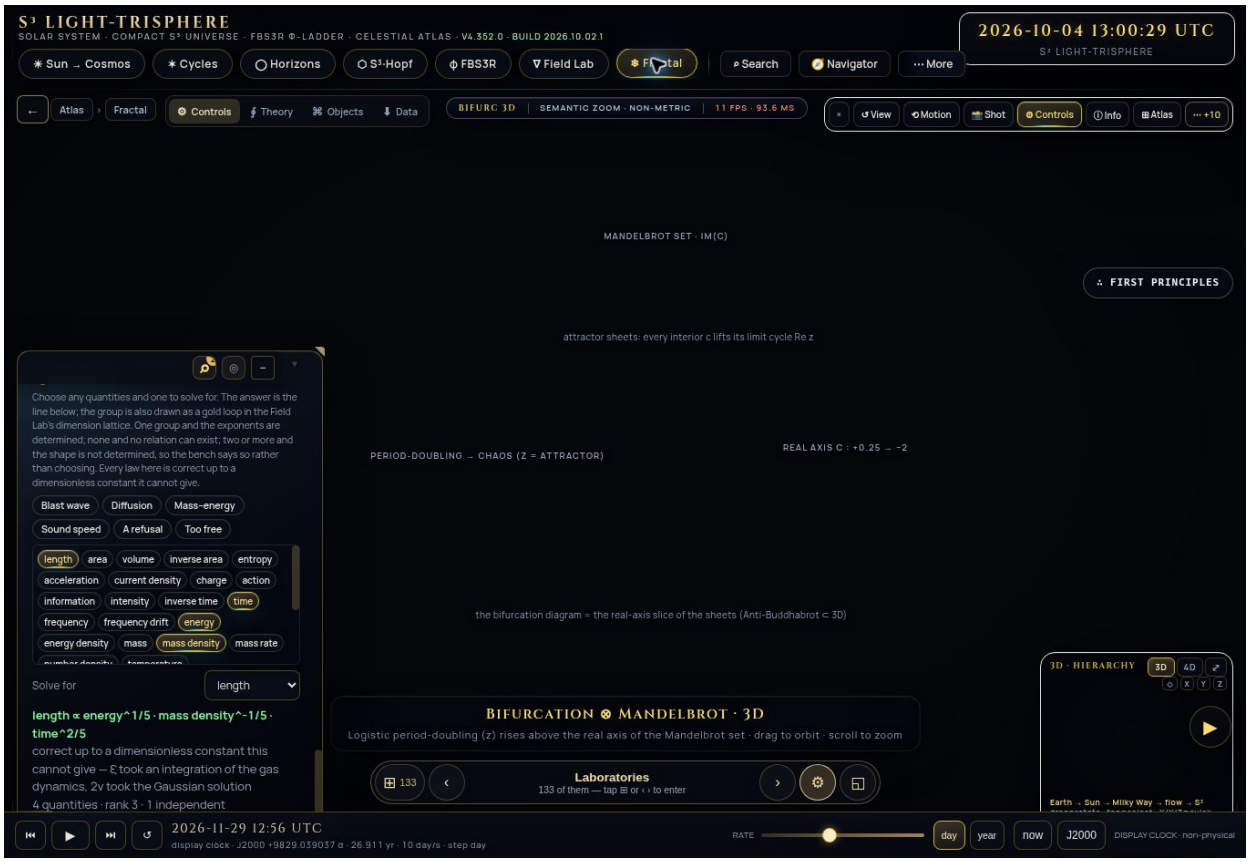


Figure 18: Fractal Explorer interface: mathematical construction labels and mode controls.

23 Rendering and Visual Semantics

The renderer is an instrument interface, not an independent source of scientific evidence. Scene meshes, projected curves, screen-space labels, native DOM inspectors and mathematical calculation have different responsibilities. A marker's colour may encode a category, temperature proxy or selection; its on-screen radius can be exaggerated for visibility. These encodings must be read from the current legend and scale contract.

The premium pipeline adds presentation effects and has a quality/performance cost. A glow is not a radiometric prediction unless an instrument explicitly defines and calibrates that quantity. Camera exposure, tone mapping, bloom and line thickness can

change the image without changing the scientific state. A reproducible image therefore records both model parameters and presentation settings. The source's quality controls and performance adaptation make this separation operationally necessary.

Native calculations produce quantities and diagnostics that are mapped into scene geometry, labels and inspectors. Projection and camera settings control how that scientific state is presented. This separation supports numerical export, interactive inspection and device-specific rendering while retaining a common model state.

Projected singularities, clipping, label overlap, scale exaggeration and camera-dependent apparent intersections affect the interpretation of a scene. Stereo presentation adds eye-specific projection and viewing-scale requirements. Native metrics and diagnostics provide the quantitative reference for interpreting these visual encodings.

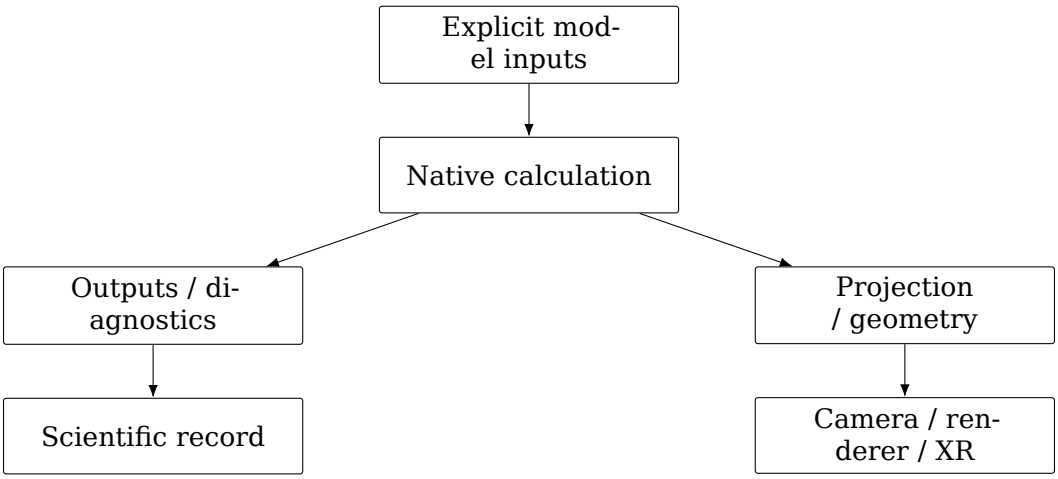


Figure 19: Scientific-state and display-state dependencies: model inputs feed native calculations, numerical outputs, projection and device-specific presentation.

24 Desktop and Mobile Interaction

The common workflow is world selection, laboratory selection, parameter inspection, object selection and evidence inspection. Desktop panels provide catalogue search, category filters, formula access, relation traversal and export actions. Camera orbit, pan and zoom manipulate a view; they do not change a physical frame automatically. A selected object is a semantic focus with its own inspector and provenance, distinct from the current laboratory and from a camera target.

Mobile layouts condense the same catalogue and inspection work into touch-accessible panels. Dragging a slider changes its declared parameter; dragging the scene changes the camera according to the active mode. Panel scrolling must be distinguished from scene gestures. A parameter reset is a model-state action; a camera reset is a view action. Deep links make a route portable between devices, but a route by itself need not serialize every slider, seed or time value.

For reproducibility, start with a reset, record the complete input object, and retain the exported record rather than relying on a remembered gesture. Where a source selector exists, include it. Mobile interaction uses responsive panels, touch gestures and device-specific layout controls.

25 WebXR Control and Immersive Access

WebXR is an additional interaction layer over the Atlas. Secure context, the presence of `navigator.xr`, session support, reference-space creation and successful rendering are separate gates. Merely exposing the browser API does not imply that immersive VR or AR is available. The XR diagnostic panel reports these gates separately, including session support and reference-space information.

Controller rays support pointing and selection; grip/locomotion actions and in-world panels expose navigation and controls without assuming a desktop pointer. The source's XR support includes presentation adaptations and mode guards. A mono raymarch scene cannot be accepted as stereo simply by entering a session: the Fractal world switches unsupported modes to supported geometry. Such a switch changes what the observer sees and belongs in the experiment record.

A meaningful XR acceptance test checks session entry/exit, reference-space and handedness assumptions, selection at multiple distances, readable panels, comfortable scale, controller disconnects and restoration of desktop state. It also checks that each eye receives the intended geometry and that model values survive session transitions. VR/AR should be treated as ways to inspect an existing model, not as evidence that its embedding is physically literal.

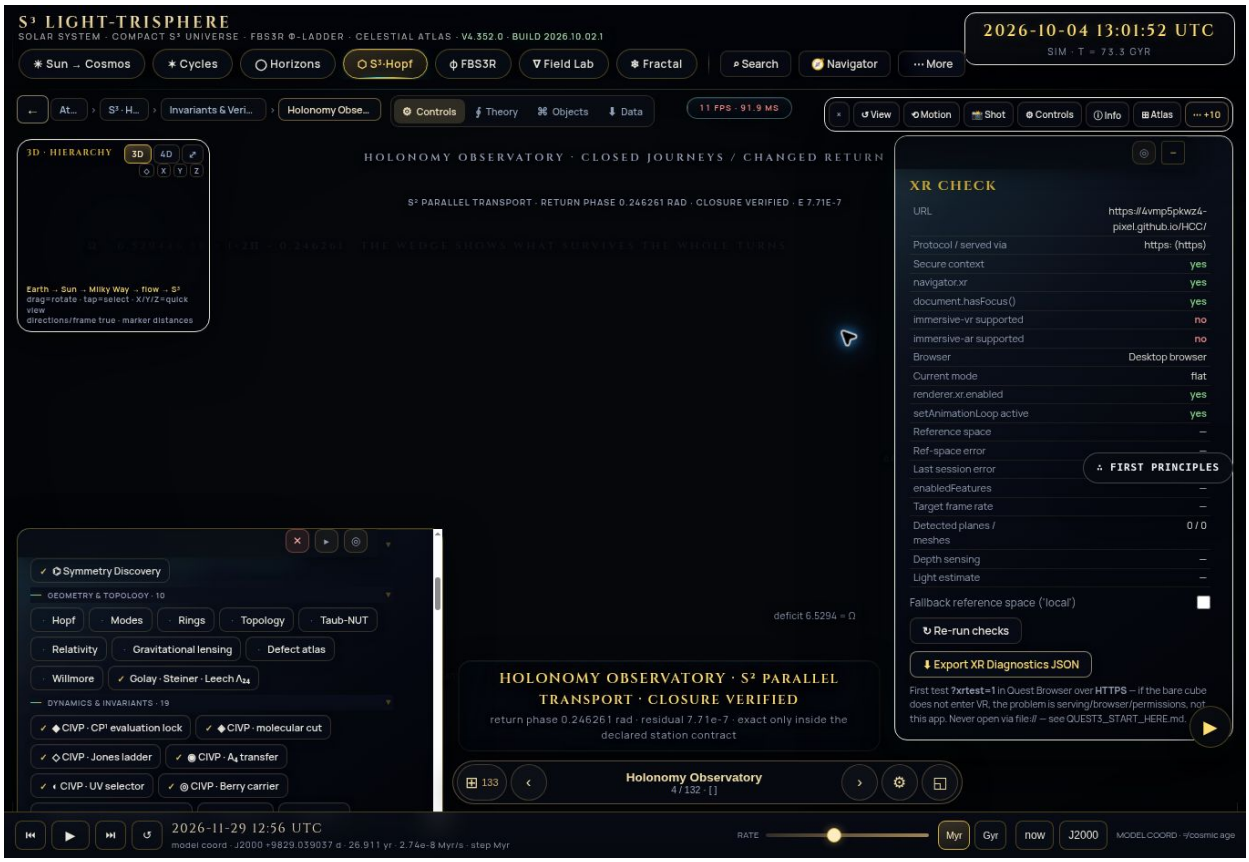


Figure 20: XR diagnostic panel: secure-context status, API availability, immersive-session support and reference-space controls.

26 Validation Architecture and Scientific Evidence

Validation is layered. Registry checks test identifiers, routes and coverage. Contract checks test declared dimensions and semantic completeness. Numerical checks test residuals, invariants, limits or independent references. Browser tests inspect navigation and rendering. Device tests inspect touch and XR. Empirical validation compares a model's observables with appropriately calibrated observations. Passing one layer does not imply passing the others.

The release manifest reports 133 contracts, but only two laboratory native dimensions are explicitly declared. Its first-principles audit counts 527 parameters, all 527 undeclared in the strict parameter-contract sense. This is consistent with usable UI labels and typed instrument inputs: the stricter laboratory contract is a separate layer. The two overrides are `anyzoo` and `jonesq`. The contract represents missing declarations explicitly rather than inferring native dimensions from a rendered scene.

The repository declares quick, source, release, agent and research checks. `test:quick` runs policy validation and quick verifiers; `test:source` expands source verification with `HCC_VERIFY_ALL=1`; release checks add first-principles, Fibonacci-convention, visual and bus checks. Verifier references identify the corresponding checks; run records carry their outcomes and release identity.

Each validation layer addresses a particular scientific or software claim. Registry coverage concerns addressability; a residual concerns a specified equation; convergence concerns discretization error; and a comparison with calibrated data concerns empirical adequacy. Results retain their parameter domain, conventions, tolerance and provenance so that these different claims remain interpretable.

A useful numerical acceptance report names the quantity, domain, initial condition, step/grid, tolerance, comparison and failure criterion. For a conserved quantity, distinguish absolute from relative drift and handle a zero denominator. For a convergence claim, provide successive resolutions and observed order. For a source-derived formula, retain the exact version and convention. Appendix K maps the validation layers to their objects and scientific scope.

27 Reproducibility and Versioned Research Records

A reproducible Atlas result requires the source version, instrument identifier, complete normalized inputs, units and quantity kinds, model assumptions, numerical settings, source tags and outputs. A picture additionally needs route, selected object, clock, camera, quality settings and a statement of display transformations. A deep link is an address, not necessarily a complete experiment serialization.

The repository README documents a Node-based development workflow with dependency installation, browser installation and a CI entry point. Follow the pinned lockfile and the README's supported Node version; the looser package engine range is not sufficient evidence that every dependency runs on every listed version. The HTML imports local modules, so reproduce the repository layout and serve it over HTTP rather than treating the attachment as a fully standalone bundle.

For source-level reproduction, obtain commit `a9a219c6de92e1a9db2ad3f898ccdd626c7af3db`, install its dependencies, run the declared checks, and serve the application. Record any network-fetched catalogue or asset separately. The release manifest, contract records and source identifiers provide the reference for comparing runs

across installations.

Use a result envelope containing the release identity, an instrument contract snapshot, input and output objects, timestamp, verifier references and environment capabilities. Save refusal records as evidence too: an incompatible quantity transfer is a valid, reproducible outcome. If a later release changes an input domain or default, compare explicit input objects rather than assuming the default experiment is unchanged.

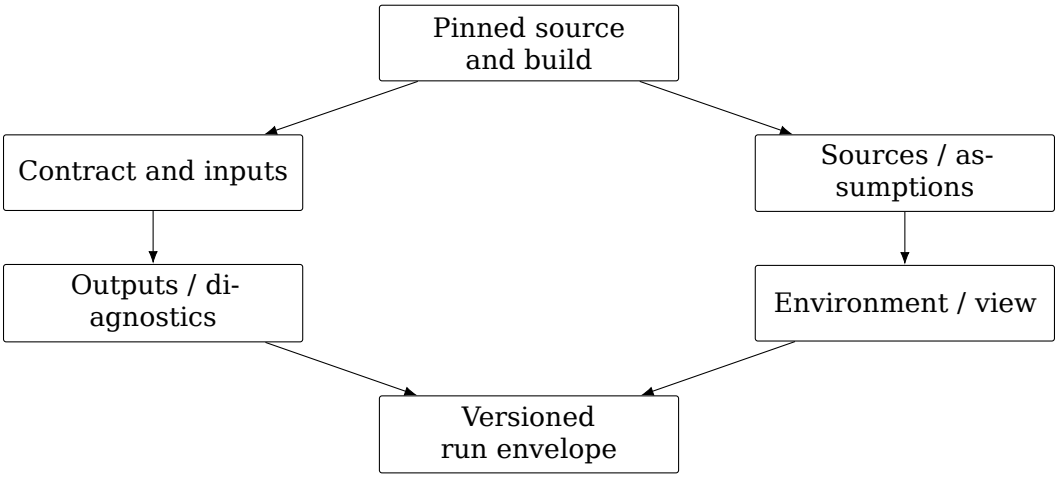


Figure 21: Reproducibility requires a run envelope, not only a route or image. This is a documentation pattern, not a claim of a universal Atlas export schema.

28 A Human Research Workflow

Consider comparing a spinor phase construction with a geometric holonomy. Begin in the catalogue and open the Spin laboratory. Read the native construction and its declared inputs before interpreting the Bloch sphere. Reset the state, choose a controlled loop and record the returned phase and norm diagnostic. Open the formula panel to identify conventions, then export the instrument report.

Move through a typed Nexus relation to Holonomy. Read the edge's claim: a representation or common invariant is narrower than physical identity. Choose the intended transport station, record its loop and compare the return map with that station's analytic reference. A sphere-transport angle and a quantum phase can have related geometry while belonging to different bundles and observables. Multiview can hold the two contexts side by side, but synchronized screen time alone does not align their physical clocks.

Before attempting numerical transfer, ask the quantity bus for an admissible link. If none exists, preserve the refusal or absence; do not copy a number merely because it has the same unit. Complete the record with routes, inputs, outputs, source references and an interpretation restricted to the chosen models. A subsequent researcher should be able to reconstruct both computations without reproducing the camera gesture.

29 An AI-Agent Research Workflow

An agent should discover the site through the static agent description, establish the build and capabilities, and fetch the manifest before selecting an instrument. It should not infer an executable HTTP service from a discoverable JSON file. The public agent descriptor explicitly reports no public HTTP compute service and no public MCP endpoint; the documented SDK/local server are separate deployment choices. [5]

After readiness, list instruments and describe the chosen exact identifier. Inspect every required input, default, domain, quantity kind and unit. Construct an explicit input object and evaluate only within the declared domain. Retain the full response and verifier references. If a calculation requires a source-qualified case, the case identifier is part of the scientific input, not an optional display label.

For a multi-instrument workflow, query bus compatibility and apply the declared conversion exactly once. Check output validity before feeding a successor. Keep the connection claim, conversion, provenance and refusal alternatives in the run record. A Nexus path can suggest an investigation but cannot authorize a computational edge. A ranked structural candidate remains an unadmitted hypothesis until a typed scientific claim, source and review exist.

An agent result should identify the executed instrument, explicit inputs and returned outputs. Missing scientific fields remain unknown. An agent can summarize a reduced model and its numerical output while refusing to promote that output into an observational finding. This distinction is the principal safeguard against a machine-readable atlas becoming a machine-amplified source of overclaiming.

30 Nonclaims and Present Limitations

The Atlas is not a proof of a closed universe, a measurement of global topology, a replacement for standard cosmology, a full radiation-MHD code, or a universal solver for all represented laboratory families. The FBS ladder is a specified construction with internal scale anchors; numerical proximity on that ladder is not by itself causal or statistical evidence. The conditional S^3 radius is contingent on a curvature likelihood and branch conditioning.

Scientific labels, formula membership and verifier filenames do not certify empirical accuracy. A deterministic educational simulation can be scientifically useful while remaining reduced, illustrative or conditional. The lack of a public compute endpoint does not imply that the browser instruments are absent; conversely a static discovery surface does not make those instruments remotely executable.

Instrument domains and declared metadata also constrain automated use. Several first-principles metadata fields are explicitly undeclared, while typed instruments retain their own input/output contracts. A missing scientific declaration remains unknown until it is supplied by the relevant model or source; it cannot be inferred from the dimensionality or appearance of a scene.

31 Conclusion

S³ LIGHT-TRISPHERE organizes heterogeneous scientific models into addressable worlds, laboratories and typed instruments. Its strongest architectural distinction is between representation, computation and evidence: spatial scenes support inspection, instrument contracts delimit operations, source records preserve lineage, and typed relationships state what kind of comparison is admissible.

Release 4.352.0 contains seven worlds, 133 laboratories and 145 instruments. Its 408 Nexus edges and 57 quantity-bus links serve different purposes. A complete contract object does not imply complete metadata, and an internal verifier does not imply empirical validation. Retaining these distinctions makes the Atlas useful as a research interface and an agent-accessible catalogue without converting its shared visual language into a claim of shared physical ontology.

This preprint supplies a stable reading map and a versioned documentary record. The executable Atlas remains necessary for parameter exploration, calculation, spatial inspection and device-specific interaction. Versioned contracts and run records allow subsequent releases to be compared without conflating changes in scientific models, computational methods and presentation.

A World Registry

Table 17: All seven current world routes.

ID / title	Route and declared role
`solar` Solar	Open world. the Solar System as an ephemeris rather than a diagram: bodies on computed orbits, the clock driving them, and the scale bar that says how far the picture has been compressed
`cyc` Cycles	Open world. the periods a body actually keeps — precession, nutation, the Milankovitch terms and the ancient calendars — each one compared against its own source rather than against a consensus
`obs` Horizons	Open world. the observable universe at the scales light can reach: the distance ladder, the horizons, and the boundary between what has been measured and what has been extrapolated, drawn as a boundary
`s3` S^3 · Hopf	Open world. the three-sphere and the laboratories that live on it: the Hopf fibration, the curl shells and eigenmodes of the round metric, and the bench from which every other laboratory of the atlas is reachable
`fbs` FBS3R	Open world. the φ-ladder taken seriously enough to be falsified: its rungs, its graded capacity, and its own Planck anchors published beside the CODATA ones so the disagreement can be read rather than resolved by fiat
`field` Field Lab	Open world. fields you can put your hands on: configurations integrated in front of you, their conserved charges counted from the configuration itself, and every parameter live rather than baked into a picture
`fractal` Fractal	Open world. scale invariance measured instead of asserted: dimensions estimated by box counting on the object in view, with the fit window shown, because a fractal dimension without its window is a number without a claim

B Complete Laboratory Registry

The manifest is authoritative for these 133 records. The third column gives world, category, registry kind and assigned instrument. A host assignment is not a claim that the instrument implements every scene operation. The six parametric entries have no assigned instrument. Descriptions are the English release descriptions.

Table 18: Complete laboratory catalogue with clickable routes.

No. / ID	Laboratory and scope	Host / kind / instrument
1. `he3`	Helium-3 superfluid · A and B the p-wave gap drawn as the function on the Fermi surface that it is, its exactly-two-thirds angular average in the A phase, the closed-form density of states with its point-node tail, and the two nodes as Berry monopoles of charge plus and minus one	s3 / quant computational `he3`
2. `nuosc`	Neutrino oscillation · PMNS the PMNS matrix, the three flavour-transition probabilities that sum to exactly one at every baseline, the unitarity triangle whose area is the Jarlskog invariant, and the flavour content of the three mass states	s3 / quant computational `nuosc`
3. `adisk`	Accretion disk · Shakura–Sunyaev the Shakura–Sunyaev temperature profile, its exact maximum at 49/36 of the inner edge, the Kerr ISCO and the efficiency it fixes, and the multicolour spectrum summed from the annuli	s3 / rel computational `adisk`
4. `elens`	Einstein ring · the lens equation the lens equation, its two roots, the ring they close into and the identities that hold at every offset	s3 / rel computational `elens`
5. `emb`	Embadon laboratory · Sym^N(CP¹) the atoms of the molecular cut, the four-dimensional space of unordered configurations, its moment polytope and torus fibre, the collision locus in R⁴ under SO(4), and the braid that exchanges two embadons	s3 / inv computational `emb`
6. `civplock`	CIVP · CP¹ evaluation lock the residual evaluation complex on CP¹: acyclic exactly when the molecular divisor degree equals the index-space dimension, decided by two routes that share no code	s3 / inv computational `civplock`
7. `civpcut`	CIVP · molecular cut the atomic null area measure: q_can = Σγ_i, the bosonic quotient, double-null gluing, and the gate that separates a support count from a capacity	s3 / inv computational `civpcut`
8. `civpidx`	CIVP · Jones ladder the small-index Jones spectrum, the strict Pimsner–Popa window, and the four things a finite-index defect cannot be	s3 / inv computational `civpidx`

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No. / ID	Laboratory and scope	Host / kind / instrument
9. `civpa4`	CIVP · A_4 transfer the exact integral transfer BB^T = N_τ² = [[1,1],[1,2]] and the Fibonacci fibre over molecular configuration space	s3 / inv computational `civpa4`
10. `civpsel`	CIVP · UV selector the invariant UV selector calculus: κ = Δ²log I_q as a complete affine invariant, unique-sector crossing and topology-stability bounds	s3 / inv computational `civpsel`
11. `civpcar`	CIVP · Berry carrier the Berry–Kirillov off-by-one and the Capelli transfer, with the Stinespring leakage identity as a finite reduction diagnostic	s3 / inv computational `civpcar`
12. `civpclo`	CIVP · conditional closure the conditional closure: the exact de Sitter capacity map, additive entropy typing, the BMS shape quotient and five undischarged certificates	s3 / inv computational `civpclo`
13. `anyon`	Fibonacci anyon braiding Fibonacci anyons, braided by hand: worldlines you weave and the quantum state they compute	s3 / quant computational `anyon`
14. `nexus`	Invariant Nexus typed, non-metric cross-laboratory relation universe with morphing disciplinary and invariant embeddings	s3 / inv parametric none
15. `nul`	Spinor & Light-Cone Observatory exact CP^1, SL(2,C), null-tetrad, spin-weight and radial conformal-infinity identities	s3 / rel computational `nul`
16. `act`	Contact & Action Observatory contact dynamics on an S^3 energy shell, exact Reeb return, Legendrian curves, KS regularization and Maslov action	s3 / contact computational `act`
17. `hol`	Holonomy Observatory closed-path return maps across round-sphere transport, Berry phase, SU(2), Lorentz boosts and SL(2,R) Möbius actions	s3 / inv computational `hol`
18. `syd`	Symmetry Discovery Chamber finite-orbit candidate testing across SO(3), Lorentz, U(1), symplectic and Möbius transformation families	s3 / inv parametric none
19. `sec`	section a moving 2-sphere section of S^3, the elementary way to see a three-sphere at all	s3 / geom computational `ebk`
20. `osc`	Frequency generators · phase coincidence a bank of frequency generators and the number theory that decides when they fall back into phase: the continued fraction of the ratio, its convergents against the Hurwitz bound 1/sqrt5, the three-distance theorem counted on the circle, and the Fibonacci waiting times a golden ratio produces	s3 / maps computational `osc`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
21. `scales`	Scale panorama · Planck to Hubble every typed instrument evaluated at its own declared defaults, every output whose unit can be converted put into SI, and the whole atlas laid on one logarithmic axis per kind of quantity — with the quantity bus underneath, one declared coupling at a time, driven end to end so the influence can be watched rather than read about	s3 / inv computational `scales`
22. `hopf`	Hopf fibres the Hopf fibration drawn as linked fibres, every pair with linking number one	s3 / hopf computational `anyon`
23. `leech`	Golay · Steiner · Leech Λ_{24} the extended Golay code spanned from twelve generator rows, its 759 octads, the Steiner system S(5,8,24) checked by walking all 42504 five-point choices, and the 196560 minimal vectors of the Leech lattice projected along three of its own orthogonal vectors onto 107 exact integer sites	s3 / geom computational `leech`
24. `eig`	modes β the Laplace eigenmodes of the round S^3 and their $(k+1)^2$ degeneracy	s3 / hopf parametric none
25. `red`	redshift redshift along a great circle of the compact S^3 universe	s3 / geom computational `red`
26. `ring`	rings the ring family of S^3 and how it foliates the sphere	s3 / hopf parametric none
27. `lab`	centre lab the central laboratory bench, the entry point of the S^3 world	s3 / geom parametric none
28. `bell`	Bell lune the Bell lune and the spherical excess that measures its curvature	s3 / geom computational `bell`
29. `drd`	DRD info the declared reconstruction domain, and what it is and is not	s3 / cosmo parametric none
30. `sel`	selector the selector: every registered object of the atlas in one field	s3 / geom computational `tomography`
31. `hopfion`	Hopfion lab a hopfion field configuration and its conserved Hopf charge	s3 / hopf computational `hopfion`
32. `skyrmion`	Skyrmion lab a skyrmion and the topological degree that stabilises it	s3 / hopf computational `skyrmion`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
33. `quasi`	quasiparticles phonon, magnon, exciton and collective-mode reference models	s3 / quant computational `quasi`
34. `atom`	elements Aufbau occupations, nuclear radius law and approximate Slater effective radii	s3 / quant computational `atom`
35. `grav`	Gravitas 3-body symplectic reduced-unit few-body dynamics with invariant monitoring	s3 / dyn computational `grav`
36. `qm`	Quantum wave lab periodic 512-point split-step Fourier TDSE reference solver	s3 / quant computational `qm`
37. `qcd`	Chromodynamics lab Cornell potential, colour-neutrality and one-loop running-coupling demonstrators	s3 / quant computational `qcd`
38. `topo`	Topology atlas Gauss linking, Poincaré–Hopf index and Hopf-flow stations	s3 / geom computational `topo`
39. `sc`	Superconductivity lab Meissner screening, Josephson phase and Abrikosov flux lattice	s3 / quant computational `sc`
40. `taub`	Taub-NUT lab Berger-sphere fibres and reduced monopole-like orbit demonstrator	s3 / rel computational `taub`
41. `eot`	Extraordinary transmission phenomenological hole-array plasmon resonance model	s3 / maps computational `eot`
42. `wave`	Wave optics lab Huygens superposition through configurable slit arrays	s3 / maps computational `wave`
43. `spin`	Quantum spin lab Bloch-sphere and Rabi two-level dynamics	s3 / hopf computational `spin`
44. `rel`	Relativity lab Lorentz kinematics, simultaneity and causal-cone benchmarks	s3 / rel computational `rel`
45. `chaos`	Strange attractors Lorenz, Rössler and related deterministic nonlinear flows	s3 / dyn computational `chaos`
46. `kin`	Kinetic theory Maxwell–Boltzmann reference distribution and collisions	s3 / dyn computational `kin`
47. `sh`	Spherical harmonics real Y_lm angular modes and orthogonality diagnostics	s3 / geom computational `sh`

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No. / ID	Laboratory and scope	Host / kind / instrument
48. `kdv`	Soliton (KdV) lab dispersive nonlinear-wave propagation with invariant checks	s3 / dyn computational `kdv`
49. `ret`	Resonant energy transfer coupled-oscillator energy exchange and detuning	s3 / maps computational `ret`
50. `dip`	Dipole radiation retarded angular radiation pattern demonstrator	s3 / maps computational `dip`
51. `cp`	Charged particle Boris integration in prescribed electric and magnetic fields	s3 / maps computational `cp`
52. `rd`	Reaction–diffusion Gray–Scott pattern formation on a reduced grid	s3 / dyn computational `rd`
53. `he`	Heat engine ideal Carnot-cycle state and efficiency relations	s3 / maps computational `he`
54. `bb`	Blackbody radiation Planck spectrum, Wien peak and Stefan–Boltzmann scaling	s3 / maps computational `bb`
55. `nuc`	Nuclear binding semi-empirical mass-formula landscape	s3 / quant computational `nuc`
56. `lens`	Gravitational lensing Schwarzschild deflection and capture-threshold demonstrator	s3 / rel computational `lens`
57. `te`	Thermoelectric Seebeck/Peltier relations and finite-ZT efficiency	s3 / maps computational `te`
58. `gw`	Gravitational waves leading-order compact-binary chirp relations	s3 / rel computational `gw`
59. `pv`	Photovoltaic single-diode reference curve and maximum-power point	s3 / maps computational `pv`
60. `su2`	SU(2) spinor unit-quaternion double cover, belt trick and SLERP	s3 / hopf computational `su2`
61. `bht`	Black-hole thermodynamics Kerr horizon radius, ergoregion, area entropy, angular velocity and a separate Schwarzschild-only evaporation worldtube	s3 / rel computational `bht`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
62. `rpd`	RPD transformer the RPD transformer and its impedance transformation law	s3 / maps computational `rpd`
63. `imp`	Impedance · Möbius · Smith fractional-linear impedance maps, reflection coefficient and resonance matching	s3 / maps computational `smith`
64. `pole`	Complex-frequency poles poles of a complex-frequency response and the modes they name	s3 / quant computational `pole`
65. `disp`	Dispersion · v_p vs v_g phase against group velocity, and where they part company	s3 / quant computational `disp`
66. `berry`	Berry curvature · Chern Berry curvature and the Chern number it integrates to	s3 / quant computational `berry`
67. `cusp`	Catastrophe · caustics catastrophe theory: the cusp, its caustics and their stability	s3 / dyn computational `cusp`
68. `kam`	KAM · standard map the standard map, KAM tori and the last barrier to break	s3 / dyn computational `kam`
69. `rsh`	Riemann sheet · poles a Riemann sheet, its branch cut and the poles on the second sheet	s3 / quant computational `rsh`
70. `and`	Anderson · hyperbolic drift Anderson localisation seen as hyperbolic drift	s3 / dyn computational `and`
71. `psp`	Phase space · S³ shell · camel canonical area, non-squeezing and Hamiltonian phase-space geometry	s3 / contact computational `psp`
72. `dfx`	Defect atlas · π_n defect classification by the homotopy groups π_n of the order parameter	s3 / geom computational `dfx`
73. `noe`	Noether · momentum maps continuous-symmetry charges, momentum maps and controlled symmetry breaking	s3 / contact computational `noe`
74. `cau`	Causality · Kramers–Kronig causality as the Kramers–Kronig relation between the parts of a response	s3 / quant computational `cau`
75. `wil`	Willmore · Clifford torus Willmore energy and the Clifford torus that minimises it	s3 / geom computational `wil`
76. `gate`	Prediction gate · refinement the prediction gate: a refinement that must be passed before a claim counts	s3 / quant computational `gate`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
77. `cps`	Covariant phase space · Wald charges covariant phase space and Wald charges	s3 / contact computational `cps`
78. `poin`	Poinsot · Dzhanibekov the Poinsot construction and the Dzhanibekov flip of the intermediate axis	s3 / dyn computational `poin`
79. `qcrys`	Quasicrystal · cut-and-project a quasicrystal built by cut-and-project from a higher-dimensional lattice	s3 / geom computational `qcrys`
80. `cmb`	CMB low multipoles deterministic Gaussian sky synthesis, Y_lm inversion, cosmic variance and masked mode-coupling diagnostics	s3 / cosmo computational `cmb`
81. `mimg`	S³ multiple imaging conditional round-S³ geodesic paths and observable-topology caveats	s3 / rel computational `mimg`
82. `bhr`	Relativistic black hole pixelwise Schwarzschild null-geodesic integration with explicit capture and step diagnostics	s3 / rel computational `bhr`
83. `ns`	Neutron star (TOV) reduced Tolman–Oppenheimer–Volkoff interior with declared equation-of-state dependence	s3 / rel computational `ns`
84. `sn`	Supernova Bateman abundances, grey gamma escape, stable Arnett diffusion, a conditional photosphere, a smooth ejecta/Sedov match and a radius-continuous radiative phase switch	s3 / rel computational `sn`
85. `psr`	Pulsar magnetic-dipole lighthouse geometry and spin-down reference relations	s3 / rel computational `psr`
86. `qso`	Quasar Eddington scaling, thin-disk efficiency and relativistic Doppler-beaming reference model	s3 / rel computational `qso`
87. `rmhd`	Radiation–MHD & Helicity five reduced analytic and kinematic stations for flux freezing, magnetic helicity, reconnection, shocks and magnetized Rayleigh–Taylor instability	s3 / rel computational `rmhd`
88. `wd`	White-dwarf degeneracy Nauenberg zero-temperature composition-dependent mass–radius family and Chandrasekhar endpoint	s3 / rel computational `wd`
89. `s3lock`	The terminal lock · a $\mathcal{J}$-paired Hopf filament, and the one number the programme has not closed a Hopf filament and its $\mathcal{J}$-conjugate on the Clifford family, the drift that winds them, and the terminal invariant lattice measured beside them — with the lock itself left open	s3 / inv computational `s3lock`

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No. / ID	Laboratory and scope	Host / kind / instrument
90. `tri`	The Trisphere · everything that leans on S³ the three-sphere whole — Clifford tori, Hopf fibres, and every laboratory placed on the torus of its distance from S³, joined by the geodesics of S³ along its declared chain	s3 / hopf computational `tri`
91. `bbh`	The last merger of the Local Group · Sgr A* and M31*, inspiral to ringdown Sgr A* and M31* spiralling together in the gravitational-wave regime, the wave on the orbital plane at its retarded time, and the remnant ringing down	s3 / rel computational `bbh`
92. `s3gamma`	Space forms S³/Γ · their spectra, and whether the sky could show them the Poincaré, octahedral, tetrahedral, prism and lens spaces: their exact Laplace spectra from Spin(4) characters, the images of the observer (the 600-cell), and whether matched circles could appear in the sky on this atlas's own curvature posterior	s3 / geom computational `s3gamma`
93. `discover`	The Discovery Explorer · what this atlas has found, confirmed and corrected the ledger of what this atlas has found, confirmed and corrected, as a constellation: a star per claim shaped by its kind, arcs where two share a laboratory, the atlas's corrections of itself in red, and the open leads	s3 / inv computational `discover`
94. `lgmerge`	The Local Group's future, by law · the Milky Way and Andromeda from today's data the Milky Way and Andromeda from today's data: two haloes under dynamical friction calibrated to the N-body merger, two discs of test-particle stars, the black holes riding the centres — and the probability of a merger	s3 / rel computational `lgmerge`
95. `nsgal`	The fluid of the three-sphere · Navier–Stokes beyond the exact solutions Navier–Stokes solved in the Spin(4) curl shells: a hundred modes exchanging energy through the Lamb vector, drawn in the trisphere's four-dimensional turn with its streamlines and tracers, beside the phase space of the invariants and the rigid rotation it must end in	s3 / dyn computational `nsgal`
96. `nsblow`	The anatomy of a forced blow-up · the 2026 exponents and the window every theorem leaves The September 2026 exponents of the forced Navier–Stokes blow-up put through every classical regularity theorem: the collapsing swirl core in its own frame, the physical cores stretching as `τ^{−h}`, the plane of all collapses in the space of logarithms, and the ladder of unstable singularities	s3 / dyn computational `nsblow`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
97. `strobe`	The Stroboscope · resonances across space, time and parameters Resonances across space, time and parameters: the exact Navier–Stokes solution on S³ strobed on its torus knots, the atlas's clocks strobed against a null, and every closing frequency ratio with the exact carriers on it	s3 / dyn computational `strobe`
98. `vstretch`	Vortex stretching is a Vandermonde determinant · the curl levels of S³ The total vortex stretching of any sum of curl levels on S³ as a Vandermonde determinant of the curl eigenvalues: S³ coloured by where vortex lines stretch and shorten, the triads that carry it, and the two numbers that must agree	s3 / dyn computational `vstretch`
99. `triad`	The tennis racket inside the fluid · every triad of S³ is a free rigid body Every triad of Beltrami modes of S³ as a free rigid body: the energy sphere cut into orbits by the helicity levels, the top turning with the triad as its angular momentum, the lever rule of the unstable leg, a strobe at Jacobi's period, the double Clebsch–Gordan rule of which triads exist, and the crowd of the whole fluid around the unstable leg	s3 / dyn computational `triad`
100. `omnistrobe`	The stroboscope of everything · phases from the electron to the Galaxy Every period the atlas knows, from the electron's Compton oscillation to the galactic year, on one ladder of frequencies with the band the display can show truly sliding along it; two phases on a torus, three in a cube where a resonance is a plane; the resonance finder, blind, with its null	s3 / dyn computational `omnistrobe`
101. `formulas`	The Formula Portal · every formula of the S³ programme, and the ones the atlas can check Every formula of the S³ Navier–Stokes programme as a star — 1801 distinct of 5201 displayed, in ten constellations — searchable, typeset, linked to its laboratory, and checked here where the atlas can	s3 / dyn computational `formulas`
102. `nsflow`	A Navier–Stokes flow on S³, integrated · the exact solution the equation itself confirms an exact non-stationary solution put back into the equation and its residual measured, thousands of tracers carried by it on S³ measuring its energy, and the ledger of what is proven, conditional and open	s3 / dyn computational `nsflow`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
103. `s3kb`	The chiral Killing–Beltrami law · one identity, and the asymmetry that separates two shells the exact polarized intertwining law, the two chirality coefficients it gives every signed shell, and the quadratic Casimir that turns viscous dissipation into six convective response norms	s3 / dyn computational `s3kb`
104. `s3escape`	Spectral escape on S³ · the rank a bounded-energy peak has to outgrow the exact band-rank staircase against the two lower bounds a bounded-energy peak forces on it, with the Clebsch–Gordan gate that decides which shell pairs can feed a given output at all	s3 / inv computational `s3escape`
105. `s3tube`	Geodesic-axis coordinates on S³ · where the curvature actually enters the exact round metric around a great circle, its determinant identity, the incompressibility transfer it forces and the single Euler polynomial that carries the whole curvature of a pure swirl	s3 / geom computational `s3tube`
106. `s3torus`	The maximal torus of S³ · where this fluid becomes two scalar heat equations the complete two-scalar invariant family, its Jacobi diagonalisation, the pressure that absorbs the whole nonlinearity and the chiral identity that lands mode n in curl shell 2n	s3 / dyn computational `s3torus`
107. `s3shell`	Navier–Stokes on the round three-sphere · curl shells, Beltrami solutions and spectral escape signed curl eigenspaces built from harmonic polynomials on $S^3 \subset \mathbb{H}$, with the exact Beltrami amplitude law and the sharp finite-band projector constants measured rather than quoted	s3 / dyn computational `s3ns`
108. `cosmicweb`	Cosmic web laboratory · from primordial modes to the largest structures one deterministic Zel'dovich displacement field linking primordial modes to voids, sheets, filaments and nodes	s3 / cosmo computational `nesting`
109. `jeans`	Jeans collapse · when a cloud stops holding itself up the mass above which a gas cloud stops holding itself up and starts becoming a star	s3 / rel computational `jeans`
110. `origins`	Where the elements came from · three minutes, ten billion years, and one second the atlas could give the electron configuration of gold and had nothing to say about where gold came from: an elements laboratory with every configuration, a nuclear laboratory that finds the binding peak, a supernova that watches nickel decay, and nothing between them about the half of the periodic table heavier than iron	s3 / cosmo computational `origins`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
111. `rungs`	The ladder · from a triangle to two Hubble constants seven outputs in this atlas are quoted in astronomical units and nothing consumed one, while the cosmological-constant bench offers a button for Planck's 67.66 and a button for SH0ES's 73.04 and never says why there are two	s3 / cosmo computational `rungs`
112. `mainseq`	The main sequence · where stars live, and the ratio a galaxy hangs on three laboratories in this atlas publish a luminosity in watts and across a hundred and six nothing had ever consumed one, while the galaxy bench declares in as many words that its largest uncertainty is a stellar mass-to-light ratio it had no way to compute	s3 / cosmo computational `mainseq`
113. `galrot`	Galaxy rotation · the mass you can see is not the mass that is there an atlas that computes the cosmological constant to a hundred and twenty-three orders and inverts Kepler's third law on an exoplanet to nine figures had never drawn a galaxy, where the same law fails by a factor of three	s3 / cosmo computational `galrot`
114. `exo`	The planets of other stars · a transit, a wobble, and a zone an atlas of a hundred and four laboratories that had measured a black hole horizon, a Jeans mass and a distance ladder contained the word exoplanet zero times, while six thousand of them are known	s3 / cosmo computational `exo`
115. `seis`	Asteroseismology · the star rings, and the ringing gives its mass the main sequence turns a mass into a radius through a MODEL and the transit bench measures a planet against a star it has to ASSUME, so neither can weigh a star it has not already decided about; the topic census over this atlas returned the word asteroseismology zero times	s3 / cosmo computational `seis`
116. `wind`	Stellar wind · what a star loses while it burns every lifetime in this atlas is computed as if a star kept its mass, and a hot star does not; the topic census returned the phrase mass loss zero times, so the nuclear clock, the turnoff mass and the coincidence built on it all rest on an assumption nothing here stated	s3 / cosmo computational `wind`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
117. `gyro`	Gyrochronology · how old is a star, and why a lifetime is not an age the atlas could compute how long a star LIVES and never how old one IS, which are different numbers by a factor of 2.27 for the Sun; and the spin-down calibration is a relation in three variables that every textbook draws as a curve, hiding the colour dependence that carries its whole error budget	s3 / cosmo computational `gyro`
118. `lam`	The cosmological constant, as a length · and the horizon that is three horizons four publications in this atlas carry a cosmological constant in inverse square metres and nothing has ever consumed one, while the distance ladder takes a Hubble constant and a matter density and never takes Lambda, though it is the same cosmology	s3 / cosmo computational `lam`
119. `cosmo`	Distance ladder · redshift to distance the FLRW integral from a redshift you measure to the three distances you need, and the redshift past which further away means larger on the sky	s3 / cosmo computational `cosmo`
120. `phot`	Photometry · what a telescope actually sees the inverse square law and the magnitude scale: how much of a published luminosity actually arrives, and what that is in the units an observing log is written in	s3 / rel computational `phot`
121. `resolve`	Angular resolution · can it be seen at all the diffraction floor under every angle this atlas computes, and the aperture that would be needed to get under it	s3 / rel computational `resolve`
122. `clock`	What is a clock · the exponent per WHAT, and when two clocks can be exchanged the five clocks this atlas measures rates against and has nowhere to compare: per unit time, per iteration, per lattice site, per unit of volume time and in years — with the one exchange rate that can be measured, measured, and the two that cannot, refused in writing	s3 / quant computational `clock`
123. `phorizon`	The predictability horizon · how long a forecast lasts, and the five rates that cannot say the one thing a positive separation rate is for, which four laboratories published eight times in inverse time and no instrument in a hundred ever consumed: how long a forecast lasts, and which of those eight rates may say so	s3 / quant computational `phorizon`

Continued from previous page

No. / ID	Laboratory and scope	Host / kind / instrument
124. `circ`	The same theorem twice · circulation in a superfluid, flux in a superconductor the join between two quantisation laws this atlas has held apart since it was written: five publications carry a circulation in square metres per second and nothing consumed one, while the superconductivity laboratory counts flux quanta and was never told it is computing the same integer	s3 / quant computational `circ`
125. `infolab`	What does one bit cost · entropy, information and the price of forgetting the join between entropy and information, which five laboratories in three clusters published and none consumed: what a bit costs to erase, how many a region can hold, and how fast energy can be made to compute	s3 / quant computational `infolab`
126. `qregime`	Where does quantum begin · the nine conditions the one question sixteen quantum laboratories never answer: when quantum mechanics is the answer at all, computed as nine ratios rather than described	s3 / quant computational `qregime`
127. `jonesq`	From a knot to a computer · the Jones chain, walked the seven links from a knot invariant to universal quantum computation, each one computed and checked against the next rather than cited	s3 / quant computational `jonesq`
128. `anyzoo`	The anyon zoo · every model and its modular data the classification of two-dimensional statistics, computed rather than described: thirteen modular tensor categories with their dimensions, spins, S-matrices and central charges, and every consistency condition checked	s3 / quant computational `anyzoo`
129. `sbright`	Surface brightness · why a bigger telescope does not help the invariant of the whole observational chain: a flux divided by the solid angle it covers, the theorem that a larger aperture cannot raise it, and the exponent the Tolman test is written in	s3 / rel computational `sbright`
130. `tscale`	Timescales · how long does it last the three clocks that run in every self-gravitating body, their ordering, and what an accretion rate or an age means once there is something to measure it against	s3 / rel computational `tscale`
131. `eos`	Equation of state · what holds it up the one integral between a density and a pressure, with the ideal gas and electron degeneracy as its two measured limits rather than two terms added together	s3 / rel computational `eos`

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No. / ID	Laboratory and scope	Host / kind / instrument
132. `photon`	Photon budget · how long must you look the last rung of the observational chain: quanta arriving at random, the Poisson noise that is their own square root, and the exposure any detection actually costs	s3 / rel computational `photon`
133. `chronometry`	Ancient Chronometry Observatory Source-locked comparative chronometry: textual definitions, historical astronomy, correspondence tests and explicit falsification boundaries.	cyc / inv computational `chronometry`

C Complete Typed-Instrument Registry

All 145 typed instruments are indexed below with their complete release status text. The final column counts inputs and outputs. Full field names, defaults, bounds, units, quantity kinds, descriptions and verifier references are retained in `data/atlas-manifest.json`; source formulas and limits are in `data/source-extract.json`. The host link is reproduced from the instrument contract even where it is not a separately registered laboratory. Status text records the instrument's declared scientific classification.

Table 19: Complete instrument registry.

No. / ID	Title, host and declared status	I / O
1. `chronometry`	**Ancient Chronometry Observatory · source-locked cases**. cyc/chronometry. SOURCE-LOCKED + FALSIFICATION-GATED; astronomy comparisons remain PENDING_EPOCH_CORRECTION	2 / 13
2. `ladder`	**Zero-point ladder — one rung of R_N = ℓ_P φ^N**. fbs/zpt. MODEL (declared: round S³, conformally coupled massless real scalar, one renormalization scheme, structural ansatz R_N = l_P phi^N) + VERIFIED arithmetic	4 / 14
3. `ebk`	**EBK quantisation — the semiclassical prediction against the exact spectrum**. s3/sec. DERIVED (action integral) + VERIFIED against exact diagonalisation, including the 1/j^2 convergence rate; the licence to use EBK here was MEASURED in work package C2, not assumed	5 / 4
4. `creation`	**Particle creation — what an anisotropic universe makes, and what expansion does not**. s3/sec. DERIVED (exact factorisation + exact Bogoliubov system) + VERIFIED; conformal coupling and zero mass are ASSUMPTIONS, not results	9 / 7
5. `spectrum`	**Spectral operator — H(t) = ½ Σ a_i⁻² K_i² on an anisotropic S³**. s3/sec. DERIVED (operator and matrix elements) + EXACT (Jacobi diagonalisation, machine precision) + VERIFIED; the adiabatic interpretation is explicitly NOT claimed	4 / 7
6. `bianchi`	**Bianchi IX — constraint-preserving evolution and trajectory class**. s3/sec. DERIVED (volume-time lapse makes the Hamiltonian polynomial) + VERIFIED; the classification is a MEASUREMENT of the integrated trajectory, not a theorem about it	8 / 16
7. `smith`	**Smith–Möbius unrolling — one SU(2) rotation of the Riemann sphere**. s3/imp. THEOREM (the map is a rigid rotation; both routes agree identically) + VERIFIED to the chordal metric on CP^1	3 / 6
8. `bht`	**Kerr horizon thermodynamics — the four exact constants of a rotating hole**. s3/bht. EXACT (closed form for the Kerr family; the evaporation time is the Schwarzschild Page result and is labelled as such)	2 / 7

Continued from previous page

No. / ID	Title, host and declared status	I / O
9. `sh`	**Spherical harmonics — the orthonormal basis the multipole sky is written in**. s3/sh. EXACT (Condon-Shortley associated Legendre functions with the standard normalisation)	4 / 6
10. `nuc`	**Semi-empirical mass formula — the five terms that shape the valley of stability**. s3/nuc. REFERENCE_MODEL (a standard fitted model, useful as a benchmark; it is NOT a measurement and it is NOT a nuclear structure calculation)	2 / 5
11. `bb`	**Planck spectral radiance — and the two laws that follow from integrating it**. s3/bb. EXACT (the Planck law in closed form; Stefan-Boltzmann and Wien are recovered by quadrature and by root-finding rather than quoted)	2 / 5
12. `infolab`	**What does one bit cost — entropy, information and the price of forgetting**. s3/infolab. EXACT (closed-form bounds: Landauer, Bekenstein, the horizon entropy, Margolus-Levitin and Bremermann. Every one of them is a LIMIT and none is a prediction of what a device achieves, which is the distinction the whole laboratory turns on)	8 / 15
13. `qregime`	**Where does quantum begin — the nine conditions, all evaluated at once**. s3/qregime. EXACT (nine dimensionless ratios in closed form; the phenomena a system can host are READ from the ratios rather than listed, and every one of the criteria is an ideal-gas or single-particle estimate, which is stated rather than hidden)	9 / 16
14. `jonesq`	**From a knot to a computer — the Jones chain, walked**. s3/jonesq. EXACT (the Kauffman bracket over Temperley-Lieb diagrams with exact integer Laurent arithmetic, closed by the Markov trace; the gate hunt is EXHAUSTIVE to a declared depth and sampled beyond it, and reports which it is)	6 / 15
15. `anyzoo`	**The anyon zoo — every model, and its modular data computed**. s3/anyzoo. EXACT (the modular data of thirteen models, computed from their fusion rules and quadratic forms; the quantum dimensions come from power iteration on the fusion matrix rather than from the closed form, and every consistency condition is checked rather than assumed. Which models are universal for quantum computation is an EXTERNAL theorem, cited and not derived, and its consequence is measured instead)	3 / 16

Continued from previous page

No. / ID	Title, host and declared status	I / O
16. `sbright`	**Surface brightness — how bright per patch of sky, and why a bigger telescope does not help**. s3/sbright. EXACT (radiance is a flux divided by a solid angle and the image-plane invariant is the conservation of etendue; the Tolman exponent is assembled from its three powers rather than typed, and the sky brightness is a declared INPUT because a sky is a site and a night, not a constant)	7 / 14
17. `tscale`	**Timescales — how long does it last**. s3/tscale. EXACT (three closed-form timescales and their ordering; the nuclear efficiency is derived from the proton and helium masses rather than quoted, and the burning fraction is a declared input because it is a convention and not a measurement)	7 / 13
18. `eos`	**Equation of state — what holds it up**. s3/eos. EXACT (the relativistic Fermi-Dirac pressure integral at finite temperature, with the ideal gas and the Chandrasekhar T=0 form as its measured limits rather than as separate terms added together)	5 / 18
19. `photon`	**Photon budget — how long must you look**. s3/photon. EXACT (Poisson statistics on a declared flux; the background is an input and not a model, so a zero background reports the photon-limited floor)	5 / 9
20. `resolve`	**Angular resolution — can it be seen at all, and with what**. s3/resolve. EXACT (the Airy pattern from Bessel J1; the 1.22 coefficient is BISECTED out of the first zero rather than quoted)	3 / 9
21. `phot`	**Photometry — how much of it actually arrives, and what you write in the log book**. s3/phot. EXACT (the inverse square law and the IAU 2015 bolometric zero point, which is a definition and is reproduced rather than quoted)	2 / 8
22. `cosmo`	**Distance ladder — the rung between a redshift you measure and a distance you need**. s3/cosmo. EXACT (the FLRW distance integrals by Simpson quadrature; the age and the angular-diameter turnover are DERIVED here, not quoted)	3 / 10
23. `jeans`	**Jeans collapse — the mass above which a cloud stops holding itself up**. s3/jeans. EXACT (the isothermal Jeans criterion in closed form; both mass conventions published rather than one chosen)	3 / 8
24. `wil`	**Willmore energy — and the torus that minimises it**. s3/wil. EXACT (an exactly convergent quadrature on a surface of revolution; the Clifford value 2 pi^2 is recovered, not quoted)	2 / 4

Continued from previous page

No. / ID	Title, host and declared status	I / O
25. `cau`	**Kramers-Kronig — causality as an identity between the two halves of a response**. s3/cau. EXACT (the Lorentz oscillator has a closed form for both halves; the dispersion integral is a converged discrete Hilbert transform and its residual is returned)	4 / 7
26. `lens`	**Exact light deflection — the Schwarzschild integral, not its first term**. s3/lens. EXACT (the deflection is the full quadrature alpha = 2 int du/sqrt(f) - pi; the weak-field 4GM/c^2b is recovered as its limit and is not the definition)	1 / 5
27. `poin`	**Poinsot — the free rigid body, solved in the elliptic functions it needs**. s3/poin. EXACT (Jacobi elliptic solution of Euler's equations; both quadratic invariants are returned so the caller can see them held)	6 / 7
28. `kam`	**The standard map — a Lyapunov exponent measured, not asserted**. s3/kam. NUMERICALLY_VERIFIED (the exponent is a finite-time average over a finite ensemble; its convergence is the caller's to judge and the sample size is an input)	3 / 5
29. `topo`	**Linking number — the Gauss integral, returned with its distance to an integer**. s3/topo. EXACT (the linking number is an integer by theorem; what is computed is a discretisation of the Gauss double integral, and its distance to that integer is returned)	4 / 5
30. `dfx`	**Winding and degree — the two integers a defect cannot lose continuously**. s3/dfx. EXACT (both are integers by theorem; the phase march and the solid-angle sum are their discretisations and both defects are returned)	4 / 8
31. `berry`	**Chern number — an integer that changes, at exactly the gap closings**. s3/berry. EXACT (the Fukui–Hatsugai–Suzuki plaquette sum is gauge invariant and returns the integer on a finite lattice, not a limit of one; the Berry curvature is the closed-form Pontryagin density of the two-band map)	4 / 6
32. `kdv`	**KdV: three invariants that a soliton collision is not allowed to spend**. s3/kdv. MEASURED (the invariants are exact constants of the PDE; what is returned is their drift under the same integrating-factor RK4 the sheet is drawn with, which is the honest number — a claim of exact conservation would be a claim about the integrator, not about the equation)	4 / 8
33. `su2`	**The double cover, measured: 2 pi is minus one and only 4 pi is one**. s3/su2. EXACT (quaternion algebra in closed form; every residual below is a floating-point residual and is returned rather than asserted away)	5 / 8

Continued from previous page

No. / ID	Title, host and declared status	I / O
34. `cusp`	**The cusp catastrophe: where three solutions become one, and the discriminant that says where**. s3/cusp. EXACT (the equilibrium set is the cubic x^3 + a x + b = 0; the roots are Newton-polished to below 1e-8 and each residual is returned, and the discriminant is closed form)	2 / 8
35. `wd`	**White-dwarf degeneracy: the mass–radius relation that runs the wrong way, and its limit**. s3/wd. MODEL (Nauenberg 1972 zero-temperature analytic fit to the Chandrasekhar equation of state; the limiting mass is exact for the model and the radius is a fit, so the status is MODEL and not EXACT)	2 / 8
36. `hol`	**Holonomy — five closed journeys, each returning the angle it was owed**. s3/hol. EXACT (each station has a closed form; the discrete transport, the 257-vertex Wilson loop and the matrix products are its discretisations and every residual is returned rather than asserted away)	3 / 8
37. `te`	**Thermoelectric efficiency — the exact maximum, and where the device stops existing**. s3/te. EXACT (the maximum-efficiency and maximum-COP expressions are closed form for a constant-property module; that the module has constant properties is the model, and the status of the MODULE is what the limits state)	3 / 7
38. `gw`	**The inspiral: a chirp mass, a time to coalescence, and a polarization that is not a choice**. s3/gw. EXACT (leading post-Newtonian quadrupole formulae in closed form; the eccentric merger time is an integration of Peters 1964 and is labelled MEASURED in its own output)	4 / 10
39. `rd`	**The Turing threshold — where a stable chemistry becomes unstable only because it diffuses**. s3/rd. EXACT (the homogeneous states, the Jacobian, the dispersion relation and the critical wavenumber are all closed form for Gray-Scott; the fastest-growing mode is found by a declared scan and is labelled separately)	4 / 10
40. `disp`	**Phase and group velocity — six dispersion relations, and the ratios are exact rationals**. s3/disp. EXACT (each dispersion relation is closed form and each group velocity is its analytic derivative, not a finite difference; the finite difference is computed too and returned as a residual)	2 / 8
41. `noe`	**Noether, and the symmetry that is hidden: L for every central force, A only for one**. s3/noe. MEASURED (the orbit is integrated by velocity Verlet, which is symplectic, so the drifts returned are the integrator; that L is conserved for every s and A only at s = 1 is exact and is what the drifts demonstrate)	3 / 9

Continued from previous page

No. / ID	Title, host and declared status	I / O
42. `atom`	**The Aufbau order, its twenty exceptions, and the charge an outer electron actually feels**. s3/atom. REFERENCE_MODEL (the Madelung order and Slater's screening rules are empirical rules that reproduce measured ground states, not solutions of the many-electron problem; the twenty exceptions are tabulated from measurement because the rule does not predict them)	1 / 10
43. `pole`	**One pole and one zero: where a resonator reflects everything, and where it absorbs everything**. s3/pole. EXACT (the single-pole single-zero reflection coefficient is closed form; every number below is an evaluation of it and none is fitted)	4 / 9
44. `rel`	**The boost that leaves one number alone**. s3/rel. EXACT (the Lorentz transformation in closed form; every residual below is a floating-point residual and is returned rather than asserted away)	4 / 10
45. `pv`	**Detailed balance: the efficiency limit that is an integral, not a rule of thumb**. s3/pv. MODEL (Shockley-Queisser detailed balance with a 5778 K blackbody source: every photon above the gap is absorbed, every recombination is radiative, one junction. The integral is exact for that model; whether the Sun is that blackbody is the model and not the mathematics)	3 / 9
46. `chaos`	**Strange attractors: exact fixed points, exact contraction, and one exponent that has to be measured**. s3/chaos. MIXED (the fixed points and the divergence of the flow are closed form and are returned as such; the largest Lyapunov exponent is a finite-time Benettin estimate and is labelled separately, because there is no closed form for it and pretending otherwise would be the lie)	2 / 8
47. `phorizon`	**The predictability horizon: the one thing a separation rate is for, and the five rates that are not one**. s3/phorizon. MEASURED (the spectrum is a finite-time Benettin estimate with Gram-Schmidt on three tangent vectors and is labelled as such; the sum rule against the divergence, the zero exponent, the Kaplan-Yorke branch and the horizon against an ensemble of twins are all CHECKS on that estimate and are returned as residuals rather than claims)	6 / 22

Continued from previous page

No. / ID	Title, host and declared status	I / O
48. `clock`	**What is a clock? The exponent per WHAT, and when two clocks can be exchanged**. s3/clock. MEASURED (the return map, its exponent and the mean return time are all measured from one trajectory by routes that share nothing but that trajectory: the map exponent never touches a tangent vector and the return time never touches a slope. Their quotient is compared with a number nobody here chose)	3 / 12
49. `circ`	**The same theorem twice — circulation in a superfluid, flux in a superconductor**. s3/circ. MEASURED (the quanta are closed form from SI defining constants; the vortex lattice is BUILT and its velocity field SUMMED over every line, so the agreement with solid-body rotation is a measurement of a sum and not a restatement of Feynman)	6 / 19
50. `lam`	**The cosmological constant, as a length — and the horizon that is four horizons**. s3/lam. EXACT (every quantity is a closed form in Lambda, G, c, hbar and k; nothing here is fitted. The three horizons are three DEFINITIONS and the laboratory computes all of them rather than choosing one)	3 / 19
51. `origins`	**Where the elements came from — three minutes, ten billion years, and one second**. s3/origins. EXACT WHERE IT IS ARITHMETIC ON MASS EXCESSES (every binding energy and Q-value here is a difference of AME2020 mass excesses and carbon-12 is exactly zero by the definition of the atomic mass unit); the freeze-out temperature and the deuterium-formation time are QUOTED, and the two-regime picture of neutron capture is a simplification with a written cost	6 / 20
52. `rungs`	**The ladder — from a triangle to two Hubble constants**. s3/rungs. EXACT WHERE IT IS GEOMETRY, QUOTED WHERE IT IS CALIBRATION (the parsec is derived from the astronomical unit and is a definition; the parallaxes, the Leavitt coefficients, the type Ia absolute magnitude and both Hubble constants are QUOTED. What is COMPUTED is the size of the disagreement and what it would take to remove it)	7 / 20
53. `mainseq`	**Where stars live — and the ratio a galaxy hangs on**. s3/mainseq. EXACT WHERE IT COMPUTES, BOUNDED WHERE IT MATTERS (the nuclear clock and the Eddington ceiling are closed forms; the mass-luminosity relation and the radii are QUOTED empirical fits; the mass-to-light ratio is main-sequence-only and is therefore an UPPER BOUND, and the laboratory declines to model the giant branch rather than invent parameters for it)	7 / 15

Continued from previous page

No. / ID	Title, host and declared status	I / O
54. galrot	**The mass you can see is not the mass that is there**. s3/galrot. EXACT WHERE IT COMPUTES, UNDECIDED WHERE IT MATTERS (the visible-mass curve is Freeman's closed form and is exact; the DISCREPANCY is a measurement and is not in dispute; the halo profiles and the modified acceleration are both FITTED and this laboratory refuses to choose between them, computing and pricing each)	7 / 21
55. exo	**The planets of other stars — a transit, a wobble, and a zone**. s3/exo. EXACT (every quantity is a closed form in the measured period, radii and masses; the overlap of two discs is the exact lens area rather than a ramp. The systems are QUOTED from other people's measurements and nothing here is fitted)	4 / 20
56. cp	**The Boris pusher: an integrator that never lets the particle gain energy**. s3/cp. EXACT (in a pure magnetic field the Boris rotation is an orthogonal transformation of the velocity, so the speed is conserved to floating point at ANY step size and for ANY number of steps; the drifts are the physics and are measured against their closed forms)	7 / 9
57. sc	**Two constants the SI made exact, and one gap that is only a ratio**. s3/sc. MIXED (Phi_0 = h/2e and K_J = 2e/h are EXACT by the 2019 SI, and are reproduced here from h and e rather than quoted; Delta(0) = 1.764 k_B T_c is weak-coupling BCS, which is a theory with a regime, and the status of each output says which it is)	4 / 10
58. ns	**The TOV equation, integrated: a maximum mass that nothing but relativity puts there**. s3/ns. MEASURED (a real fourth-order integration of the Tolman-Oppenheimer-Volkoff equation for a Gamma = 2, K = 100 polytrope; the equation is exact and the numbers are its converged solution, verified against the exactly known Newtonian limit)	2 / 8
59. he	**Carnot: an efficiency that depends on two temperatures and nothing else at all**. s3/he. EXACT (the four legs are closed form and the two isothermal heats are nR T ln(V_b/V_a); the efficiency comes out 1 - T_c/T_h to twelve digits and is not asserted anywhere in the code)	5 / 10
60. eot	**A hole smaller than the wavelength, and the resonance that lets the light through anyway**. s3/eot. MODEL (the surface-plasmon resonance condition of a square hole array with a declared Drude-like permittivity fit; the resonance is solved exactly for that fit, and the fit is the model)	3 / 7

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No. / ID	Title, host and declared status	I / O
61. `qm`	**Split-step: an integrator that cannot lose a particle, because every step is a phase**. s3/qm. EXACT in norm (each half step multiplies by a unit-modulus phase, so the propagator is unitary by construction and the norm is conserved to floating point); SECOND ORDER in the dynamics (Strang splitting, and the error in the observables is the commutator term)	6 / 8
62. `spin`	**One point of the Bloch sphere is a whole circle of states**. s3/spin. EXACT (the fibre, the Hopf map and the Rodrigues rotation are all closed form; every residual below is floating point and is returned)	4 / 9
63. `psr`	**Two numbers off a pulsar, and four you can compute from them**. s3/psr. MODEL (the magnetic dipole braking model with n = 3, R = 10 km, I = 1e45 g cm^2 and orthogonal rotation; every formula is exact for that model and the model is what the limits are about)	4 / 9
64. `qso`	**The luminosity at which a quasar starts blowing itself apart**. s3/qso. EXACT for the limit (L_Edd = 4 pi G M m_p c/sigma_T is closed form and depends only on the mass); MODEL for the accretion efficiency, which is the Schwarzschild ISCO value and assumes a thin disc around a non-spinning hole	3 / 8
65. `rsh`	**Unitarity as an identity: a pole, its mirror, and a modulus that cannot leave one**. s3/rsh. EXACT (the two-pole S-matrix satisfies |S| = 1 on the real axis identically — analyticity puts the mirror pole exactly where unitarity needs it, and no tolerance is involved; the effective-range results are a two-parameter model and say so)	6 / 9
66. `rpd`	**An information ledger whose entries are integers and whose total never moves**. s3/rpd. MODEL (a declared bookkeeping model of what a horizon hides and re-expresses; the counts are exact integers of that model and it is NOT a derivation of black-hole information recovery, which is an open problem this atlas does not claim to have solved)	1 / 9
67. `rmhd`	**Four things a magnetised plasma cannot do, and the numbers that bound them**. s3/rmhd. EXACT (the Rankine-Hugoniot jump conditions, the Alfven speed, the Sweet-Parker rate and the magnetised Rayleigh-Taylor growth rate are all closed form; they are reduced analytic stations and not a self-consistent radiation-GRMHD evolution, which is what the limits are about)	8 / 11

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No. / ID	Title, host and declared status	I / O
68. `and`	**Anderson localisation, and the one energy where the textbook formula is wrong**. s3/and. MEASURED (the Lyapunov exponent is a transfer-matrix product over a deterministic pseudorandom disorder realisation, reported with the standard error of its block means; the Thouless formula it is compared against is a weak-disorder expansion and is exact only in that limit)	4 / 8
69. `sn`	**Nickel to cobalt to iron: three fractions that always sum to one**. s3/sn. EXACT (the Bateman solution of a two-step decay chain is closed form, and the three fractions sum to 1 identically at every time; the half-lives and the decay energies are measured quantities and are quoted as such)	2 / 10
70. `bhr`	**A geodesic integrator, and the exact quadrature that says how well it did**. s3/bhr. MEASURED (a leapfrog integration of the null geodesic equation, returned ALONGSIDE the exact deflection by quadrature so the two can be compared on every call; the exact one is the same lensAlpha the lensing laboratory uses)	2 / 8
71. `psp`	**Determinant one is necessary and not sufficient, and here is the map that shows it**. s3/psp. EXACT (the symplectic condition S^T J S = J is checked entrywise and the determinant is a cofactor expansion; both are closed form and every residual below is floating point)	3 / 8
72. `cps`	**The first law of black-hole mechanics, differentiated analytically and checked to the last bit**. s3/cps. EXACT (the Kerr-Newman horizon quantities and the derivatives of its area are closed form — no finite differences anywhere in the first-law check — and the instrument REFUSES past M^2 = a^2 + Q^2 rather than extrapolating into a naked singularity)	4 / 12
73. `grav`	**A symplectic integrator: momentum exact, energy bounded, and neither of those is an accident**. s3/grav. MEASURED (fourth-order Yoshida composition of leapfrog; linear and angular momentum are conserved to rounding by Newton's third law in the force loop, and the energy oscillates without drifting because the method is symplectic)	4 / 8
74. `qcd`	**A potential whose slope never turns over, which is what confinement means**. s3/qcd. MODEL (the Cornell potential and a one-loop running coupling with a declared Lambda; both are standard phenomenological forms with fitted constants, and the one-loop alpha_s misses the measured value at the Z mass by 14 per cent, which the limits state rather than hide)	2 / 10

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No. / ID	Title, host and declared status	I / O
75. `nul`	**A spinor is a light ray, and a phase is not a direction**. s3/nul. EXACT (every station is a representation identity in flat spacetime, in closed form; the residuals below are floating point and are returned rather than asserted away)	6 / 10
76. `act`	**The Reeb flow is the Hopf flow, and a Legendrian knot is tangent to the plane everywhere**. s3/act. EXACT (the contact identities are evaluated in R^4 in closed form and hold to the last bit; the linking number is a finite Gauss sum after stereographic projection and carries its own discretisation, which is returned separately)	6 / 10
77. `gate`	**The refinement gate: nine claims, graded by whether they converge and whether they are right**. s3/gate. EXACT (the grading rubric is arithmetic on a three-level refinement sequence; what it grades is measured, and the verdicts below are recomputed on every call rather than cached)	6 / 12
78. `bell`	**A bound quantum mechanics saturates and cannot pass**. s3/bell. EXACT (the singlet correlation, the lune solid angle and the CHSH combination are closed form; the Tsirelson bound is saturated to the last bit and the search for a violation is exhaustive over the sampled grid rather than a spot check)	5 / 10
79. `cmb`	**The lowest multipoles, where a mask costs more than the signal**. s3/cmb. MEASURED (the synthesis is exact given the spectrum; the recovery is a midpoint quadrature whose residual falls as the square of the grid and is returned rather than assumed away, and the masked recovery is an f_sky-rescaled pseudo-C_l with NO mode-coupling deconvolution)	6 / 11
80. `ret`	**Two oscillators, one spring, and the exact height of a resonance**. s3/ret. EXACT for the normal modes, the transfer time and the wireless efficiency (all closed form); MEASURED for the trajectory, which is a velocity-Verlet leapfrog whose convergence order against the exact two-mode solution is returned rather than assumed	10 / 20
81. `wave`	**Where the fringes actually are, which is not where lambda L over d says**. s3/wave. EXACT for the order positions (the equal-path locus is a hyperbola and it is solved in closed form, with no small-angle step anywhere) and for the discrete-aperture minima; MEASURED for the peak positions of the actual Huygens sum, whose departure from the ideal locus is returned rather than hidden	7 / 15

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No. / ID	Title, host and declared status	I / O
82. `kin`	**A gas that is not told what distribution to have**. s3/kin. MEASURED. The molecular dynamics is exact hard-sphere mechanics and conserves energy to the last bit, but everything thermodynamic here — the approach to Maxwell-Boltzmann, the pressure, the compressibility factor, the coarse-grained entropy — is a finite sample over a finite time and is returned with the statistic that decides whether it means anything	8 / 18
83. `dip`	**A pattern that is exactly sin squared, and a knot of light that is exactly null**. s3/dip. EXACT for the angular pattern, its normalisation and the omega-to-the-fourth law (all closed form); MEASURED for the pattern integral, the null-field angles of the Rañada knot and its self-linking, all of which are quadratures and are returned with their residuals	9 / 17
84. `emb`	**The atoms of the cut, and the four-manifold they live in**. s3/emb. EXACT. Sym^N(CP1) = CP^N is a theorem and the instrument VERIFIES it by round trip rather than quoting it; the moment polytope, the discriminant and the braid monodromy are closed form; the root finding is a Durand-Kerner iteration whose residual is returned, and the capacity gate is the molecular cut kernel unchanged	10 / 25
85. `qcrys`	**Five-fold symmetry, and the integer matrix that permits it**. s3/qcrys. EXACT for the arithmetic — the orthogonal splitting, the five-fold integer matrix, its order, its trace and the golden equation the inflation satisfies are all closed form and all verified rather than quoted; MEASURED for the patch, which is a finite cut whose atom count and minimum separation are returned as what they are	5 / 21
86. `taub`	**A cone drawn before the orbit, and the orbit that stays on it**. s3/taub. EXACT for the conserved vector and the cone it defines (both closed form); MEASURED for the trajectory, which is a fourth-order Runge-Kutta whose convergence order and whose drift off the cone are returned rather than assumed	7 / 15
87. `skyrmion`	**A charge that is a lattice sum, and a bound the map saturates**. s3/skyrmion. EXACT for the Belavin-Polyakov map, the Bogomolny bound and the Thiele solution (all closed form); MEASURED for the charge and the energy, which are lattice sums over a FINITE window and are returned with the deficit that window costs rather than rounded to the integer they are approaching	11 / 16

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No. / ID	Title, host and declared status	I / O
88. `hopfion`	**An integer read back out of the field that was built from it**. s3/hopfion. EXACT for the field, the preimage curve and the Derrick minimum (all closed form); MEASURED for the Hopf invariant, which is a Gauss linking integral over discretised preimages, and for the Faddeev-Niemi energies, which are finite differences on a finite box and are returned as such	9 / 13
89. `red`	**A kernel that is a surface-to-volume ratio, and the one exponent that is a choice**. s3/red. EXACT for the geometry — the ball volume, its boundary area, their ratio, the closed-form integral and every limiting value are identities on the round three-sphere and are verified rather than quoted; DECLARED MODEL for the redshift itself, which is an ansatz with a free exponent and is NOT a cosmological prediction	4 / 17
90. `mimg`	**Two ways round, and why the far side looks larger**. s3/mimg. EXACT for the geometry — the two arc lengths, their sum, the geodesic sphere area, the angular-diameter distance and the antipodal focusing are identities on the round three-sphere; CONDITIONAL for observability, which is not geometry and is returned as a stated budget rather than as a prediction	4 / 15
91. `quasi`	**Five excitations, and a circulation that does not care what loop you measure it round**. s3/quasi. EXACT — every dispersion here is a closed form of its declared model: the diatomic chain, the nearest-neighbour Heisenberg ferromagnet, the hydrogenic exciton, the weak-coupling Frohlich polaron and the irrotational vortex. What is MODELLED is which model applies to any real material, and that is stated rather than assumed	11 / 22
92. `he3`	**Two phases of one condensate, and an average that is exactly two thirds**. s3/he3. EXACT for the order parameters named. The angular averages are exact rationals, the density of states is a closed form obtained by substitution rather than by sampling, and the node charges are integers. What is MODELLED is the pairing: weak-coupling BCS with a temperature-independent gap magnitude, and only the two bulk phases	5 / 20
93. `nuosc`	**One unitary matrix, three flavours, and a sum that is exactly one**. s3/nuosc. EXACT for vacuum oscillation. Every probability is the closed-form three-flavour expression evaluated at the supplied mixing parameters; the unitarity residual, the triangle closure and the two independent routes to the Jarlskog invariant are returned as measured residuals rather than asserted. What is MODELLED is the propagation: plane waves in vacuum, no matter effect, no wave-packet decoherence	10 / 21

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No. / ID	Title, host and declared status	I / O
94. `flow`	**Dark energy read as a flow of horizon capacity**. s3/flow. MODEL over a declared construction: CPL w(a) = w_0 + w_a(1−a), and N_∂ taken as the reciprocal of Λ_eff by construction rather than by a second formula that could drift from the first. The null and the exchange current are computed, and the null is returned as a measured residual	3 / 12
95. `imp`	**The reflection coefficient, and the two things it secretly is**. s3/imp. EXACT. Every quantity is a closed form in the normalised load impedance: the Cayley map and its inverse, the standing-wave ratio, the rapidity, and the lossy-line transformation. The two cross-laboratory identities are returned as residuals rather than asserted. What is MODELLED is the line: uniform, single-mode, with a real characteristic impedance	5 / 20
96. `adisk`	**A maximum at 49/36, and it is not a fit**. s3/adisk. EXACT for the model. The peak radius, the ISCO, the efficiency and the Eddington luminosity are closed forms with no fitted constant anywhere; the spectrum and its index are quadratures over the disk with the residual reported. What is MODELLED is the disk: geometrically thin, optically thick, locally blackbody, zero-torque inner edge — Shakura and Sunyaev exactly, and nothing beyond it	6 / 18
97. `elens`	**Two images, always, and a difference that is exactly one**. s3/elens. EXACT. The lens equation is a quadratic and every quantity here is one of its closed-form roots or a ratio of them; the three identities are algebraic consequences and are returned as residuals rather than asserted. What is MODELLED is the lens: a point mass, or a singular isothermal sphere, each stated	7 / 16
98. `capacity`	**Capacity selector — which integer sector fixes Λ**. fbs/zpt. DERIVED (Preece & Batenin) + VERIFIED internally; the three gates are CONDITIONAL and are not re-derived here	1 / 11
99. `wot`	**Wheels of Time — where ten traditions stand at an epoch**. cyc/cyc. SOURCE-LOCKED arithmetic over declared eras and turn lengths; the correlations are STIPULATED and are not derived here	1 / 12
100. `s3ns`	**Curl shells of the round three-sphere — dimension, decay, helicity and the band a peak must outgrow**. s3/s3shell. INTRINSIC and measured: every output is computed from a shell BUILT here out of harmonic polynomials on $S^3 \subset \mathbb{H}$. The relative singular branch of the source paper is NOT evaluated and is not an output.	6 / 18

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No. / ID	Title, host and declared status	I / O
101. `s3torus`	**The maximal torus of S³ — two scalar heat equations, and the shell they belong to**. s3/s3torus. INTRINSIC and measured: the reduction is exact for the G_0-invariant sector, and the three identities are published as residuals this laboratory differenced rather than as claims it makes.	5 / 8
102. `tri`	**The Trisphere — the three-sphere whole, and every laboratory that leans on it**. s3/tri. EXACT for the geometry (every invariant in closed form, the multiplicity checked against a count of harmonic polynomials); the census is read from the relations the atlas declares, breadth first from the ten core stations.	3 / 10
103. `nsflow`	**A Navier–Stokes flow on S³, integrated — the exact solution the equation itself confirms**. s3/nsflow. EXACT solution, MEASURED against the partial differential equation: the residual $\partial_t u + \nabla_u u + \nu A u$ is computed by finite differences on S³ and its curl — zero exactly when it is a pressure gradient — is published beside the curl of the convective term alone.	6 / 11
104. `bbh`	**The last merger of the Local Group — Sgr A* and M31*, inspiral, merger and ringdown**. s3/bbh. EXACT at leading order for the gravitational-wave inspiral (Peters 1964, quadrupole waveform); FIT for the end state (numerical-relativity fits for non-spinning holes) and the l = m = 2 ringdown. The galaxies' merger, dynamical friction and stellar hardening are NOT computed.	3 / 12
105. `lgmerge`	**The Local Group's future, by law — Milky Way and Andromeda under dynamical friction**. s3/lgmerge. NUMERICAL: the relative orbit of two extended galaxies (Hernquist haloes, Miyamoto–Nagai discs, Hernquist bulges) integrated by leapfrog with Chandrasekhar friction whose Coulomb logarithm is CALIBRATED once to the N-body merger of van der Marel et al. (2012); the stars in the laboratory are test particles. M33, the LMC, tidal stripping and self-gravity are not included.	3 / 5
106. `discover`	**The Discovery Explorer — the ledger of what the atlas has found**. s3/discover. EXACT as a ledger: counts of the atlas's own registry of discoveries, each tied to a verifier that runs on every build.	1 / 3
107. `nsgal`	**The fluid of the three-sphere — Navier–Stokes in the Spin(4) shells**. s3/nsgal. NUMERICALLY_VERIFIED: a Galerkin truncation to shells $k \le 2$ (52 modes) with an exact Hopf-coordinate quadrature; energy and helicity conserved without viscosity to the time step, the six Killing components to rounding; reproduces the exact solution to 1e-9.	4 / 5

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No. / ID	Title, host and declared status	I / O
108. `nsblow`	**The anatomy of a forced blow-up — the 2026 exponents and the window of the theorems**. s3/nsblow. EXACT_ARITHMETIC on published exponents: the scaling exponents of the September 2026 forced Navier–Stokes construction (A = ½ + h, D = ½ − h, 0 < h < 1/100) put through the classical regularity theorems; the core profile drawn is illustrative, not the construction's.	2 / 7
109. `strobe`	**The Stroboscope — resonances across space, time and parameters**. s3/strobe. EXACT: the carrier frequencies and knots of the exact Navier–Stokes solution on S³ from its own Killing weights, the windings counted on the real flow; the clock resonances scored against a jittered-period null.	2 / 5
110. `vstretch`	**Vortex stretching is a Vandermonde determinant**. s3/vstretch. MEASURED: the total vortex stretching of a finite sum of curl levels on S³ against the Vandermonde law, on the exact Hopf quadrature of the Galerkin basis, with the atlas's own orientation.	2 / 4
111. `triad`	**The tennis racket inside the fluid**. s3/triad. EXACT on the atlas's Galerkin basis of S³: a triad of Beltrami modes is Euler's free top; its coupling S is measured by the exact Hopf quadrature, its period by Jacobi's formula checked against integration, which triads exist by the double Clebsch–Gordan rule of Spin(4).	2 / 8
112. `omnistrobe`	**The stroboscope of everything**. s3/omnistrobe. EXACT arithmetic on stated periods (SI-defined, measured or derived, each with its precision): the display as a stroboscope, continued fractions stopped at the data, an exhaustive lattice for triples and a random-frequency null.	3 / 6
113. `formulas`	**The Formula Portal — the S³ programme's formulas and their checks**. s3/formulas. AUDITED: the corpus (data/formula-atlas.json) with the checks the atlas can run on its own kernels; statements about limits are not checkable and are shown as untouched, never as true.	1 / 2
114. `s3gamma`	**Space forms S³/Γ — Laplace spectra and matched circles**. s3/s3gamma. EXACT: multiplicities by the character sum over Γ's conjugacy classes; the matched-circle criterion θ_min R < 2χ_LSS is geometry; the probability is this atlas's own curvature posterior, interpolated between its five quantiles.	2 / 3
115. `s3tube`	**Geodesic-axis coordinates on S³ — the exact metric, the determinant identity and the curvature that is left**. s3/s3tube. INTRINSIC and exact: the metric, its determinant and the correction operators are proved in the source without any transfer assumption, and the residuals here are measured off the embedding.	4 / 16

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No. / ID	Title, host and declared status	I / O
116. `s3escape`	**Spectral escape on S³ — the rank a bounded-energy peak has to outgrow**. s3/s3escape. INTRINSIC where the projector identities and the closure algebra are concerned; every bound is a NECESSARY condition on an externally supplied peak, and the peak is not evaluated here.	7 / 19
117. `s3kb`	**The chiral Killing–Beltrami law — two coefficients, and the Casimir that makes dissipation six numbers**. s3/s3kb. INTRINSIC and measured. Every statement here is proved on the round sphere without any transfer assumption, and the pointwise identities are re-evaluated by differences rather than recited.	4 / 10
118. `s3lock`	**The terminal lock — three probes, one Casimir gap, a paired Hopf filament and the number that is still open**. s3/s3lock. MEASURED where it can be (the frame bound, its condition number, the Casimir gap, the involution, the Gaudin involution, the winding and the capacity threshold are all computed and compared against their closed forms); SCORED where it cannot be (the two measure functionals are evaluated on the configuration supplied, not on a limit). THE TERMINAL LOCK IS OPEN and this laboratory does not close it.	14 / 39
119. `cyc`	**Cycles of the sky — the beats the sky already has, and one you can set**. cyc/cyc. DERIVED from declared periods; the perihelion advance is the general-relativistic term ALONE and is not the observed rate	3 / 8
120. `nesting`	**The nesting, measured — which structure actually contains which**. obs/cosmicweb. MEASURED over the atlas's own catalogue: no new coordinate, no new extent, only the containment arithmetic that had never been run. A containment is ASSERTED only where the container has a known centroid and a shape a sphere can honestly stand for; every other overlap is returned with the reason it is not a claim.	1 / 10
121. `tomography`	**Shell tomography — what a sky map can and cannot tell you about a closed geometry**. s3/sel. IMPLEMENTED FROM THE DERIVATIONS, NOT QUOTED FROM THE RESULT. Preece & Batenin, 9 September 2026. Every number here is computed from the paper's equations by this atlas, and the verifier reproduces the paper's own coefficient table to the last printed digit, its exact shell ranks for 0 <= L <= 8, its degeneracy test at chi = 0.9, and its two-shell inversion at d1 = 3, d2 = 6, R = 10.	4 / 9

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No. / ID	Title, host and declared status	I / O
122. `localflow`	**The expansion, fitted from this atlas's own galaxies**. obs/rungs. MEASURED. The atlas has quoted a late-time Hubble constant up its own distance ladder for releases, as a calibration, and has held forty-five galaxies at exact positions since v4.213.0. The two had never met. Twenty-four of those galaxies now carry a measured radial velocity, and a velocity against a distance is the Hubble law with nothing else in it.	2 / 7
123. `samelength`	**The same length, written twice — what the three scale tables disagree about**. fbs/zpt. MEASURED, then repaired where the repair was provable. Twelve rows stopped being typed and are computed from the authority the atlas already held; five disagreements are left standing with written reasons, because each is one authority against one measuring a different thing.	1 / 8
124. `localgroup`	**The mass of everything we are bound to — the missing mass one rung above a rotation curve**. obs/galrot. EXACT WHERE IT COMPUTES AND IN TENSION WHERE IT MATTERS. The cycloid is the closed-form two-body radial solution and the bisection on it is exact; the zero-velocity integral is checked against its own closed form in the Lambda=0 limit to ten decimal places at boot. The two estimators disagree by a factor of two and a half and the ratio is PUBLISHED rather than reconciled.	4 / 9
125. `seis`	**Asteroseismology — a mean density from the ringing alone**. obs/seis. SCALED from the solar values; the scaling relations are empirical and are stated as such	3 / 6
126. `gyro`	**Gyrochronology — an age from a rotation period, and where the clock stops**. obs/gyro. EMPIRICAL (Barnes colour term, Ballesteros colour-temperature); the Kraft break comes out 630 K above the measured one and is reported as it comes out	2 / 6
127. `wind`	**The radiative wind — the ceiling photons can push, and what a star spends of it**. obs/wind. DERIVED from single-scattering momentum conservation; the ceiling is a ceiling for SINGLE scattering and real winds may exceed it	3 / 5
128. `fermi`	**The Fermi bubbles — one outflow, dated four ways**. cyc/fermi. KINEMATIC; the total energy is an ORDER OF MAGNITUDE and everything downstream of it inherits that	2 / 9
129. `earth`	**The axis, over two hundred millennia — obliquity, the pole's arc and the equinox drift**. solar/earth. MODEL: Vondrak long-term precession, fitted over +/-200 kyr; the legacy linear engine is offered and is honest only within a few centuries	2 / 4

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No. / ID	Title, host and declared status	I / O
130. `zod3d`	**The zodiac at its measured distances — every figure is deeper than it is wide**. solar/zod3d. MEASURED: 528 stars in the IAU boundaries of the twelve zodiac constellations and Ophiuchus, one self-consistent Gaia DR3 / Hipparcos / SIMBAD-2018 astrometric solution per star by a stated rule, the distance error bar taken from the disagreement of two independent catalogues where both exist, and positions moved with the clock by a port of ERFA eraStarpm checked against the C library to 4e-7 mas. Linear space motion is declared valid to ±1 Myr.	3 / 17
131. `edge`	**Recursion operator $\mathcal{R}$ and the edge-Hamiltonian admissibility kernel**. fbs/zpt. IMPLEMENTED AS WRITTEN + VERIFIED, and the headline result is the manuscript's own NO-GO: the naive closure gives N ≈ 9, not 292. $\mathcal{R}$ is a registry consistent with rung 292; it does not produce it.	8 / 20
132. `anyon`	**Fibonacci anyons — the fusion algebra the golden ladder runs on**. s3/hopf. VERIFIED CATEGORY (F, R, braid, S, T and c = 14/5 all checked against each other). It is NOT a claim that the manuscript's boundary sector IS a Fibonacci liquid — a shared φ is a hint, not an identification.	2 / 11
133. `civplock`	**CIVP CP¹ evaluation lock — a station of the trace-free de Sitter locking chain**. s3/civplock. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 8
134. `civpcut`	**CIVP molecular cut — a station of the trace-free de Sitter locking chain**. s3/civpcut. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 6
135. `civpidx`	**CIVP Jones ladder — a station of the trace-free de Sitter locking chain**. s3/civpidx. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 6
136. `civpa4`	**CIVP A_4 transfer — a station of the trace-free de Sitter locking chain**. s3/civpa4. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 7
137. `civpsel`	**CIVP UV selector — a station of the trace-free de Sitter locking chain**. s3/civpsel. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 6
138. `civpcar`	**CIVP Berry carrier and Capelli transfer — a station of the trace-free de Sitter locking chain**. s3/civpcar. EXACT (closed form, or an identity that holds to machine precision by construction)	3 / 7

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No. / ID	Title, host and declared status	I / O
139. `civpclo`	**CIVP conditional closure — a station of the trace-free de Sitter locking chain**. s3/civpclo. CONDITIONAL (exact kinematics; the six physical certificates are NOT derived here)	4 / 7
140. `polytope`	**Finite subgroups of S³ — the 24-cell, the 600-cell and E8**. s3/hopf. MEASURED (the groups are generated from coordinates and the group axioms, the conjugacy classes and the E8 root system are all computed, never tabulated)	1 / 14
141. `leech`	**The Golay code and the Leech lattice — twenty-four dimensions**. s3/hopf. MEASURED (the code is constructed from a circulant and every invariant — the weight enumerator, self-duality, the Steiner system, the kissing number, the covolume — is counted out of it; the first construction attempted here was WRONG and the weight enumerator is what said so)	1 / 18
142. `scales`	**The atlas on one axis — what it spans, and how one end reaches the other**. s3/scales. MEASURED (every typed instrument that declares a convertible unit is evaluated at its own declared defaults and every such output is converted to SI; nothing here is tabulated, and the Planck anchors are built from hbar, G, c and k_B rather than quoted)	3 / 23
143. `osc`	**Frequency generators — and when they fall back into phase**. s3/osc. MEASURED (the continued fraction, its convergents, the Lagrange number, the three-distance count and the first coincidence are all computed from the ratio; the Hurwitz bound is swept rather than quoted and no ratio has yet fallen below it)	7 / 24
144. `divalg`	**Division algebras and all four Hopf fibrations**. s3/hopf. MEASURED (the tower is built by Cayley-Dickson and every property is measured at each level; nothing is read from a table)	2 / 15
145. `mixmaster`	**Mixmaster — symplectic Misner evolution and the Lyapunov doubling test**. s3/bianchi. MEASURED (eighth-order Yoshida composition; order of convergence, bounded Hamiltonian and the C3 symmetry of the wall potential are all measured, not asserted)	8 / 15

D Relationship Taxonomy and Quantity-Bus Register

Table 20: Operational Nexus relationship classes.

Type	Interpretive contract
`exact`	theorem or exact identity inside the declared mathematical model
`representation`	two representations or bases of a stated structure; not necessarily the same physical system
`invariant`	the same typed invariant is measured or conserved in distinct models
`limit`	one model is a stated limit or effective reduction of another under explicit assumptions
`coupling`	an output, parameter or physical context can feed another model; no automatic unit-safe solver coupling is implied
`causal`	a directed physical succession under the stated scenario; graph traversal remains bidirectional for comparison
`analogy`	shared mathematics or morphology without physical identity
`contrast`	distinguishable explanations, regimes or observables that must not be conflated

The scientific contract has eleven canonical relation types: (1) exact identity; (2) unit, coordinate or observer transformation; (3) isomorphism or equivalent representation; (4) limiting reduction; (5) shared symmetry; (6) shared invariant or conservation law; (7) scale transformation; (8) empirical correlation; (9) causal hypothesis; (10) analogy; (11) contradiction or competing explanation. The source's mapping is explicit: `exact` maps to 1; `representation` to 2–3; `invariant` to 5–6; `limit` to 4 and explicitly 7; `coupling` to 7–8 or a declared model input; `causal` to 9; `analogy` to 10; and `contrast` to 11. This mapping retains the original claim and status. The complete 408 edge records are in `data/nexus-relations.json`.

D.1 Explicit quantity-bus links

These are a different registry. Displayed conversion factors may be rounded; the JSON manifest retains their full precision. A scale converts the source quantity into the target unit only under the declared quantity-kind match. No row authorizes a chain beyond its explicit endpoints.

Table 21: All 57 manifest-reported bus links.

From	To	Kind / target unit	Scale
`gyro.teff`	`seis.teff`	temperature / K	1
`photon.limiting_magnitude`	`rungs.limiting_magnitude`	limiting_magnitude / mag	1

Continued from previous page

From	To	Kind / target unit	Scale
quasi.circulation	circ.circulation	circulation_quantum / m^2/s	1
chaos.lyapunov_max	phorizon.lambda	lyapunov_exponent / 1/time	1
capacity.n_phi	ladder.N	golden_ladder_rung / rung	1
capacity.q	edge.q	boundary_capacity / boundary degrees of freedom	1
bianchi.alpha_final	spectrum.alpha	misner_alpha / ln(volume)/3	1
bianchi.beta_plus_final	spectrum.beta_plus	misner_beta_plus / 1	1
bianchi.beta_minus_final	spectrum.beta_minus	misner_beta_minus / 1	1
bianchi.alpha_final	creation.alpha	misner_alpha / ln(volume)/3	1
bianchi.beta_plus_final	creation.beta_plus	misner_beta_plus / 1	1
bianchi.beta_minus_final	creation.beta_minus	misner_beta_minus / 1	1
bianchi.beta_plus_final	ebk.beta_plus	misner_beta_plus / 1	1
bianchi.beta_minus_final	ebk.beta_minus	misner_beta_minus / 1	1
bianchi.p_plus_final	creation.p_plus	misner_momentum_plus / momentum	1
bianchi.p_minus_final	creation.p_minus	misner_momentum_minus / momentum	1
cosmo.angular_diameter_distance	elens.D_source	angular_diameter_ distance / Mpc	1
qso.luminosity	phot.luminosity	luminosity / erg/s	1
elens.einstein_radius	resolve.angular_size	angular_size / rad	1
phot.flux	photon.flux	flux / W m^-2	1
cosmo.luminosity_distance	phot.luminosity_distance	luminosity_distance / Mpc	1
cosmo.scale_factor	flow.scale_factor	scale_factor / 1	1
adisk.peak_temperature	jeans.temperature	temperature / K	1
jeans.jeans_mass	bht.M	mass / kg	1
kin.pressure	eos.pressure	pressure / Pa	1
eos.number_density	jeans.number_density	number_density / m^-3	1
eos.number_density	qregime.number_density	number_density / m^-3	1

Continued from previous page

From	To	Kind / target unit	Scale
bht.area	infolab.area	horizon_area / m^2	1
anyzoo.topological_entanglement_entropy	infolab.entropy_nats	entropy / nat	1
adisk.peak_temperature	eos.temperature	temperature / K	1
wd.mean_density	eos.density	mass_density / kg m^-3	1
ns.mass	tscale.mass	mass / M_sun	1
qso.luminosity	tscale.luminosity	luminosity / erg/s	1
qso.accretion_rate	tscale.accretion_rate	accretion_rate / M_sun/yr	1
bht.t_evap_yr	tscale.age	duration / yr	1
bb.exitance	sbright.flux	flux / W m^-2	1
bell.solid_angle	sbright.solid_angle	solid_angle / sr	1
phot.apparent_magnitude	sbright.apparent_magnitude	apparent_magnitude / mag	1
ns.mass	wd.mass	mass / M_sun	1
ns.mass	qso.mass	mass / M_sun	1
ns.mass	adisk.mass	mass / M_sun	1
ns.mass	elens.lens_mass	mass / M_sun	1
bht.T_H	bb.T	temperature / K	1
bb.lambda_max	wave.wavelength	wavelength / m	1
bhr.b_critical	lens.b	impact_parameter / r_s	1
ns.radius_km	nuosc.baseline	path_length / km	1
ladder.omega	dip.omega	angular_frequency / rad/s	1
ladder.R	red.radius	curvature_radius / m	1.0570×10^{-25}
ladder.R	mimg.radius	curvature_radius / m	1.0570×10^{-25}
imp.beta	rel.beta	speed_ratio / 1	1
imp.gamma_real	smith.z_re	reflection_coefficient_real / 1	1
imp.gamma_imag	smith.z_im	reflection_coefficient_imag / 1	1
civpsel.q_star	civpclo.q	selected_sector / 1	1
bianchi.beta_plus_final	mixmaster.beta_plus	misner_beta_plus / 1	1
bianchi.beta_minus_final	mixmaster.beta_minus	misner_beta_minus / 1	1
bianchi.p_plus_final	mixmaster.p_plus	misner_momentum_plus / momentum	1
bianchi.p_minus_final	mixmaster.p_minus	misner_momentum_minus / momentum	1

E Principal Formula Register

This is the complete effective principal registry: 126 initial entries and seven appended entries. Expressions below preserve the source's plain-text notation so that a reader can match a formula identifier exactly. Mathematical exposition, symbols, units and scope for central formulas appear in sections 15 and 18 to 21. An internal source tag is not necessarily a bibliographic key. Some formula source strings have no matching key in `SOURCE_MAP`; they remain internal identifiers. A formula's current producer is obtained through `formulaSite(id)` rather than inferred from a prefix. The historical radius expression states an absolute-curvature scale; the closed-branch reconstruction is the conditional procedure in section 18.

Table 22: Principal formula registry, preserving implementation expressions.

ID / source	Source expression / units
`s3.radius` S3	R = (c/H0)/sqrt(|Omega_K|) *Units:* Gly
`s3.volume` S3	V = 2*pi^2*R^3 *Units:* Gly^3
`s3.causal` S3	D_causal = pi*R *Units:* Gly
`s3.circle` S3	C = 2*pi*R *Units:* Gly
`s3.area` S3	A(chi) = 4*pi*R^2*sin(chi)^2 *Units:* Gly^2
`s3.gamma` S3	Gamma(chi) = (chi - sin(2chi)/2)/pi *Units:* 1
`s3.eigen` S3	lambda_beta = (beta^2-1)/R^2 *Units:* Gly^-2
`s3.degen` S3	g_beta = beta^2 *Units:* 1
`s3.curl` S3NS	curl phi = sigma (k+2)/R phi on E_{k,sigma} *Units:* Gly^-1
`s3.shelldim` S3NS	dim E_{k,sigma} = (k+1)(k+3) = (k+2)^2 - 1 *Units:* 1
`s3.stokes` S3NS	A = curl^2 - 4/R^2, so A = k(k+4)/R^2 on E_{k,sigma} *Units:* Gly^-2
`s3.beltrami` S3NS	u(t) = exp(-nu k(k+4) t/R^2) u(0); H = 2 mu E *Units:* 1
`s3.bandrank` S3NS	N_K = (K+1)(K+2)(2K+9)/3 *Units:* 1
`s3.projector` S3NS	||Pi_K||_{1->inf} = N_K/(3 V); ||Pi_K||_{2->inf} = sqrt(N_K/(3 V)) *Units:* Gly^-3
`s3.killing` S3NS	b_s = k^2(k+4)/(4(k+2)R^2 V); b_o = k(k+4)^2/(4(k+2)R^2 V); b_s+b_o = kappa_k/(2V) *Units:* Gly^-5
`s3.gap` S3NS	2 R^2 V / 5 = 1/(b_s(1)+b_o(1)) *Units:* Gly^5
`s3.torusjac` S3NS	lambda_n = 4 nu n(n+2)/R^2 on P_n^{(1,0)}(1-2s) and P_n^{(0,1)}(1-2s) *Units:* Gly^-2
`s3.toruspress` S3NS	p_s = R^2 alpha^2/2 - beta^2/2 absorbs the whole convective term *Units:* 1
`s3.chiral` S3NS	curl e_n^theta = 2(n+1) e_n^z, so e_n^theta +/- R e_n^z lies in E_{2n,+/-} *Units:* Gly^-1
`s3.tubemetric` S3NS	ds^2 = dr^2/a + r^2 dtheta^2 + a dz^2, a = 1 - r^2/R^2, det g = r^2 *Units:* Gly^2
`s3.swirlop` S3NS	L_R^theta = L_0^theta + R^-2 P(E), P(x) = -(x-1)(x+3) *Units:* Gly^-2

Continued from previous page

ID / source	Source expression / units
`s3.eulertower` S3NS	L_theta r^lambda = (lambda^2-1)r^{lambda-2} - (lambda-1)(lambda+3)R^-2 r^lambda *Units:* Gly^-2
`s3.escape` S3NS	N_K >= 3 V eta^2 ||u||_inf^2/||u||_2^2 and N_K >= 3 eta V/V_eff *Units:* 1
`s3.resonant` S3NS	c_* = k/(k+4); sum_{l=1}^{k+1}(2l+1) = (k+1)(k+3) *Units:* 1
`s3.intertwine` S3KB	2 Q(K_tau,v) = ((k+2-2 tau sigma)/(k+2)) [K_tau,v] *Units:* 1
`s3.polarized` S3KB	grad_u v + grad_v u = grad <u,v> - u x curl v - v x curl u *Units:* 1
`s3.curlcross` S3KB	curl(u x v) = -[u,v] for divergence-free u, v *Units:* Gly^-1
`s3.casimir` S3KB	A = -2 V sum_{tau,a} (T_a^tau)^2; Delta_iso = 2 curl^2; A = Delta_iso/2 - 4/R^2 *Units:* Gly^-2
`s3.sixresp` S3KB	2 nu ||Def u||_2^2 = 2 nu V sum_{tau,a} ||Q(K_ a^tau,u)||_2^2 *Units:* 1
`s3.embed` S3KB	||v||_inf <= (R^{s-3}/(3 pi^2)) [zeta(s-2)-zeta(s)] ||Delta_1^{s/2} v||_1, s>3 *Units:* Gly^-3
`s3.zetapole` S3KB	lim_{s->3+} (s-3) C_s(R) = 1/(3 pi^2), independent of R *Units:* 1
`s3.uncert` S3KB	||w||_2^2/||w||_inf^2 >= 3V/dim W, attained by the reproducing-kernel field *Units:* Gly^3
`s3.grass` S3KB	dim Gr(d, E_{k,sigma}) = d((k+1)(k+3) - d) *Units:* 1
`ring.eigen` RING	lambda_l = l(l+2)/R_eff^2; g_l=(l+1)^2 *Units:* Mpc^-2
`ring.zonal` RING	Xi_l(Theta)=(l+1)/(2pi^2) sin[(l+1)Theta]/sin Theta *Units:* 1
`ring.corr` RING	C(Theta)=SUM_l P_l Xi_l(Theta) *Units:* model dependent
`ring.chord` RING	d_l=2 R_eff sin[pi/(l+1)] *Units:* Mpc
`ring.fit` RING	Rhat_eff=SUM_i d_obs_i s_i / SUM_i s_i^2 *Units:* Mpc
`ring.project` RING	r_proj=sqrt(rho^2-h^2), h<rho *Units:* Mpc
`ring.arc` RING	Theta_arc≈2 acos[(rho^2-h^2-Delta r^2)/(2 rho Delta r)] *Units:* rad
`fbs.ladder` FBS	R(N) = l_P*phi^N *Units:* m
`fbs.inverse` FBS	N(L) = ln(L/l_P)/ln(phi) *Units:* 1
`fbs.density` FBS	rho(N)/rho_P = phi^(-3N) *Units:* 1
`fbs.fib` FBS	F_N ~ phi^N/sqrt(5) *Units:* 1
`fbs.coscale` COSCALE	t(N) = t_P*phi^N; R(N)/t(N) = c *Units:* s
`zpe.planck` ZPE	epsilon_bar=0.5*hbar*omega+hbar*omega/(exp(hbar* omega/kBT)-1) *Units:* J
`zpe.spectrum` ZPE	omega_beta=beta*c/R; g_beta=beta^2 on conformal scalar R x S^3 *Units:* s^-1
`zpe.shell` ZPE	E0_beta=0.5*hbar*c*beta^3/R *Units:* J

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ID / source	Source expression / units
zpe.cutoff ZPE	E_bare(M)=hbar*c*M^2*(M+1)^2/(8R) *Units:* J
zpe.casimir ZPE	E_C=(hbar*c/2R)*zeta(-3)=hbar*c/(240R) *Units:* J
zpe.density ZPE	rho_C=E_C/(2*pi^2*R^3)=hbar*c/(480*pi^2*R^4) *Units:* J/m^3
zpe.action ZPE	epsilon_0*(R/c)=hbar/2; E_C*(R/c)=hbar/240 *Units:* J*s
zpe.compact ZPE	2G*epsilon_0/(c^4 R)=(l_P/R)^2=phi^(-2N) *Units:* 1
zpe.fractal ZPE	rho_N proportional phi^(-(d_eff+z)N); d_eff=q-z *Units:* 1
zpe.noether ZPE	d(rho*V)+p*dV=0; p=rho/3 => rho*R^4=constant *Units:* J*m
eph.kepler JPL	M = E - e*sinE (Newton-iterated) *Units:* rad
eph.elements JPL	a,e,I,L,varpi,Omega = value_J2000 + rate*T *Units:* AU/rad/deg
eph.anchor JPL	Earth orbital row = Earth-Moon barycenter approximation *Units:* reference point
eph.season DERIVED	event: lambda_sun(t) = k*90 deg *Units:* date
cycle.phase DERIVED	phase_i(t) = frac((t-t0_i)/P_i) *Units:* turn
cycle.beat DERIVED	P_beat = 1/|1/P_A - 1/P_B| *Units:* days
cycle.commens DERIVED	minimize |n_A P_A - n_B P_B| over bounded integers *Units:* days
gcs.kernel GCS	K(chi) = 2 sin^2 chi / (chi - sin chi cos chi) *Units:* 1
gcs.vobs GCS	V_obs = 2 pi R^3 (chi - 1/2 sin 2chi) *Units:* Gly^3
gcs.redshift GCS	ln(1+z) = INT p(V) K(chi)/R dr = p ln[m(chi)/m(chi0)] *Units:* 1
gcs.hubble GCS	H(ell) = 3 c p / ell (windowed slope) *Units:* 1/s
gcs.tension GCS	H_late/H_CMB = (p_loc/p_rec)(l_rec/l_loc) *Units:* 1
gcs.track GCS	d rho_obs/dt = -Gamma (rho_obs - rho_tot), Gamma = c/(R chi_L) *Units:* 1/s
gcs.twosector GCS	R(t) = R_frac (1-sigma) + R_light sigma *Units:* m
bell.corr HOPF	E(a,b) = -a.b = -cos(theta) = Re exp(i(pi-theta)) *Units:* 1
bell.lune HOPF	Omega[Gamma]=2(pi-theta); U=exp(i Omega/2)= exp(i(pi-theta)) *Units:* sr
bell.chern HOPF	E_q=cos(q(pi-theta))=-cos theta iff q=+-1 (Chern rigidity) *Units:* 1
bell.chsh HOPF	S=E(ab)+E(ab2)+E(a2b)-E(a2b2); |S|<=2 sqrt2 (Tsirelson) *Units:* 1
bell.fs HOPF	d_FS=theta/2; E=-cos(2 d_FS); rho_A=rho_B=I/2 (no signalling) *Units:* 1
bh.hawking BHTREF	T_H = hbar c^3/[4 pi G M k_B] sqrt(1-chi^2)/(1+sqrt(1-chi^2)); Schwarzschild chi=0 *Units:* K
bh.horizon BHTREF	r_+ = GM/c^2 [1+sqrt(1-chi^2)]; A=8 pi (GM/c^2)^2[1+sqrt(1-chi^2)] *Units:* m / m^2

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ID / source	Source expression / units
bh.rotation BHTREF	Omega_H = chi c^3/[2 G M (1+sqrt(1-chi^2))] *Units:* rad/s
bh.entropy BHTREF	S_BH = k_B c^3 A / (4 G hbar) *Units:* J/K
bh.luminosity BHTREF	L_ideal = hbar c^6 / (15360 pi G^2 M^2) *Units:* W
bh.evap BHTREF	t_ideal = 5120 pi G^2 M^3 / (hbar c^4); M(t)=M0(1-t/t_ideal)^{1/3} *Units:* s
qft.unruh BHTREF	T_U = hbar a / (2 pi c k_B) *Units:* K
tov.pressure DERIVED	dP/dr = -(eps+P)(m+4 pi r^3 P) / (r(r-2m)) *Units:* geometrized
tov.mass DERIVED	dm/dr = 4 pi r^2 eps *Units:* geometrized
tov.eos DERIVED	P = K rho0^Gamma; eps = rho0 + P/(Gamma-1) (Gamma=2, K=100 benchmark) *Units:* geometrized
tov.buchdahl DERIVED	2GM/(R c^2) < 8/9 for any static star *Units:* 1
wd.chandrasekhar WDREF	M_Ch = 5.816/mu_e^2 M_sun *Units:* M_sun
wd.nauenberg WDREF	R = (0.0225/mu_e) R_sun sqrt[1-(M/M_Ch)^(4/3)] / (M/M_Ch)^(1/3) *Units:* km
rmhd.flux RMHDREF	dPhi_B/dt=0 in ideal MHD; E+v cross B=0; div B=0 *Units:* normalized
rmhd.helicity RMHDREF	H=int A dot B dV; for two closed tubes H_mut=2 Lk Phi_1 Phi_2 *Units:* flux^2
rmhd.sweet_parker RMHDREF	S=L v_A/eta; delta/L=v_in/v_A=S^(-1/2) *Units:* 1
rmhd.shock RMHDREF	rho_2/rho_1=(gamma+1)M^2/[(gamma-1)M^2+2] *Units:* 1
rmhd.rt RMHDREF	omega^2=A g k - B_parallel^2 k^2/[2 pi (rho_1+rho_2)] (declared normalized planar model) *Units:* rate^2
sn.bateman DERIVED	X_Ni=e^{-t/tNi}; X_Co=tCo/(tCo-tNi)(e^{-t/tCo}-e^{-t/tNi}); X_Fe=1-X_Ni-X_Co *Units:* 1
sn.radioactive DERIVED	Q_radio=M_Ni[e_Ni e^{-t/tNi}+e_Co X_Co(t)] *Units:* erg/s
sn.gamma DERIVED	f_gamma=1-exp[-(t0_gamma/t)^2]; t0_gamma^2= kappa_gamma M_ej/(4 pi v_ej^2) for the declared q=1/3 grey column *Units:* 1 / days
sn.diffusion DERIVED	tau_m=[2 kappa M_ej/(beta c v_ej)]^{1/2}; beta=13.8 *Units:* days
sn.arnett DERIVED	L(t)=2 exp(-x^2) integral_0^x z exp(z^2) Q_dep(tau_m z) dz; x=t/tau_m *Units:* erg/s
sn.photosphere DERIVED	R_ph=v_ph t; T_eff=[L_bol/(4 pi sigma_SB R_ph^2)]^{1/4}; valid while tau_opt>=2/3 *Units:* cm / K
sn.sedov DERIVED	R_ST=xi(E t^2/rho)^{1/5}; xi=1.15167 for gamma=5/3, continuously matched to R_free=v_ej t *Units:* cm

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ID / source	Source expression / units
sn.cooling DERIVED	t_sf=3.61e4 E_51^{3/14} n_0^{-4/7} yr; radius-continuous pressure-driven shell R proportional t^{2/7} *Units:* yr / cm
sn.morphology DERIVED	D_box,N = slope[ln N_occupied(epsilon), ln(1/epsilon)] on the finite rendered ejecta sample *Units:* finite-sample diagnostic
psr.bfield DERIVED	B = 3.2e19 sqrt(P dP/dt) *Units:* G
psr.spindown DERIVED	L = 4 pi^2 I (dP/dt) / P^3; tau = P/(2 dP/dt) *Units:* erg/s
qso.eta DERIVED	eta_Schw = 1 - sqrt(8/9) = 0.05719 (ISCO) *Units:* 1
qso.eddington DERIVED	L_Edd = 4 pi G M m_p c / sigma_T *Units:* erg/s
cmb.harmonics CMBREF	Delta T(n) = sum_lm a_lm Y_lm(n); D_l=l(l+1)C_l/(2 pi) *Units:* microK / microK^2
cmb.estimator CMBREF	C_l_hat = (1/(2l+1)) sum_m |a_lm|^2; D_l=l(l+1)C_l_hat/(2 pi) *Units:* microK^2
cmb.cosmic_variance CMBREF	sigma(C_l_hat)/C_l = sqrt(2/(2l+1)) full sky; approx / sqrt(f_sky) when masked *Units:* 1
cmb.mask CMBREF	a_lm_tilde = (1/f_sky) integral W(n) DeltaT(n) Y_lm(n) dOmega; W couples multipoles *Units:* microK
cmb.parseval CMBREF	integral |DeltaT|^2 dOmega = sum_lm |a_lm|^2 for an orthonormal full-sky basis *Units:* microK^2 sr
topo.s3paths TOPOREF	ell_short=chi R; ell_complement=(2 pi-chi)R; sum=2 pi R *Units:* length
hol.sphere HOLONOMY	Delta alpha = integral K dA = Omega = 2 pi(1-cos theta) (mod 2 pi) *Units:* rad
hol.berry HOLONOMY	gamma_B = -Omega/2; gamma_polygon = -arg product_j <u_j|u_(j+1)>/|<u_j|u_(j+1)>| *Units:* rad
hol.wilson HOLONOMY	C=U_x U_y U_x^-1 U_y^-1; W=Tr(C)/2; W is invariant under C -> H C H^-1 *Units:* 1
hol.wigner HOLONOMY	Omega_W = -2 atan[tanh(alpha/2)tanh(beta/2)] for orthogonal boosts *Units:* rad
hol.mobius HOLONOMY	C=A B A^-1 B^-1 in SL(2,R); det C=1; Tr(H C H^-1)= Tr C *Units:* 1
act.reeb ACTION	lambda_0=1/2 sum_j(x_j dy_j-y_j dx_j); R=2Jz on S^3; lambda_0(R)=1; i_R d lambda_0|_{TS^3}=0 *Units:* 1
act.return ACTION	E(a,b): |z_1|^2/a+|z_2|^2/b=1; phi_R^t=(e^{2it/a}z_1,e^{2it/b}z_2); theta_return=2 pi b/a *Units:* rad
act.legendrian ACTION	gamma_(p,q)=(sqrt(q/(p+q))e^{ipt},sqrt(p/(p+q))e^{-iqt}); lambda_0(gamma_dot)=0 *Units:* 1
act.ks ACTION	X(u)=(2(u1u3+u2u4),2(u2u3-u1u4),u1^2+u2^2-u3^2-u4^2); |X|=|u|^2; dt=|u|^2 ds *Units:* dimensionless
act.maslov ACTION	I=(1/2 pi)oint p dq=E/omega=(n+mu/4)hbar; mu=2 for the 1D oscillator; exp(-i pi mu/2)=-1 *Units:* action

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ID / source	Source expression / units
`nul.pauli` NULLGEO	2 psi psi^dagger = k^mu sigma_mu; det(psi psi^dagger)=0 iff k_mu k^mu=0; psi and exp(i alpha)psi give the same future null direction *Units:* 1
`nul.lorentz` NULLGEO	X=x^mu sigma_mu; X'=A X A^dagger for A in SL(2,C); det X'=det X=x_mu x^mu; A and -A induce the same Lorentz map *Units:* 1
`nul.celestial` NULLGEO	zeta=psi_1/psi_0; zeta'=(c+d zeta)/(a+b zeta); the complex cross-ratio is invariant under the induced Mobius map *Units:* 1
`nul.tetrad` NULLGEO	l=sqrt(2) o o^dagger, n=sqrt(2) i i^dagger, m=sqrt(2) o i^dagger; l.n=1, m.mbar=-1; m -> exp(i chi)m and m tensor m -> exp(2i chi)m tensor m *Units:* 1
`nul.diamond` NULLGEO	u=t-r, v=t+r; U=atan(u/L), V=atan(v/L); T=U+V, R=V-U; ds^2=(L^2/4) sec^2(U) sec^2(V)(dT^2-dR^2) *Units:* dimensionless
`predict.local_holdout` OMNI-TRANSFER	y(x+Delta) ~= y(x) + (dy/dx) Delta; validate against withheld exact y *Units:* station native
`predict.uncertainty` OMNI-TRANSFER	sigma_y ~= |dy/dx| sigma_x + |d2y/dx2| sigma_x^2/2 + epsilon_num *Units:* station native
`predict.inverse` DERIVED	find every bracketed x in [0,1] with wrap(y(x)-target)=0 *Units:* normalized control
`nexus.spectral` DERIVED	three non-trivial eigenvectors of D^(-1/2) A D^(-1/2) *Units:* non-metric interface coordinate
`nexus.resource_allocation` DERIVED	score(a,b)=SUM_common z 1/degree(z) *Units:* structural curation only
`imp.rlc_inverse` TOUCHSTONE	Z=Z0(1+Gamma)/(1-Gamma); Im Z=omega L-1/(omega C) *Units:* ohm/H/F
`bh.thermal_peak` HAWKING-REFERENCE	x=h nu/(k_B T); x_peak=2.821439372 for x^3/(exp x-1) *Units:* 1/Hz

F Exports and Reproduction Records

Literal export filenames implemented in the release HTML are listed with source line anchors. Dynamic names additionally include selected-object runtime contracts, laboratory operational forecasts and predictive-reach records. Preserve MIME type and schema when parsing an export; a CSV loses nested contract context unless accompanied by metadata.

Table 23: Literal export filenames.

Filename	HTML line
atlas_bookmarks.json	108549
cmb_multipole_experiment.json	105654
fbs3r-phi-ladder-benchmark.csv	86204
fbs3r_causal_graph.json	106288
fbs3r_civp_corner_locking.json	82750
fbs3r_contact_action_observatory.json	81136
fbs3r_development_audit.json	95659
fbs3r_device_sensor_snapshot.json	95122
fbs3r_field_snapshot.json	92875
fbs3r_holonomy_observatory.json	80669
fbs3r_invariant_nexus.json	84024
fbs3r_runtime_atlas_manifest.json	19289
fbs3r_smith_measurement_assimilation.json	105213
fbs3r_spinor_lightcone_observatory.json	81543
fbs3r_symmetry_discovery.json	80291
fbs3r_vector_layers.json	95658
formula_registry.json	94430
planck-casimir-zero-point-fractal-ladder.csv	86214
prediction_gate_audit.json	105304
radiation_mhd_helicity_experiment.json	105738
s3-lts-fbs3r-state.json	86154
s3-lts-object-relations.json	18964
s3_projected_ring_consistency_snapshot.json	86249
s3_projected_ring_mode_table.csv	86255
source_registry.json	94434
validation_report.json	94438
xr_diagnostics.json	93426
xr_interaction_debug.json	94733
xr_session_capabilities.json	94424

The common machine report and Markdown surfaces are distinct from laboratory-specific exports. Nexus exports contain claims and graph semantics; runtime manifests contain registry state; field and FBS snapshots contain model diagnostics/state; benchmark CSVs contain rows for a particular calculation. None is a universal serialization of all seven worlds. A minimal reusable record includes build, exact identifier, full inputs, outputs, assumptions, source tags and verifier references. Keep camera and rendering state as an additional layer.

G Source Tags and Bibliographic Boundaries

The following 41 records are the internal source map exactly in release 4.352.0, with English labels. These labels distinguish catalogue/standard references, published model anchors, project constructions and derived values. Incomplete external metadata has not been supplemented by guesswork. Source names and historical numeric claims inside a label retain their source scope and can be older than a later implementation patch.

Table 24: Internal source identifiers and their source-provided descriptions.

Key / visible tag	Source description
`GAIA` Gaia	ESA Gaia mission: Gaia DR3 five-parameter astrometry (Gaia Collaboration, Vallenari et al., 2023) for the zodiac and constellation stars, and Gaia DR2 parallaxes where the Stellarium 23.4 catalogue adopts them for the Hipparcos stars to V = 7
`HIP` Hipparcos	The Hipparcos and Tycho Catalogues (ESA 1997, ESA SP-1200) and Hipparcos, the new reduction of the raw data (F. van Leeuwen 2007): parallaxes, their standard errors and proper motions, with J2000 positions, V and B−V via d3-celestial 0.7.35 (BSD-3-Clause, © Olaf Frohn)
`SIMBAD` SIMBAD	SIMBAD astronomical database (CDS, Strasbourg), the 2018 compilation distributed with the Swiss Ephemeris as sefstars.txt: astrometry and radial velocities where Gaia and Hipparcos give none
`S3` S3-v38	Preece & Batenin 2026, ”Conditional Reconstruction of a Quasi-Local Observable Domain within Compact S³ Geometry” v38 — fiducial benchmark table (conditional reconstruction, NOT a topology detection)
`FBS` FBS3R	Preece & Batenin 2025, ”Fractal Bindu (S³) of Reality” — PhilArchive PREFBS / figshare 28631525 / OSF vdfwh: R(N)=l_P·φ^N, published recombination-level label N_rec=266, $\rho \propto \varphi^{-3N}$, E_n∝F_n. The N_rec label is not derived from the separate CoScale clock t(N)=t_P·φ^N.
`GCS` S3-GCS	Preece & Batenin 2026, ”Geometric–Causal–Spectral Cosmology on a Compact Three-Sphere S³: Kernel, Causal Light Front, Density Tracking, and Redshift as a Density-Gradient Integral” — conditional phenomenological framework: K(χ), V_obs, ln(1+z)=∫p·K/R dr, H(ℓ)=3cp/ℓ. NOT a ΛCDM replacement, NOT a parameter derivation.
`RING` S3-RING	Preece & Batenin 2026, ”Compact S3 Spectral Consistency Analysis of Ring-like Ultra-Large-Scale Structure at z≈0.8” — projected-shell consistency analysis. R_eff is local/effective, mode assignments are empirical, and no topology detection or Bayes factor is claimed.
`LOPEZ` Lopez+	Lopez, Clowes & Williger — OBSERVATIONAL DISCOVERY papers (the authority): ”A Giant Arc on the Sky” (2022, 4.5σ), ”A Big Ring on the Sky” (2024, 4.10σ), Lopez & Clowes ”A Giant Ring on the Sky” (2026, 4.73/4.47σ). Mg II absorbers in SDSS quasar spectra, redshift slice z=0.722–0.862, mean z=0.802.

Continued from previous page

Key / visible tag	Source description
S3KB S3-KillingBeltrami	Batenin & Preece 2026, ”Killing–Beltrami algebra and exact reductions of Navier–Stokes flow on the three-sphere” — the exact chiral intertwining 2Q(K_τ,v) = ((k+2−2τσ)/(k+2))[K_τ,v] and what follows from it: the quadratic-Casimir representation of the Stokes operator, the sharp projector and concentration constants, the global shell embedding for s > 3 with residue 1/3π², and the paper‘s own negative results — that the exact reductions form Grassmannians rather than a finite list, that Γ → 0 is not a spectral dichotomy, and that bounded energy with an unbounded peak does not force spectral entropy. Every statement is INTRINSIC to the round sphere and logically independent of the conditional singularity transfer.
S3NS S3-NavierStokes	Preece & Batenin 2026, ”Finite-Dimensional Spectral Closure and a Relative Forced Singular Branch for Navier–Stokes on the Round Three-Sphere”, with its research addendum — the INTRINSIC half is used here: the signed curl shells E_{k,σ}, the Ebin–Marsden operator A = curl² − 4/R², the exact Beltrami and maximal-torus solution families, the Killing interaction coefficients, the sharp band-projector norms and the finite-sector exclusion. The paper‘s RELATIVE singular branch depends on an external Euclidean forced-blowup package (assumptions S_0…S_4) which this atlas does not hold, does not evaluate and does not publish any number from.
HOPF Hopf-Bell	Preece & Batenin 2026, ”A Hopf-Bundle Holonomy Representation of Spin-Singlet Bell Correlations” — E(a,b)=−cosθ=Re·exp(i∮A) via a Bloch-sphere loop of solid angle Ω=2(π−θ); Chern rigidity q=±1; CHSH→Tsirelson 2√2. A geometric representation of standard QM — NOT a hidden-variable model, no new prediction.
JPL JPL-approx	JPL approximate planetary ephemerides (Standish/Williams), Keplerian elements & century rates, J2000 — valid 1800–2050 AD to ~arcminutes. The published Earth row is the Earth–Moon barycenter; this display uses it as an explicitly labelled approximation to Earth-center position.
MEEUS Meeus	J. Meeus, Astronomical Algorithms — low-precision lunar theory, principal periodic terms (~0.1–0.3°)
CODATA CODATA	CODATA 2018: Planck length 1.616255×10^{-35} m, Planck time 5.391247×10^{-44} s
IAU IAU	IAU/J2000 reference frames and NAIF PCK rotational elements used here: mean obliquity 23.4392911°; galactic pole RA 192.8595°, Dec +27.1283°; Sgr A* direction; low-precision Earth pole/W display model; Sun pole RA 286.13°, Dec +63.87° and Carrington W. NAIF cautions that high-accuracy Earth orientation requires EOP-driven ITRF93 binary PCK data.
IERS IERS	IERS Conventions 2010 / IAU 2000–2006: rigorous Earth orientation is the time-dependent ITRS→GCRS transformation Q(t)R(t)W(t), including celestial-pole motion, Earth rotation and polar motion. The cycle laboratory exposes only a labelled mean-precession diagnostic, not a precision Earth-orientation service.
CANON NASA-Canon	NASA/GSFC Five Millennium Canon of Solar Eclipses (Espenak & Meeus) — catalogue dates, types & greatest-eclipse times

Continued from previous page

Key / visible tag	Source description
ASTRO astro-std	Standard astronomy: galactic year ≈230 Myr (225–250 quoted range); axial precession 25,772 yr; bright-star positions curated to ~0.1°; Saros 6585.32 d; Metonic 19 yr
FLOW CF4-flow	Tully et al. 2014 Laniakea peculiar-velocity basin; Tully et al. 2022 Cosmicflows-4; Valade et al. 2024 probabilistic basins; Stiskalek et al. 2026 scale-dependent Great-Attractor convergence; Hollinger et al. 2026 CF4++/Vela reconstruction; Pomarède et al. 2026 Sloan BoA cosmography; Pomarède & Tully 2020 South Pole Wall; Hoffman et al. 2017 Dipole Repeller. Displayed streamlines are reconstruction-inspired orientation aids, never measured galaxy trajectories.
LSS LSS-survey	Large-scale-structure surveys (observational): Böhringer et al. 2025 CLASSIX X-ray cluster superstructures (Quipu); Bagchi et al. 2017 Saraswati supercluster (SDSS Stripe 82); Kirshner et al. 1981 Boötes Void. Published directions/extents; centroids of extended/filamentary structures are approximate and 3D shape is not asserted.
JWST_LRD JWST-LRD	Juodžbalis et al. 2026 (Nature); Abell 2744−QSO1; direct dynamical black-hole mass measurement in a little red dot at z=7.04; JWST/NIRSpec IFU; MOKA3D; MAST; Zenodo reduced products.
DRD DRD	Preece & Batenin 2026, ”Dependently Reconstructible Distinguishability (DRD): A Functorial Theory of Information Carriers, Graded Spectral Capacities, and Causal Integration on Compact S^3 Geometry” — exact graded capacities N_0,N_TT,N_D on S^3_R; PIR(T) causal integration with t_coh=πR/c; suspension chain $S^0 \to S^1 \to S^2 \to S^3$. Mode counts are NOT bit counts; the gravitational selector Σ_grav is the stated open problem; the volume–area tension is stated, not resolved.
COSCALE CoScale	Preece & Batenin 2026, ”Causal Co-Scaling of Distance and Reconstruction Time in Fractal S^3 Geometry” — axiom R_N=l_P·φ^N, t_N=t_P·φ^N $\Longrightarrow$ R_N/t_N=c at every rung. A causal-admissibility IDENTITY (t_P=l_P/c), explicitly NOT a derivation of c; $N_\star$=207 is an internal reference index (≈1 light-second), NOT recombination; the φ-log-periodicity signature tested against DESI DR2 BAO + Pantheon+ with null results.
ZPE ZP-S^3	Planck–Schrödinger–Casimir zero-point ladder added 2026-08-04 — derived mathematical synthesis, not an observational discovery: conformally coupled massless scalar on round S^3 has ω_β=βc/R and g_β=β^2; every mode carries ½ħω; the bare sum diverges, while zeta/Abel–Plana renormalization gives the scalar Casimir benchmark E_C=ħc/(240R). The φ-rung laws and action/compactness identities are algebraic consequences of this declared model. They do not identify vacuum energy with the cosmological constant and do not establish quantum gravity.

Continued from previous page

Key / visible tag	Source description
SEL S3-SEL	Preece & Batenin 2026, ”Causal–Spectral Distinguishability Selection on a Conditional Round S^3 Carrier” (Status-Preserved Full Edition v14) — closed selector theorem $0 \le \Sigma_{grav} = B_{phys}/B_{causal} \le 1$ with $B_{causal} = b_P \cdot \Gamma(T) \cdot N_P(\varepsilon)$ and ceilings B_{Bek}, B_{holo}; two-phase (τ, χ) portrait with ridge $b_P \cdot \Gamma \cdot N_P = B_{ceil}$ and resolution threshold ε_{min}; PROOF-AUDIT calculus (filtering load $\mathcal{L} = B_c/B_{ceil}$, log pressure $\mathfrak{P} = -\ln\Sigma$, universal cubic $\Sigma = \min(1, u^3)$, curvature-free invariant $\mathcal{J}_P \to 2 b_P \cdot C_P/3\pi$, distance law $p(\theta) = (2/\pi)\sin^2\theta$ with Γ as its CDF, golden φ-ladder scaling); algebraic invariant network ($\rho_{scr} = c^4/4\pi^2 G r^2$, $\Lambda_{scr} = 2/\pi r^2$, $\rho_{scr}/R_3 = c^4/24\pi^2 G$, $\Omega_{scr}(c/H_0) = 2/3\pi \approx 0.212$ — order-of-magnitude only, NOT a dark-energy derivation); trisphere-screen transform T(r); curvature-input ledger. $R_{curv} \neq r_D$; T(r) is a comparison language, NOT a topology theorem; $\Sigma \ll 1$ does not prove decoherence; τ^* is NOT a universal transition time; ε_{min} is NOT modified Heisenberg.
CURV ΩK-obs	Observational curvature constraints (posterior/model-dependent, sign open): Planck 2018 VI (CMB alone $\Omega_K = -0.044^{+0.018}_{-0.015}$; +lensing -0.011 ± 0.007; +lensing+BAO $+0.0007 \pm 0.0019$), Planck PR4/Tristram et al. 2024 (-0.012 ± 0.010), DESI 2024 VI BAO+CMB ($+0.065^{+0.068}_{-0.078}$), Zhai 2024 non-CMB ($+0.108 \pm 0.056$); Di Valentino+ 2020, Handley 2021, Efstathiou & Gratton 2020 on the closed-Universe tension. $R_{curv} = c/(H_0\sqrt{\lvert\Omega_K\rvert})$. NO row proves global topology.
PDE PDE-FDM+MC	Eight reference laboratories in reduced units: 7-point Cartesian finite differences for heat, damped waves, Gray–Scott chemistry, electrostatic/Newtonian Poisson fields, Schrödinger wave packets and FitzHugh–Nagumo excitable media; periodic Metropolis Monte Carlo for the 3D Ising model. Solver-specific boundaries and live invariants are disclosed. These are reproducible educational/reference models, not calibrated material, clinical or engineering predictions.
QMNUM TDSE-FFT	Feit, Fleck & Steiger, J. Comput. Phys. 47 (1982), symmetric split-step Fourier propagation. This app uses a finite periodic grid in reduced units; discretisation, boundary and splitting errors remain.
QCD PDG-QCD	Particle Data Group QCD review plus lattice string-breaking studies (Duncan–Eichten–Thacker; Bernard et al.). Cornell and one-loop curves are reference/phenomenological models; colour motion and fragmentation are schematic.
LABREF REF-LABS	Canonical benchmark equations and historical reference models named in each laboratory. Reduced demonstrators are not calibrated material, clinical or engineering predictions.
BHTREF BH-THERMO	Hawking and Bekenstein–Hawking relations for isolated Kerr/Schwarzschild horizons. The horizon area, angular velocity and surface-gravity temperature are analytic Kerr relations; the displayed evaporation worldtube is explicitly Schwarzschild-only. Greybody factors, particle degrees of freedom, accretion, charge, spin-down and the final quantum-gravity phase are not solved.

Continued from previous page

Key / visible tag	Source description
RMHDREF RMHD-HEL	Ideal/resistive MHD reference observatory: Alfvén frozen-flux transport, gauge-safe closed-domain magnetic helicity, Sweet–Parker scaling, Rankine–Hugoniot shocks and linear magnetized Rayleigh–Taylor growth. Every station is a reduced analytic or kinematic model, not a full radiation-GRMHD simulation.
WDREF WD-NAU	Nauenberg 1972 analytic zero-temperature white-dwarf mass–radius approximation with Chandrasekhar mass M_Ch=5.816/μ_e² M_sun. It omits finite-temperature envelopes, rotation, magnetic support, crystallization, Coulomb corrections and calibrated stellar evolution.
CMBREF PLANCK-CMB	Planck 2018 V/VII likelihood and large-angle isotropy/statistics context; HEALPix spherical-harmonic conventions. The laboratory generates a deterministic synthetic Gaussian low-ℓ realization and an explicit reduced plateau model, never a Planck map or likelihood product. Cosmic variance, masks, beam/noise, foregrounds, transfer functions and look-elsewhere choices remain separate.
TOPOREF COSMIC-TOPO	Round-S³ geodesic and cosmic-topology reference model. Multiple observable images require sufficient conformal lookback time and a detectable compact topology; no ghost image or matched-circle topology is claimed as observed.
NEXUS INV-NEXUS	Invariant Nexus — curated, typed cross-laboratory relation layer derived from the atlas contracts. Edge classes keep theorem/identity, representation, shared invariant, limiting relation, coupling, causal succession, analogy and contrast distinct. The 3D embedding and graph distance are interface coordinates, never physical distance, evidence strength or causal effect size.
SYMDISC SYM-DISC	Symmetry Discovery Chamber — deterministic finite-orbit candidate tests over exact SO(3), SO+(1,1), U(1), symplectic and Möbius actions. Mathematical anchors: PDG 2025 kinematics and NIST DLMF §1.9; recent symmetry-discovery research motivates the invariant-first instrument, but no machine learning runs in this browser lab.
HOLONOMY HOLONOMY	Holonomy Observatory — exact mathematical anchors: Simon (1983), Berry phase as line-bundle holonomy; Berry & Visser (2021), non-collinear Lorentz boosts and Wigner rotation; Rodríguez et al. (2024), spherical spin parallel transport and solid-angle holonomy; Dai et al. (2026), non-Abelian geometric cycles and gauge-invariant Wilson-loop trace. The browser stations are mathematical models and finite numerical checks, not measurements or claims of new physical effects.
ACTION CONTACT-ACTION	Contact & Action Observatory — exact mathematical anchors: Boothby & Wang (1958), contact manifolds; Kustaanheimo & Stiefel (1965), spinor regularization of Kepler motion; Arnol'd (1967), the characteristic class entering quantization conditions; Hofer, Wysocki & Zehnder (1998/2003), global disk-like sections and Reeb dynamics on sphere-like energy surfaces; Moser (1970), regularization of the Kepler problem. The five stations are declared mathematical models with deterministic finite checks, not experimental evidence, a general contact-topology solver or a proof about arbitrary Hamiltonian systems.

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Key / visible tag	Source description
NULLGEO SPINOR-NULL	Spinor & Light-Cone Observatory — exact mathematical anchors: Newman & Penrose (1962), null tetrads and spin coefficients; Penrose (1963), conformal null infinity; Penrose (1967), twistor and conformal spinor geometry; Oblak (2015), the explicit SL(2,C), Lorentz and celestial-sphere correspondence. The five stations are deterministic mathematical identities in flat spacetime or a declared radial conformal section, not observations, a curved-spacetime solver, a quantum-gravity theory or evidence that a qubit is a photon.
DERIVED derived	Derived live inside this app from the tagged formulas — no fitting, no hidden parameters

H Routes, Addresses and Catalogue Navigation

World routes take the form `#/world/<world>`; laboratory routes take `#/world/<world>/lab/<lab>`. All seven and all 133 laboratory routes are linked in the registry appendices. Exact identifiers survive translation; visible titles can change with language. A link may restore retained state, so a world default is not a guarantee that every visit starts at the same laboratory or camera.

```
https://4vmp5pkwz4-pixel.github.io/HCC/#/world/solar
https://4vmp5pkwz4-pixel.github.io/HCC/#/world/s3/lab/hol
https://4vmp5pkwz4-pixel.github.io/HCC/#/world/cyc/lab/chronometry
https://4vmp5pkwz4-pixel.github.io/HCC/agent.html
https://4vmp5pkwz4-pixel.github.io/HCC/api/agent.json
https://4vmp5pkwz4-pixel.github.io/HCC/api/manifest.json
```

Formula addresses, station addresses and object-selection keys are separate from these routes. Query the formula and station APIs to resolve their current locations. A copied world URL should not be advertised as a complete run record. The public URLs are live services; the companion data snapshots identify the documented release.

I Interaction Reference

Table 25: Interaction layers and their scientific consequence.

Action	Effect and reproduction requirement
World/lab selection	Changes context; retain the route and exact laboratory identifier.
Object selection	Changes semantic focus/inspector; retain the object key and source tag.
Orbit, pan, zoom	Changes view; no automatic change to model frame or physical scale.
Parameter control	Changes a declared model input or presentation setting; distinguish them by contract.
Time/play/speed	Changes the relevant clock; retain epoch, time variable and scale.
Reset	Record whether model, camera or both were reset.
Formula/contract lens	Exposes definitions and missing metadata; opening it does not validate a model.
Nexus focus/filter/path	Selects a typed comparison; path distance is non-physical.
Multiview	Composes contexts; synchronization does not equate native clocks or spaces.
Export/report	Produces a scoped record; preserve its schema and build.
Touch interaction	Uses responsive panels, gestures and device-specific layout controls.
XR enter/exit	May change presentation mode; record support and post-transition state.

J Machine-Interface Reference

The browser API belongs to a loaded application. The static discovery document describes resources and capabilities; it does not make every method remotely callable over HTTP. The following examples illustrate instrument discovery, evaluation and contract access.

```
await HCC_API.ready({timeout: 10000});
const health = HCC_API.health();
const capabilities = HCC_API.capabilities();
const instruments = HCC_API.list();
const contract = HCC_API.describe('hol');
// Build an explicit input object from contract.inputs before evaluation.
const result = HCC_API.evaluate('hol', input);
const site = HCC_API.formulaSite('s3.radius');
const census = HCC_API.formulaCensus();
const matches = HCC_API.formulaFind('hopf');
const stations = HCC_API.stations.of('hol');
const fp = HCC_FIRST_PRINCIPLES.lab('hol');
const audit = HCC_FIRST_PRINCIPLES.audit();
```

`ready` waits for registry readiness and can time out. It does not assert that all verifiers passed. `describe` retains the historical schema name `hcc.laboratory/1` while addressing a typed instrument. The manifest's host laboratory and instrument ID are not necessarily equal. Use `instruments.has`, `instruments.count`, `labs.list`, `labs.get`, `labs.count` and `labs.computational` to inspect the separate layers.

The public agent page documents the local ES-module SDK:

```
import { connectAtlas } from './api/agent-client.mjs';
const atlas = await connectAtlas(
  'https://4vmp5pkwz4-pixel.github.io/HCC/');
atlas.search({query: 'anyon'});
atlas.describe('anyon');
// Forecast and audit availability depends on the documented SDK contract.
```

A separately launched local server is documented as `nodeserver/server.mjs`, with MCP at `http://localhost:8974/mcp`. Its named operations include listing/describing/running laboratories, sweeps, retrieving/cancelling runs, validating runs, exporting artifacts and listing connections, measurements and open problems. These are server operations; the public site's descriptor explicitly advertises no public MCP endpoint. Authentication, deployment and network exposure cannot be inferred from a local example.

The companion `data/agent-discovery.json` preserves the release discovery document. For exact method arguments beyond the examples above, use the current descriptor or source rather than inventing a schema from a method name. Unknown domains, missing units and unavailable capabilities are reasons to stop that computation and report the boundary.

K Validation Architecture Reference

The Atlas separates registry integrity, scientific contracts, numerical verification, platform interaction and empirical validation. Each layer has a distinct object and scope; evidence from one layer does not establish the claims of another.

Table 26: Atlas validation layers, mechanisms and scope.

Layer	Object and mechanism	Scientific or operational scope
Release identity	Version, build, commit and registry hashes.	Associates contracts and outputs with a specific executable release.
Catalogue	World, laboratory and instrument registries; routes and identifiers.	Addressability and catalogue coverage.
Formula registry	Formula identifiers, producer locations and source tags.	Traceability of expressions and computed quantities.
Nexus	Relation types, endpoint identifiers, claims and direction flags.	Preserves the stated meaning of scientific comparisons.
Quantity bus	Quantity kinds, units, conversion factors, input domains and refusal rules.	Admissibility of a declared numerical transfer.
First-principles contracts	Native-space metadata, parameter semantics and explicit undeclared states.	Completeness of scientific descriptions for interpretation and machine use.
Numerical verification	Residuals, conserved quantities, limits, tolerances and convergence checks.	Correctness of the specified calculation within its domain.
Desktop and mobile	Navigation, selection, controls, responsive layouts and state transitions.	Interaction with the same scientific model across input devices.
WebXR	Capability diagnostics, session gates, stereo presentation and controller interactions.	Immersive access and preservation of model state across presentation modes.
Empirical validation	Model observables, calibrated data, uncertainty and comparison assumptions.	Adequacy of the model for a defined observational question.

L Interface Illustration Index

The interface illustrations accompany the corresponding world, laboratory and machine-access descriptions. The index identifies their source files and locations in the manuscript. The agent-access figure is a schematic overview of the three supported access modes.

Table 27: Interface illustrations and corresponding contexts.

File	Purpose / figure
`figures/01-solar.jpg`	Solar entry and epoch controls (Figure 12)
`figures/02-catalogue.jpg`	Catalogue and seven worlds (Figure 2)
`figures/03-nexus.jpg`	Nexus graph census and typed controls (Figure 8)
`figures/04-multiview.jpg`	Four-tile comparison interface (Figure 11)
`figures/05-cycles.jpg`	Cycles controls and labels (Figure 13)
`figures/06-horizons.jpg`	Horizons and domain readout (Figure 14)
`figures/07-fbs3r.jpg`	Golden-ladder level controls (Figure 16)
`figures/08-field.jpg`	Field solver controls (Figure 17)
`figures/09-fractal.jpg`	Fractal mode context (Figure 18)
`figures/10-s3.jpg`	S³ current radius and section view (Figure 15)
`figures/11-holonomy.jpg`	Holonomy station and diagnostics (Figure 4)
`figures/12-contract.jpg`	First-principles contract lens (Figure 7)
`figures/13-xr.jpg`	Desktop XR support diagnostic (Figure 20)
`figures/14-agent-en.tex`	Agent workspace access schematic (Figure 5)

References and source records

References

[1] S³ LIGHT-TRISPHERE implementation. Executable HTML, release 4.352.0, build geometry-evidence-2026.10.02.1; index.html at commit a9a219c6de92e1a9db2ad3f898ccdd626c7af3db. https://github.com/4vmp5pkwz4-pixel/HCC/blob/a9a219c6de92e1a9db2ad3f898ccdd626c7af3db/index.html

[2] Generated Atlas manifest. api/manifest.json at the pinned commit; schema and counts preserved in the data supplement. https://github.com/4vmp5pkwz4-pixel/HCC/blob/a9a219c6de92e1a9db2ad3f898ccdd626c7af3db/api/manifest.json

[3] Atlas structural map. 2026_10_03_S3_LIGHT_SPHERE.docx; structural list and descriptive prose. Internal project document.

[4] Atlas scientific contract. SCIENTIFIC_CONTRACT.md at the pinned repository snapshot; datum statuses and relation taxonomy. https://github.com/4vmp5pkwz4-pixel/HCC/blob/a9a219c6de92e1a9db2ad3f898ccdd626c7af3db/SCIENTIFIC_CONTRACT.md

[5] Atlas agent discovery. api/agent.json and agent.html at the pinned repository snapshot. https://4vmp5pkwz4-pixel.github.io/HCC/api/agent.json

External scientific anchors and incomplete publication records are retained verbatim in Appendix G. Their internal tags are not substituted with fabricated bibliographic metadata.